Supplementary methods

BAMSE cohort description

The BAMSE study (Barn [Child], Allergy, Milieu, Stockholm, Epidemiological study), a Swedish population-based birth cohort, recruited 4,089 infants from inner-city, urban, and suburban districts of Stockholm, Sweden, between February 1994 and November 1996. The cohort followed these individuals from birth through 24 years of age [1,2]. Information on demographics, lifestyle, and health conditions was collected via parental questionnaires administered at recruitment and during follow-up visits when the children were aged 1, 2, 4, 8, 12, and 16 years. The participants themselves completed the questionnaire at ages 12, 16, and 24 years [3]. The study was approved by the Regional Ethical Review Board in Stockholm (Ref 2016/1380-31/2). Parental informed consent was obtained at the time of inclusion and again when the participants were 4 and 8 years old. Participants themselves provided informed consent at ages 12, 16, and 24, all in accordance with the Helsinki Declaration. Asthma was defined as either ever doctor's diagnosis of asthma together with symptoms of breathing difficulties in the last 12 months prior to the date of questionnaire 24 or use of asthma medicine occasionally or regularly in the last 12 months. For the current manuscript only clinical characteristics and nasal gene expression data from 24-year old subjects was used. RNA sequencing quality control and processing steps are described in supplementary methods section. T2-high asthma was defined as blood eos $>0.3x10^9$/L and, when measured, FeNO > 25. T2-low was defined as blood eos $< 0.15x10^9$/L and, when measured, FeNO < 25. The intermediate group included the remaining patients with asthma. In total, we analyzed 336 samples from the BAMSE cohort: T2-high (n=6), T2-low (n=24), intermediate (n=24), and controls (n=282). Due to the small size of the T2-high group, it was combined with the intermediate group for the replication analysis, and it will be referred to as T2-high asthma in the rest of the manuscript.

RNA sequencing

Nasal brush samples were obtained from 661 subjects randomly selected among the participants in the clinical examination of the 24-year follow-up of patients in the BAMSE cohort. Total RNA was extracted using the AllPrep DNA/RNA commercial kit (Qiagen, Germany). The concentration of RNA samples was measured using a NanoDrop Spectrophotometer (Thermo Fisher Scientific, Inc.), and the quality was assessed by the estimation of the RNA integrity number (RIN) with the Agilent TapeStation (Agilent Technologies, Inc.) according to the manufacturer's instructions. Samples that did not reach the QC criteria underwent another round of quality assessment or were replaced with another nasal brushing sample from the same individual. A total of 530 samples from 528 subjects were selected for RNA sequencing (RNA-seq) after excluding 133 individuals with low-quality RNA samples.

LncRNA and mRNA 150 bp libraries were constructed using the TruSeq RNA Library Prep Kit with RiboZero Gold treatment and unique dual indexing (Illumina, Inc.). Paired-end sequencing (2 × 150 bp) was performed using the NovaSeq 6000 System (Illumina, Inc.), and default parameters to obtain 30 million read pairs per sample. Initial QC procedures were applied to raw RNA-seq data utilizing the FastQC software (version 0.11.9) [4]. The subsequent analyses described below were carried out following the nf-core/rnaseq bioinformatics pipeline [5]. Trimmed FASTQ files were generated and alignment to the GRCh38/hg38 build of the reference human genome was conducted for paired-end reads using the STAR software (version 2.6.1d) [6]. The aligned reads were sorted with SAMtools (version 1.4) [7]. Transcript expression was quantified at the gene level utilizing the

FeatureCounts v1.5.1 tool [8], and additional QC analyses were performed based on parameters obtained with Picard tools (version 2.23.9) [9].

Among the 530 samples with available RNA-seq data, four mislabeled samples and two duplicates were discarded. Additionally, seven subjects were removed due to discordances between the self-reported and chromosomal sex based on expression levels of *XIST* and other genes located at the Y chromosome. This resulted in 517 participants with high-quality RNAseq data. Based on the criteria the T2 high/low asthma only 336 samples were used for replication analysis. Finally, counts per million (CPM) were obtained using the *edgeR* package [10]. Genes with very low expression levels were excluded based on CPM values of at least 5/M in less than half of the samples, where M is the median library size in millions. A total of 21,375 expression features were retained for subsequent analyses.

U-BIOPRED cohort description

U-BIOPRED (Unbiased Biomarkers for the Prediction of Respiratory Disease Outcomes) study is a multicentre cross-sectional cohort study recruiting from 16 clinical centres in 11 European countries. Detailed description of the cohort is published by *Shaw et.al.[11]*

Sample collection

Nasal brushings were collected from one nostril (4mm plastic coated wire interdental brush (DENT.O.CARE Limited, 7 Cygnus Business Centre, Dalmeyer Road, London, UK)). Samples were embedded in phosphate buffer saline (PBS) directly after the procedure.
mRNA microarray analysis
Microarray analysis was performed with the Affymetrix HT HG-U133+ PM microarray platform (Affymetrix, Santa Clara, Ca). Pre-processing and quality control were performed previously and data was stored on the U-BIOPRED data management platfrom (tranSMART). Using multi-array average normalization (Almac, Craiganvond, UK) and the obtained CEL files were normalized. Technical outliers were excluded (chip image analysis, Affymetrix GeneChip QC, RNA degradation analysis, distribution analysis, principal components analysis, and correlation analysis) and CEL files re-normalized using the robust multi-array (RMA) method. Technical batch effects (e.g., from microarray hybridization date/lot, RNA processing batch) were adjusted in the data matrices using linear modeling of batch (as random factor).

Statistical analysis

Clinical and gene expression data was downloaded from tranSMART. We applied linear modeling and empirical Bayes moderation using the limma package to identify differentially expressed genes across asthma subtypes while adjusting for age, gender, and smoking status (current, ex-, never-smoker).

Heterogeneity assessment

In addition to pairwise Euclidean distances measurements of the highly expressed genes described in the main method section we applied two other ways to assess molecular heterogeneity of T2-high and T2-low asthma:
1. We performed principal component analysis (PCA) using top 2000 most variable genes (measured by coefficient of variation) after filtering for the highly expressed genes (at least 50 CPMs in at least 20 samples). First 11 principal components were selected since they explained more than 1% variance. We calculated pairwise distances on each of the principal component in all possible pairwise combinations within each group (T2-high, T2-low, healthy). Distances between the groups were compared using the Mann-Whitney U test.

2. Another approach was to perform clustering of the subjects to evaluate the optimal number of clusters within each asthma phenotype. We performed hierarchical clustering (using Euclidian distances and complete method) of the subjects in T2-high and T2-low asthma based on cpm normalized expression level of 2000 and 500 most variable genes within each group. The optimal number of clusters was assessed using the gap statistic (clusters k=1 to 10), computed with 100 bootstrap samples using the fviz_nbclust function from the factoextra R package.

References:

1. Kull I, Melen E, Alm J, *et al.* Breast-feeding in relation to asthma, lung function, and sensitization in young schoolchildren. J Allergy Clin Immunol 2010;**125**(5):1013–9.
2. Wang G, Hallberg J, Merid SK, *et al.* Body mass index trajectories from birth to early adulthood and lung function development. Eur Respir J 2025;**65**(1):2400298.
3. Melén E, Bergström A, Kull I, *et al.* Male sex is strongly associated with IgE-sensitization to airborne but not food allergens: results up to age 24 years from the BAMSE birth cohort. Clin Transl Allergy 2020;**10**:15.
4. Andrews S. FastQC: a quality control tool for high throughput sequence data. Available Online at: Http://WwwBioinformaticsBabrahamAcUk/Projects/Fastqc 2010.
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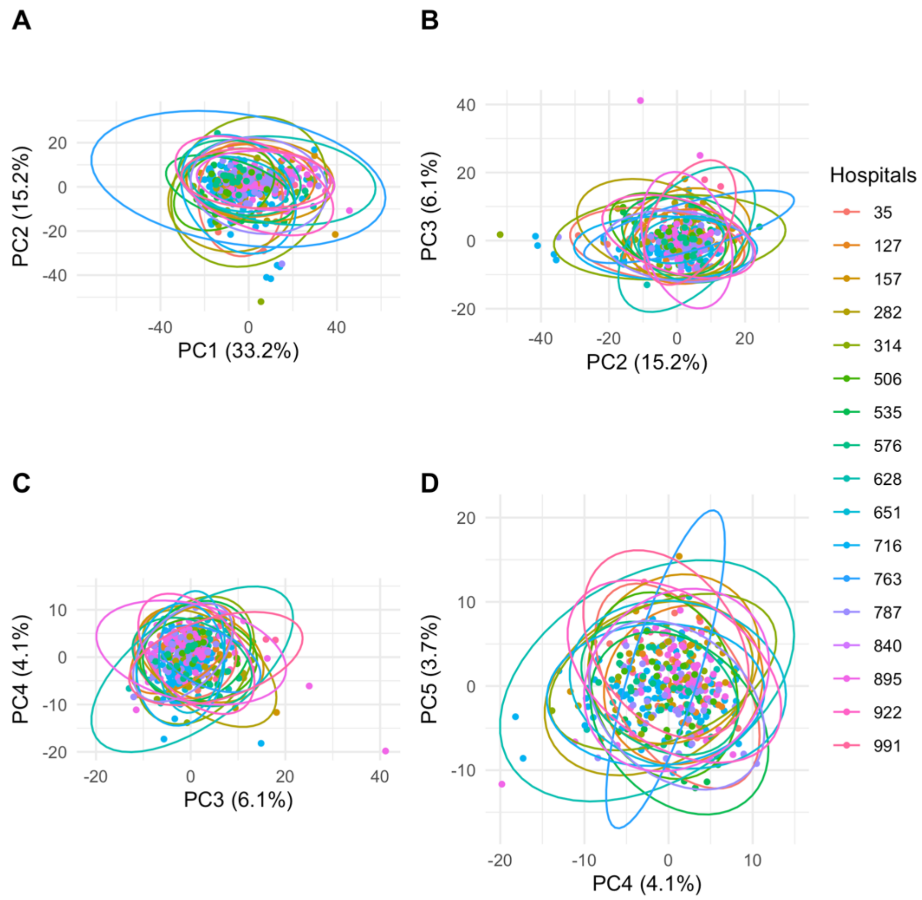
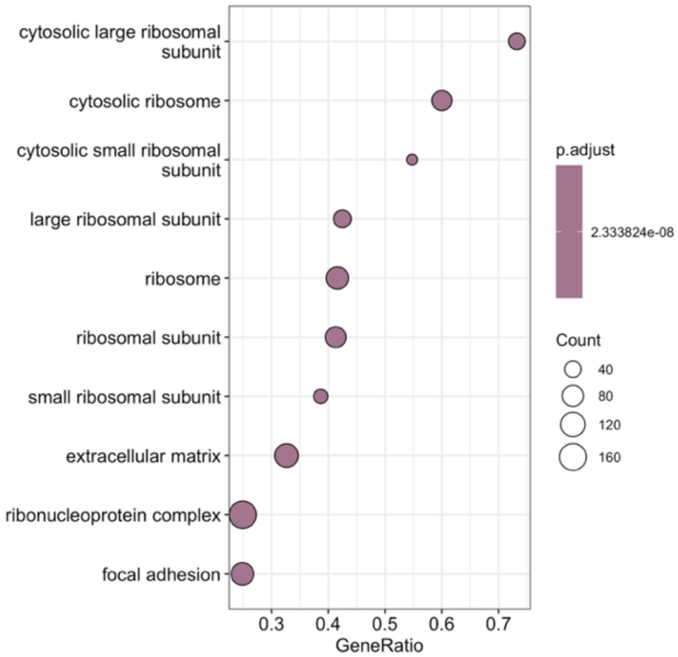
7. Danecek P, Bonfield JK, Liddle J, *et al.* Twelve years of SAMtools and BCFtools. GigaScience 2021;**10**(2):giab008.
8. Liao Y, Smyth GK, Shi W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. Bioinformatics 2014;**30**(7):923–30.
9. Broad Institute. Picard Toolkit 2018.
10. Robinson MD, McCarthy DJ, Smyth GK. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. Bioinformatics 2010;**26**(1):139–40.
11. Shaw DE, Sousa AR, Fowler SJ, *et al.* Clinical and inflammatory characteristics of the European U-BIOPRED adult severe asthma cohort. Eur Respir J 2015;**46**(5):1308–21.

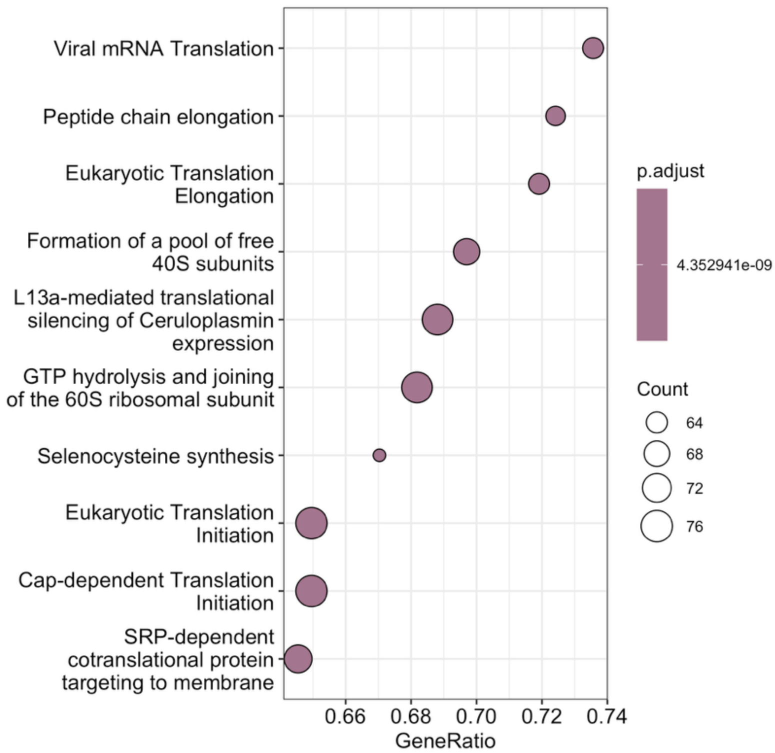
Supplementary Figures



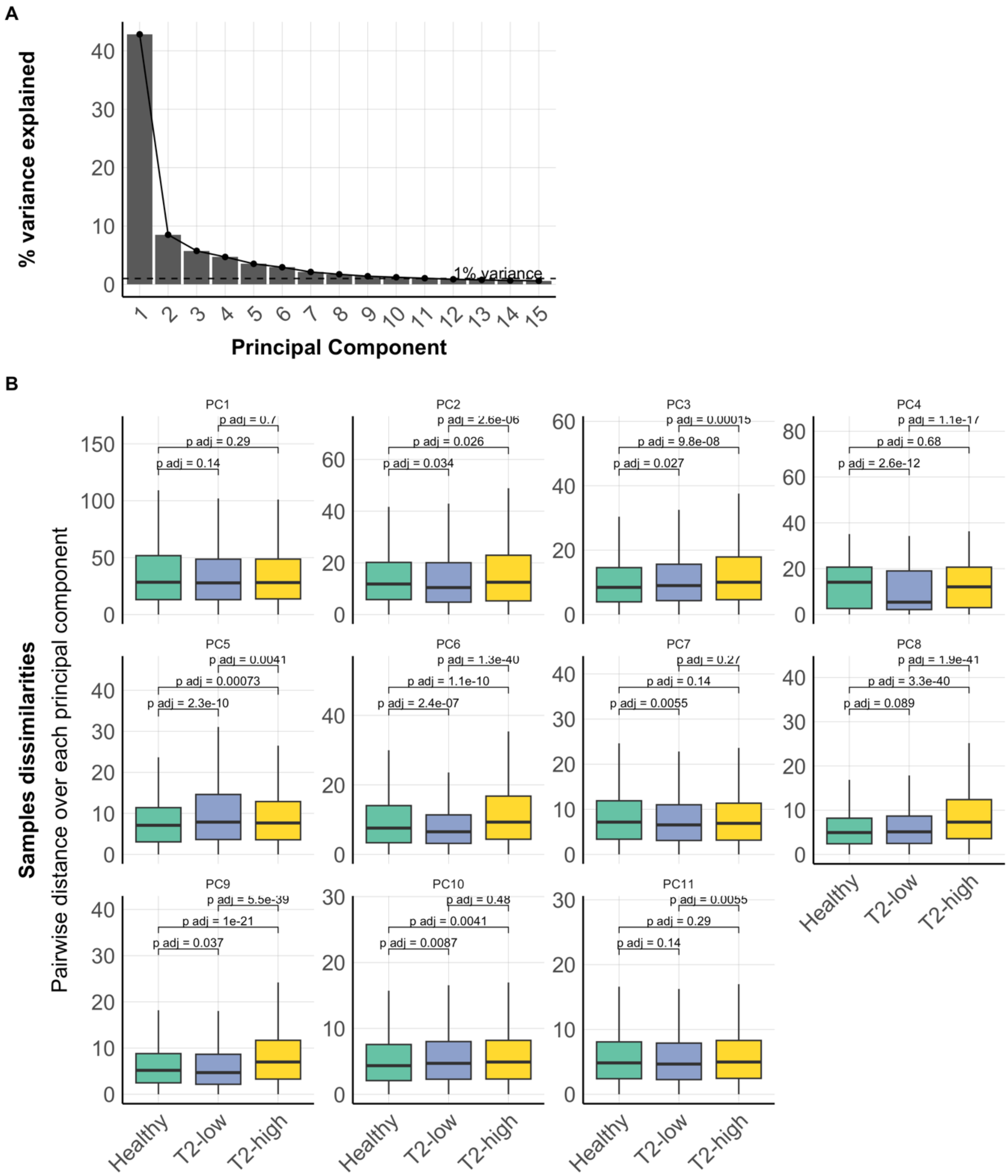
Supplementary Figure 1. *Principal component analysis of the ATLANTIS RNA-sequencing samples. The study site (hospital) of the sample collection is represented by colours. Ellipses around each hospital represent 95% confidence intervals.*



Supplementary Figure 2. *Top significant Gene Ontology pathways pathways identified by gene set enrichment using ranked gene list from the differential gene expression analysis between T2-low asthma and non-asthmatic controls. Genes were ranked by (-log10Pvalue)*± 1 (depending on the direction of logFC).*

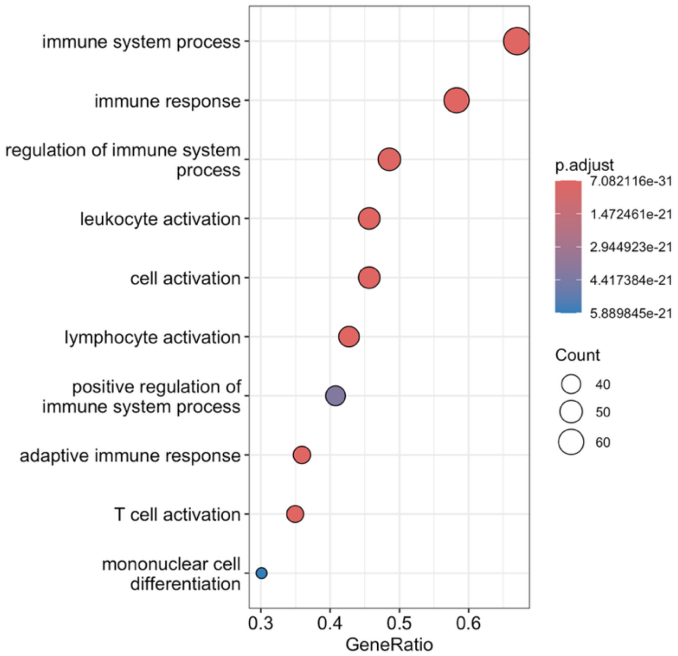


Supplementary Figure 3. *Top significant Reactome pathways identified by gene set enrichment using ranked gene list from the differential gene expression* *analysis between T2-low asthma and non-asthmatic controls. Genes were ranked by (-log10Pvalue)*± 1 (depending on the direction of logFC).*

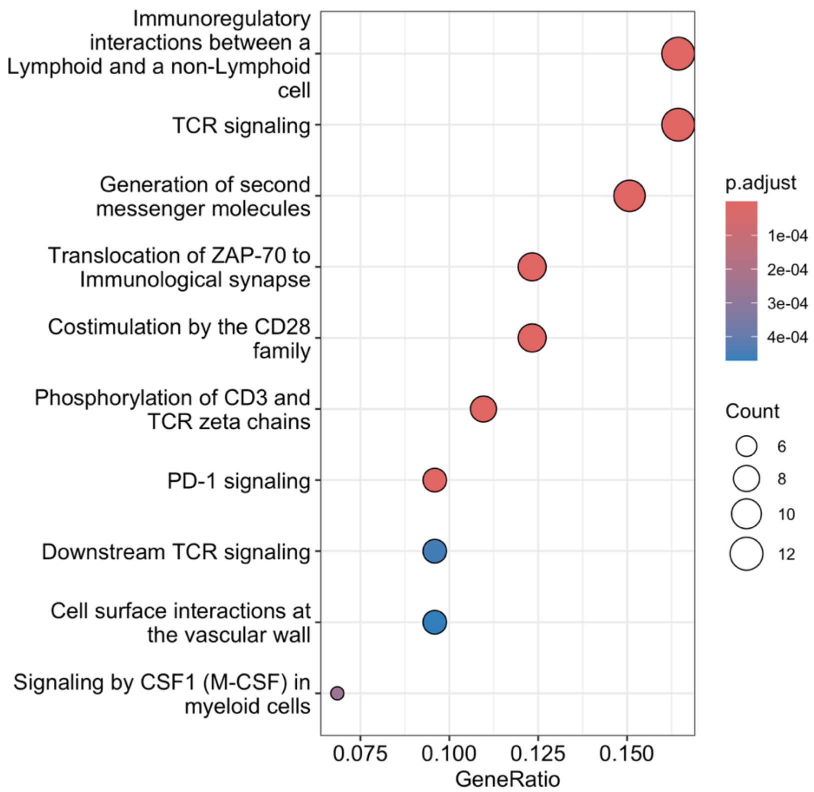


Supplementary Figure 4. ***A*** *Variance explained by each principal component. Principal component analysis was calculated using top 2000 most variable genes (measured by coefficient of variation) after filtering for the highly expressed genes (at least 50 CPMs in at least 20 samples).* ***B*** *Sample dissimilarities within T2-low, T2-high asthma and healthy participants measured by pairwise Euclidean distances of the principal component scores.*

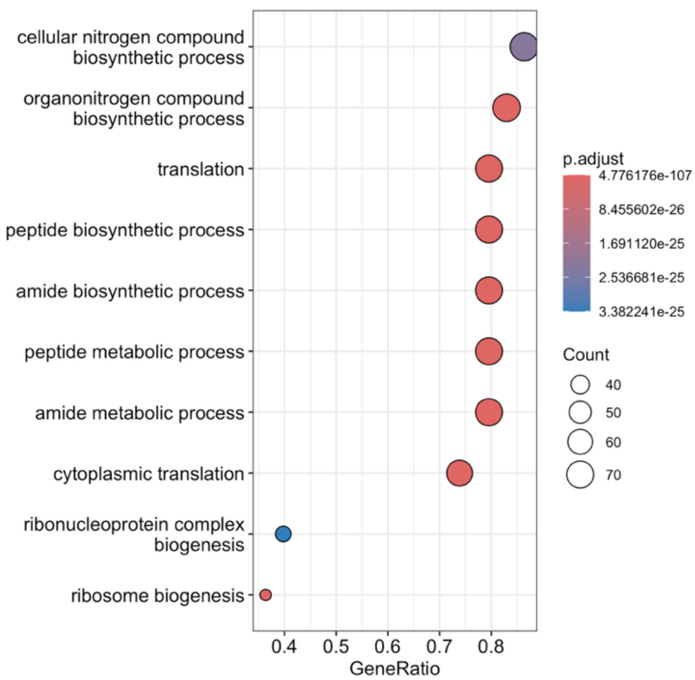
Comparison was performed using the Mann-Whitney U test. First 11 principal components were selected since they explained more than 1% variance.
Distances between T2-high subjects and T2-low subjects were not significantly different on the PC1, PC7, PC10. For the PC5 (3.52% variance explained) the distances were significantly higher in the T2-low asthma. For the other principal components (PC2, PC3, PC4, PC6, PC8, PC9, PC11) that explain 26.04 % of variance, the distances in the T2-high asthma were significantly higher than in T2-low asthma.
Gap statistics identified the same optimal number of clusters for T2-high and T2-low asthma: n = 1 when 2000 genes were used, n=2 when 500 genes were used.



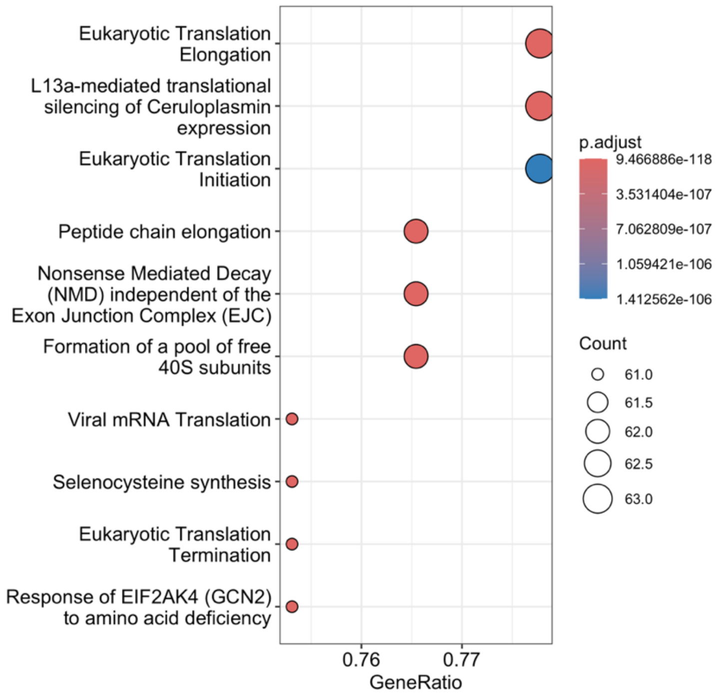
Supplementary Figure 5. *Gene ontology overrepresentation analysis. Pathways enriched in "black" gene module.*



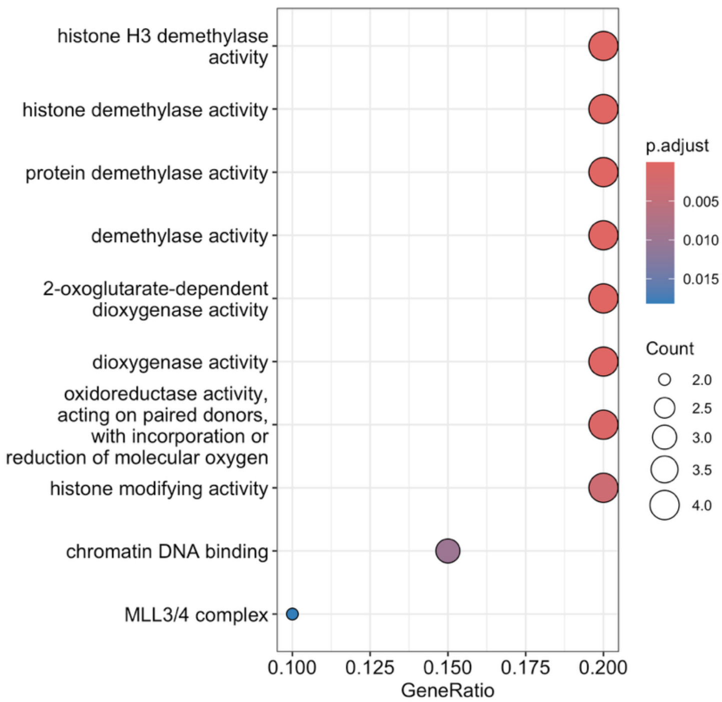
Supplementary Figure 6. *Reactome pathways overrepresentation analysis. Pathways enriched in "black" gene module.*



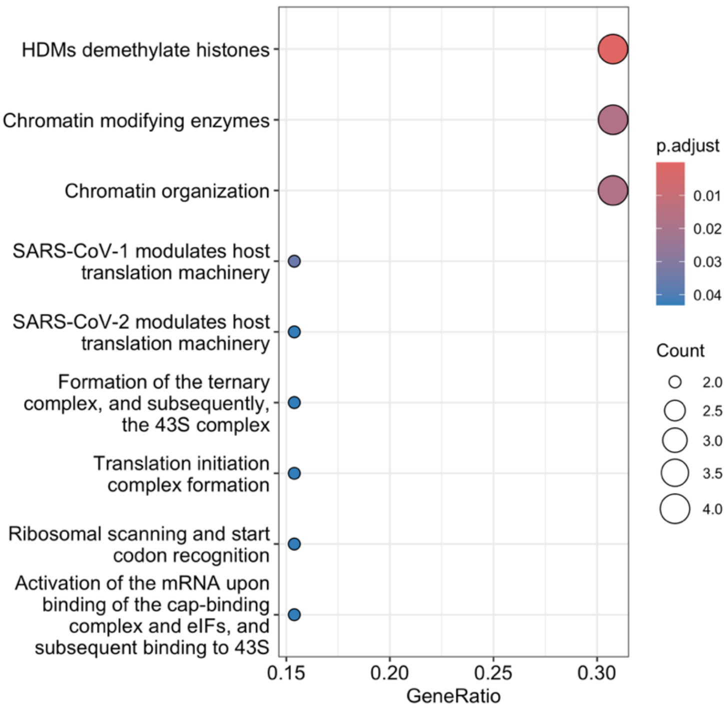
Supplementary Figure 7. *Gene ontology overrepresentation analysis. Pathways enriched in "purple" gene module.*



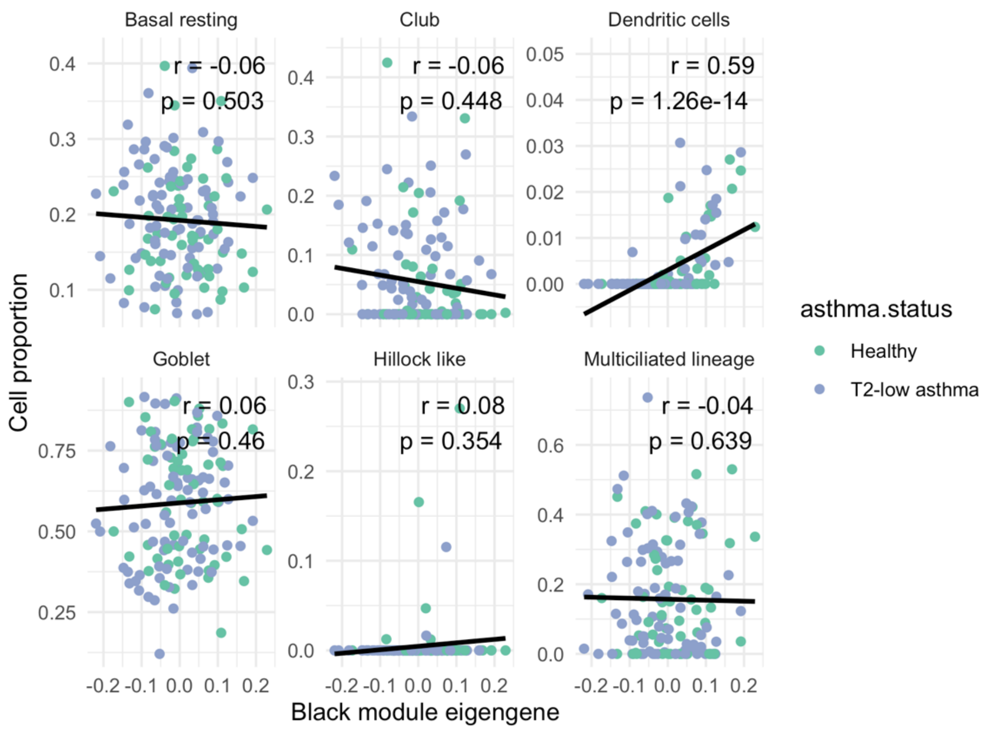
Supplementary Figure 8. *Reactome pathways overrepresentation analysis. Pathways enriched in "purple" gene module.*



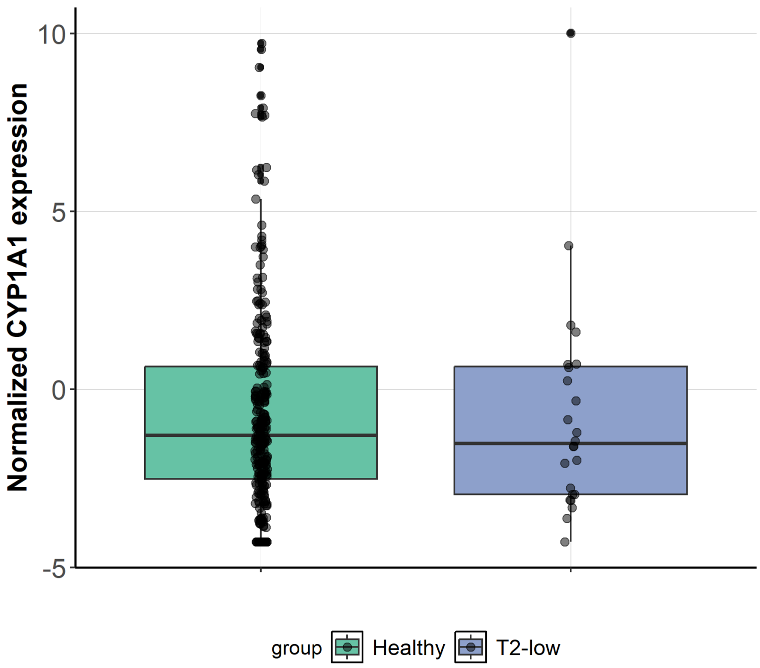
Supplementary Figure 9. *Gene ontology overrepresentation analysis. Pathways enriched in "greenyellow" gene module.*



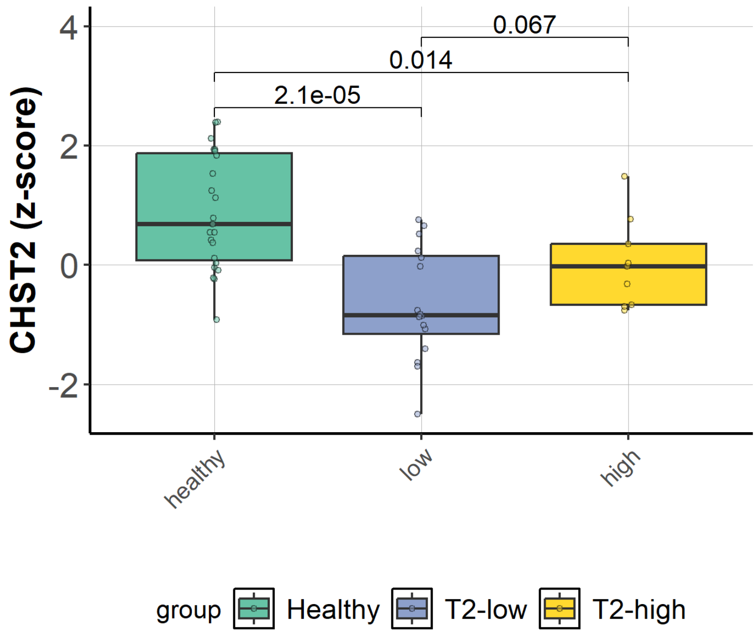
Supplementary Figure 10. *Reactome pathways overrepresentation analysis. Pathways enriched in "greenyellow" gene module.*



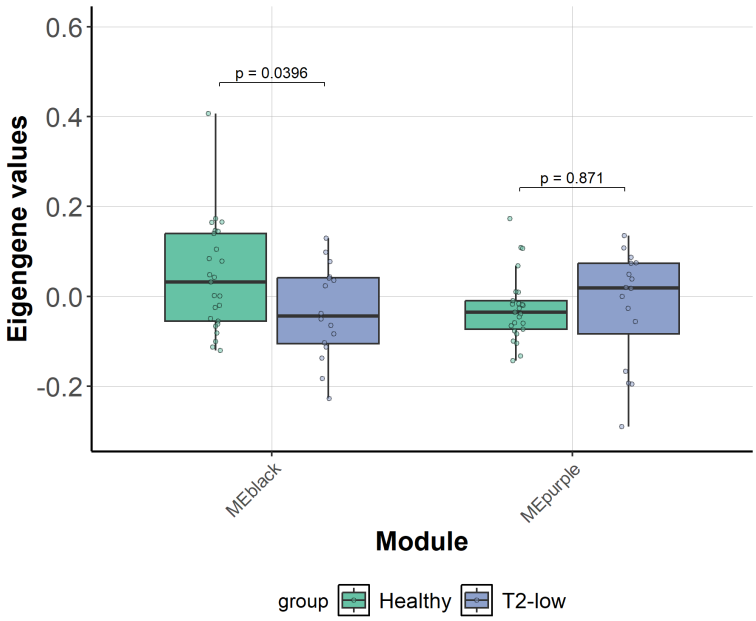
Supplementary Figure 11. *Spearman correlation between gene module black eigengene and predicted cell type proportions. Samples coloured by asthma status.*



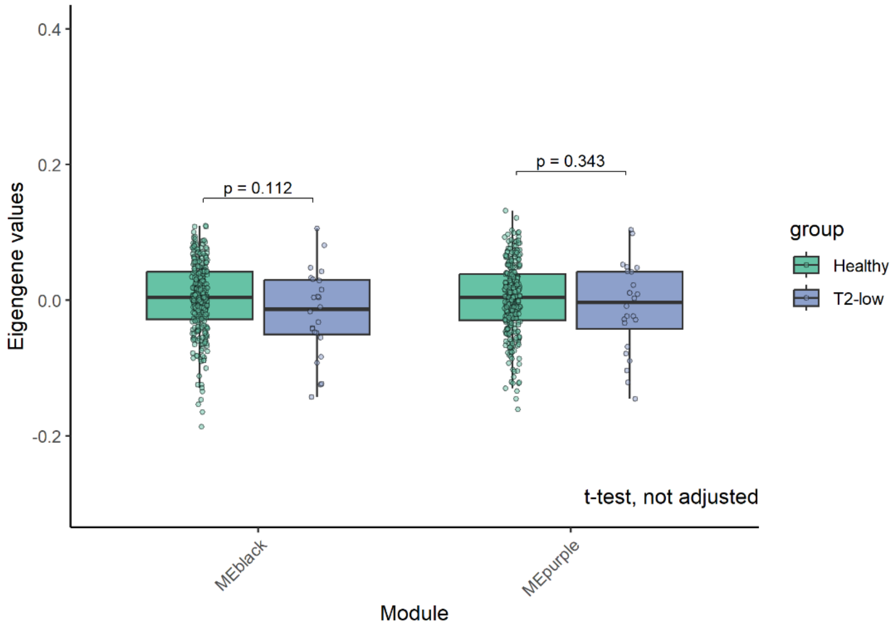
Supplementary Figure 12. *Boxplot representation of the expression of CYP1A1 gene in BAMSE cohort. Y-axis: normalized RNA-expression*



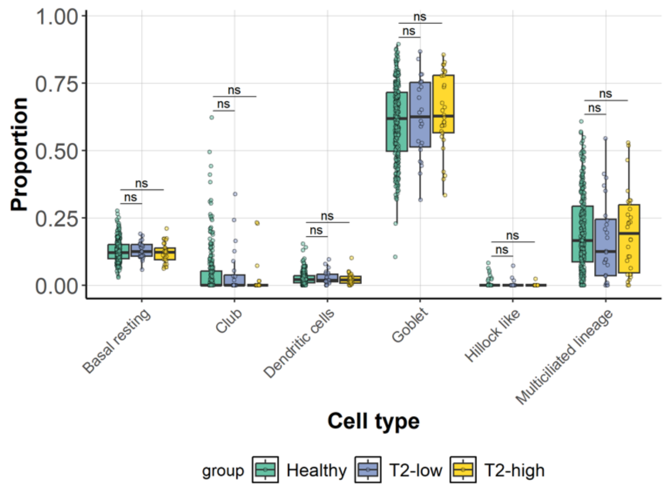
Supplementary Figure 4. *Boxplot representation of the expression of CHST2 gene in U-BIOPRED cohort. Y-axis: normalized RNA-expression*



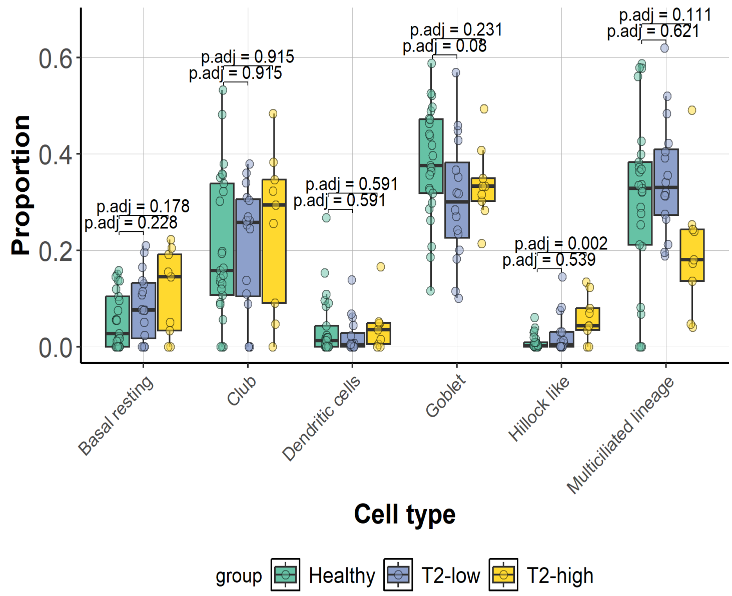
Supplementary Figure 14. *Comparison of black and purple co-expression gene modules in U-BIOPRED cohort that were identified by WGCNA in ATLANTIS. The comparison was performed with t-test.*



Supplementary Figure 15. *Comparison of black and purple co-expression gene modules In BAMSE cohort that were identified by WGCNA in ATLANTIS. The comparison was performed with t-test.*



Supplementary Figure 16. *Comparison of cell proportions between patients with T2-high and T2-low asthma and non-asthmatic controls as a reference group in BAMSE cohort. The comparison was performed with the Mann-Whitney U test. FDR adjusted p-values are shown. ns: FDR > 0.05.*



Supplementary Figure 17. *Comparison of cell proportions between patients with T2-high and T2-low asthma and non-asthmatic controls as a reference group in Ubiopred cohort. The comparison was performed with the Mann-Whitney U test. FDR adjusted p-values are shown. ns: FDR > 0.05.*

Supplementary Tables

Supplementary Tables are also available at the following GitHub repository in excel format:
https://github.com/TatiKarp/t2low_asthma_transcriptomics/blob/main/Supplementary_data/

Gene	logFC	PValue	FDR	Gene name	FDR high BEC asthma	FDR high FeNO
ENSG00000124215	1.42	1.04e-16	1.88e-12	CDH26	0.000107	0.641
ENSG00000090512	3.69	6.41e-15	5.77e-11	FETUB	9.8e-05	0.639
ENSG00000016490	4.71	4.46e-14	2.67e-10	CLCA1	6.43e-08	0.639
ENSG00000274114	1.46	1.16e-11	5.22e-08	ALOX15P1	3.75e-06	0.639
ENSG00000168542	-2.23	1.59e-11	5.73e-08	COL3A1	0.0113	0.947
ENSG00000148053	1.73	1.98e-11	5.92e-08	NTRK2	2.92e-05	0.639
ENSG00000139329	-2.7	2.8e-10	7.21e-07	LUM	0.0187	0.881
ENSG00000164692	-2.64	4.65e-10	1.05e-06	COL1A2	0.0131	0.928
ENSG00000133110	2.14	2.64e-09	5.27e-06	POSTN	0.000636	0.639
ENSG00000161905	0.9	5.64e-09	9.37e-06	ALOX15	0.000125	0.783
ENSG00000112877	0.9	5.73e-09	9.37e-06	CEP72	5.05e-05	0.645
ENSG00000007171	1.46	8.05e-09	1.12e-05	NOS2	3.15e-05	0.639
ENSG00000286811	1.53	8.06e-09	1.12e-05	nan	0.0121	0.639
ENSG00000163359	-1.78	1.11e-08	1.43e-05	COL6A3	0.0113	0.937
ENSG00000120217	1.24	1.29e-08	1.47e-05	CD274	0.000114	0.983
ENSG00000170373	3.34	1.3e-08	1.47e-05	CST1	1.94e-06	0.639
ENSG00000183196	0.49	1.92e-08	2.03e-05	CHST6	0.00168	0.639
ENSG00000225138	0.86	2.5e-08	2.36e-05	SLC9A3-AS1	3.75e-06	0.639
ENSG00000091536	1.17	2.5e-08	2.36e-05	MYO15A	0.00146	0.792
ENSG00000130822	1.65	2.65e-08	2.36e-05	PNCK	4.67e-05	0.639
ENSG00000113140	-1.74	2.76e-08	2.36e-05	SPARC	0.0121	0.796
ENSG00000275302	1.74	5.94e-08	4.65e-05	CCL4	0.0768	0.758
ENSG00000113721	-1.99	6.07e-08	4.65e-05	PDGFRB	0.00559	0.873
ENSG00000114737	1.05	6.43e-08	4.65e-05	CISH	0.000267	0.783
ENSG00000198400	1.87	6.46e-08	4.65e-05	NTRK1	0.0526	0.792
ENSG00000117616	0.33	1.03e-07	7.14e-05	RSRP1	2.92e-05	0.641
ENSG00000188559	0.25	1.14e-07	7.59e-05	RALGAPA2	0.000786	0.763
ENSG00000087245	-1.52	1.38e-07	8.85e-05	MMP2	0.00664	0.881
ENSG00000177606	0.75	1.92e-07	0.000119	JUN	0.00711	0.641

ENSG00000115602	1.68	2e-07	0.00012	IL1RL1	0.0401	0.792
ENSG00000271781	1.37	2.07e-07	0.00012	nan	0.000954	0.792
ENSG00000022567	0.5	2.27e-07	0.000128	SLC45A4	5.97e-05	0.641
ENSG00000268628	0.89	2.64e-07	0.000144	nan	0.0129	0.645
ENSG00000140479	1.07	3.65e-07	0.000193	PCSK6	0.000142	0.858
ENSG00000115325	0.84	4.13e-07	0.000212	DOK1	0.000951	0.783
ENSG00000130821	0.69	4.3e-07	0.000215	SLC6A8	5.76e-06	0.639
ENSG00000185338	1.67	4.68e-07	0.000228	SOCS1	3.4e-05	0.783
ENSG00000169245	2.66	4.99e-07	0.000231	CXCL10	0.000216	0.798
ENSG00000186340	-2.22	5e-07	0.000231	THBS2	0.0139	0.973
ENSG00000168621	1.63	5.68e-07	0.000256	GDNF	0.000164	0.639
ENSG00000102878	0.69	7.44e-07	0.000326	HSF4	1.3e-05	0.646
ENSG00000135362	0.57	7.77e-07	0.000333	PRR5L	4.26e-05	0.796
ENSG00000129566	0.35	8.21e-07	0.000343	TEP1	0.000306	0.749
ENSG00000262202	1.91	9.12e-07	0.000373	nan	NA	NA
ENSG00000164638	1.47	1.3e-06	0.000519	SLC29A4	0.00331	0.997
ENSG00000105641	1.34	1.76e-06	0.000688	SLC5A5	0.00832	0.902
ENSG00000129116	-0.36	1.96e-06	0.000752	PALLD	0.000773	0.646
ENSG00000077044	0.51	2.01e-06	0.000753	DGKD	4.26e-05	0.712
ENSG00000185633	1.61	2.06e-06	0.000756	NDUFA4L2	0.000506	0.813
ENSG00000065618	-1.1	2.36e-06	0.000849	COL17A1	0.0137	0.68
ENSG00000175164	0.64	2.67e-06	0.000942	ABO	0.00209	0.869
ENSG00000167779	-1.35	3.21e-06	0.00111	IGFBP6	0.0589	0.717
ENSG00000121898	1.3	3.44e-06	0.00117	CPXM2	0.000224	0.639
ENSG00000163638	0.88	3.69e-06	0.00123	ADAMTS9	0.141	0.975
ENSG00000277632	1.6	4.14e-06	0.00134	CCL3	0.176	0.645
ENSG00000197586	0.31	4.16e-06	0.00134	ENTPD6	0.000381	0.944
ENSG00000276070	1.73	4.27e-06	0.00135	CCL4L2	0.582	0.639
ENSG00000166126	0.71	4.34e-06	0.00135	AMN	4.8e-06	0.639
ENSG00000115414	-1.74	5.53e-06	0.00169	FN1	0.00783	0.792
ENSG00000126217	0.42	5.96e-06	0.00179	MCF2L	0.00017	0.639

ENSG00000119408	0.38	6.25e-06	0.00184	NEK6	0.0325	0.792
ENSG00000164309	0.57	6.34e-06	0.00184	CMYA5	0.0824	0.858
ENSG00000203635	1.32	6.5e-06	0.00184	nan	0.000449	0.873
ENSG00000137857	0.37	6.56e-06	0.00184	DUOX1	0.000526	0.639
ENSG00000188761	0.48	6.93e-06	0.00192	BCL2L15	0.239	0.997
ENSG00000174231	-0.2	7.13e-06	0.00194	PRPF8	0.00195	0.792
ENSG00000163751	1.6	7.69e-06	0.00206	CPA3	0.0169	0.827
ENSG00000087494	1.36	8.48e-06	0.00224	PTHLH	0.151	0.748
ENSG00000129521	0.9	8.87e-06	0.00229	EGLN3	0.00967	0.858
ENSG00000141934	0.29	8.9e-06	0.00229	PLPP2	0.000252	0.646
ENSG00000257594	0.78	9.23e-06	0.00234	GALNT4	9,00E-06	0.639
ENSG00000221978	0.26	9.48e-06	0.00236	CCNL2	3.1e-05	0.639
ENSG00000169248	2.42	9.56e-06	0.00236	CXCL11	0.000609	0.645
ENSG00000141424	-0.24	9.91e-06	0.00241	SLC39A6	0.00331	0.882
ENSG00000104365	0.24	1.01e-05	0.00243	IKBKB	0.00126	0.858
ENSG00000175538	0.38	1.03e-05	0.00243	KCNE3	0.00239	0.645
ENSG00000186395	-0.91	1.08e-05	0.0025	KRT10	0.0354	0.639
ENSG00000165125	0.59	1.08e-05	0.0025	TRPV6	0.0327	0.669
ENSG00000182606	0.24	1.2e-05	0.00273	TRAK1	0.0321	0.869
ENSG00000247095	0.72	1.32e-05	0.00297	MIR210HG	0.0177	0.758
ENSG00000165633	-1.02	1.35e-05	0.00299	VSTM4	0.243	0.669
ENSG00000148180	0.43	1.39e-05	0.00302	GSN	0.00152	0.781
ENSG00000256612	0.48	1.39e-05	0.00302	CYP2B7P	0.0111	0.669
ENSG00000223813	0.47	1.44e-05	0.00307	PRR15-DT	0.101	0.997
ENSG00000107281	0.69	1.45e-05	0.00307	NPDC1	0.000603	0.792
ENSG00000264940	2.05	1.55e-05	0.00324	SNORD3C	0.0914	0.645
ENSG00000148339	0.39	1.67e-05	0.00337	SLC25A25	0.0267	0.639
ENSG00000187609	0.49	1.68e-05	0.00337	EXD3	4.8e-06	0.639
ENSG00000162772	0.73	1.68e-05	0.00337	ATF3	0.129	0.792
ENSG00000099139	0.44	1.7e-05	0.00337	PCSK5	0.0833	0.87
ENSG00000182179	0.35	1.71e-05	0.00337	UBA7	0.000406	0.783

ENSG00000110436	-1.26	1.72e-05	0.00337	SLC1A2	0.176	0.641
ENSG00000276085	1.78	1.8e-05	0.00348	CCL3L3	0.56	0.641
ENSG00000186847	-1.74	1.82e-05	0.00348	KRT14	0.136	0.999
ENSG00000066230	1.54	1.93e-05	0.00366	SLC9A3	0.0103	0.639
ENSG00000172554	1.14	1.98e-05	0.00367	SNTG2	0.011	0.924
ENSG00000172236	1.52	1.98e-05	0.00367	TPSAB1	0.00771	0.993
ENSG00000176928	0.73	2.01e-05	0.0037	GCNT4	0.00948	0.781
ENSG00000168411	-0.24	2.07e-05	0.00377	RFWD3	0.0229	0.639
ENSG00000215853	-1.36	2.17e-05	0.0039	RPTN	0.102	0.639
ENSG00000255587	1.01	2.2e-05	0.00392	RAB44	0.032	0.783
ENSG00000095066	0.39	2.24e-05	0.00395	HOOK2	5.03e-06	0.639
ENSG00000174669	0.44	2.47e-05	0.00431	SLC29A2	0.00129	0.639
ENSG00000130827	0.37	2.55e-05	0.00442	PLXNA3	0.000136	0.639
ENSG00000137968	-0.92	2.68e-05	0.0046	SLC44A5	0.142	0.639
ENSG00000279476	0.6	2.86e-05	0.00485	nan	0.000801	0.669
ENSG00000166348	0.26	2.91e-05	0.00489	USP54	0.83	0.944
ENSG00000139631	0.37	2.99e-05	0.00494	CSAD	3.52e-05	0.639
ENSG00000204859	0.33	2.99e-05	0.00494	ZBTB48	0.000294	0.758
ENSG00000136108	-0.47	3.15e-05	0.00515	CKAP2	0.00022	0.639
ENSG00000167617	0.62	3.2e-05	0.00518	CDC42EP5	0.00219	0.858
ENSG00000011465	-1.42	3.23e-05	0.00518	DCN	0.048	0.983
ENSG00000182601	1.17	3.43e-05	0.00546	HS3ST4	0.0371	0.823
ENSG00000112294	-0.33	3.53e-05	0.00556	ALDH5A1	0.0188	0.873
ENSG00000130653	0.56	3.79e-05	0.00592	PNPLA7	0.00331	0.781
ENSG00000006062	0.39	3.9e-05	0.006	MAP3K14	0.00944	0.999
ENSG00000196139	-0.36	3.91e-05	0.006	AKR1C3	0.00175	0.792
ENSG00000136156	-0.23	3.93e-05	0.006	ITM2B	0.011	0.639
ENSG00000279602	1.29	3.98e-05	0.00601	nan	0.000435	0.639
ENSG00000113013	-0.22	4.03e-05	0.00604	HSPA9	4.26e-05	0.639
ENSG00000147488	1.07	4.07e-05	0.00605	ST18	0.2	0.916
ENSG00000164038	0.88	4.13e-05	0.0061	SLC9B2	0.266	0.947

ENSG00000105321	0.5	4.23e-05	0.00618	CCDC9	9,00E-06	0.639
ENSG00000128274	0.39	4.33e-05	0.00628	A4GALT	0.00154	0.783
ENSG00000126456	0.31	4.42e-05	0.00636	IRF3	9,00E-06	0.758
ENSG00000149021	-1.19	4.6e-05	0.00654	SCGB1A1	0.0176	0.881
ENSG00000174353	0.38	4.62e-05	0.00654	STAG3L3	1.94e-06	0.645
ENSG00000099953	-1.05	4.68e-05	0.00658	MMP11	0.206	0.915
ENSG00000258839	0.64	4.82e-05	0.00669	MC1R	5.76e-06	0.639
ENSG00000214331	0.36	4.83e-05	0.00669	PDPR2P	0.00422	0.641
ENSG00000101445	0.72	4.94e-05	0.00679	PPP1R16B	0.438	0.928
ENSG00000155659	1.32	5.49e-05	0.00748	VSIG4	0.408	0.858
ENSG00000275183	0.55	5.63e-05	0.00762	LENG9	0.0122	0.905
ENSG00000196123	0.38	5.7e-05	0.00765	KIAA0895L	2.88e-05	0.639
ENSG00000182752	-1.07	5.79e-05	0.00772	PAPPA	0.064	0.858
ENSG00000151835	-0.4	5.87e-05	0.00776	SACS	0.00121	0.723
ENSG00000138755	1.89	6.03e-05	0.00791	CXCL9	0.00325	0.881
ENSG00000179698	0.74	6.08e-05	0.00793	WDR97	0.000366	0.639
ENSG00000164176	-0.68	6.17e-05	0.00799	EDIL3	0.000308	0.954
ENSG00000140287	0.92	6.22e-05	0.008	HDC	0.0119	0.645
ENSG00000142173	-1.36	6.3e-05	0.00803	COL6A2	0.111	0.994
ENSG00000124570	0.24	6.34e-05	0.00803	SERPINB6	0.157	0.639
ENSG00000164877	0.42	6.44e-05	0.00806	MICALL2	3.75e-06	0.639
ENSG00000167615	0.36	6.5e-05	0.00806	LENG8	1.05e-05	0.669
ENSG00000139044	0.35	6.56e-05	0.00806	B4GALNT3	0.00102	0.645
ENSG00000122786	-0.79	6.58e-05	0.00806	CALD1	0.048	0.645
ENSG00000129173	-0.42	6.59e-05	0.00806	E2F8	0.0327	0.645
ENSG00000086991	-1.05	6.72e-05	0.00816	NOX4	0.73	0.639
ENSG00000130635	-0.97	7.08e-05	0.00855	COL5A1	0.0815	0.858
ENSG00000147650	-0.37	7.29e-05	0.00874	LRP12	8.42e-05	0.87
ENSG00000177943	0.73	7.7e-05	0.00913	MAMDC4	0.000953	0.68
ENSG00000157873	0.36	7.72e-05	0.00913	TNFRSF14	3.89e-05	0.712
ENSG00000085662	-0.52	7.89e-05	0.00922	AKR1B1	0.0833	0.665

ENSG00000136824	-0.34	7.89e-05	0.00922	SMC2	2.92e-05	0.781
ENSG00000171163	0.37	8.09e-05	0.00939	ZNF692	0.00092	0.748
ENSG00000129946	0.86	8.34e-05	0.00962	SHC2	0.000216	0.639
ENSG00000198089	0.3	8.52e-05	0.00968	SFI1	1.59e-05	0.639
ENSG00000179477	-1.5	8.6e-05	0.00968	ALOX12B	0.101	0.639
ENSG00000213626	0.49	8.63e-05	0.00968	LBH	0.243	0.639
ENSG00000185274	1.03	8.63e-05	0.00968	GALNT17	0.000381	0.639
ENSG00000172927	-0.72	8.68e-05	0.00968	MYEOV	0.0281	0.653
ENSG00000142765	0.34	8.72e-05	0.00968	SYTL1	8,00E-05	0.639
ENSG00000130508	0.99	8.86e-05	0.00973	PXDN	0.563	0.647
ENSG00000084463	-0.24	8.87e-05	0.00973	WBP11	0.000945	0.645
ENSG00000176532	0.33	9.19e-05	0.01	PRR15	0.00821	0.954
ENSG00000134243	-0.24	9.46e-05	0.0103	SORT1	0.0054	0.749
ENSG00000128159	0.31	9.55e-05	0.0103	TUBGCP6	2.92e-05	0.647
ENSG00000119922	1.18	0.0001	0.0107	IFIT2	0.00111	0.907
ENSG00000154175	-0.97	0.000101	0.0108	ABI3BP	0.195	0.901
ENSG00000138448	-0.23	0.000102	0.0108	ITGAV	0.00487	0.999
ENSG00000280789	0.4	0.000106	0.0112	PAGR1	0.00265	0.798
ENSG00000205356	0.31	0.000109	0.0114	TECPR1	0.000146	0.641
ENSG00000250920	-1.74	0.000112	0.0116	nan	0.0466	0.639
ENSG00000143368	-0.28	0.000116	0.012	SF3B4	0.0212	0.639
ENSG00000125740	1.45	0.000117	0.012	FOSB	0.113	0.942
ENSG00000071859	0.32	0.000119	0.0121	FAM50A	0.00113	0.905
ENSG00000129933	0.22	0.000119	0.0121	MAU2	0.000128	0.783
ENSG00000118680	-0.2	0.00012	0.0121	MYL12B	7.41e-05	0.873
ENSG00000070770	-0.21	0.000122	0.0123	CSNK2A2	0.00292	0.858
ENSG00000104936	0.43	0.000124	0.0124	DMPK	2.18e-05	0.641
ENSG00000056736	0.78	0.000125	0.0124	IL17RB	0.123	0.916
ENSG00000072858	0.34	0.000126	0.0125	SIDT1	0.0377	0.881
ENSG00000244474	0.65	0.000129	0.0126	UGT1A4	0.000228	0.712
ENSG00000134871	-0.9	0.00013	0.0126	COL4A2	0.0105	0.999

ENSG00000115365	-0.33	0.00013	0.0126	LANCL1	0.00033	0.902
ENSG00000169228	0.86	0.000133	0.0128	RAB24	5.56e-05	0.639
ENSG00000010438	-1.17	0.000135	0.0128	PRSS3	0.205	0.639
ENSG00000204539	-1.75	0.000135	0.0128	CDSN	0.0316	0.655
ENSG00000166454	-0.23	0.000135	0.0128	ATMIN	0.0033	0.775
ENSG00000270571	1.15	0.000137	0.0129	nan	0.177	0.792
ENSG00000237720	1.07	0.00014	0.0132	nan	0.044	0.905
ENSG00000164078	0.29	0.00014	0.0132	MST1R	0.000389	0.639
ENSG00000002587	0.39	0.000143	0.0132	HS3ST1	0.0775	0.858
ENSG00000276312	1.66	0.000143	0.0132	nan	0.00382	0.639
ENSG00000198804	0.5	0.000144	0.0132	MT-CO1	0.562	0.655
ENSG00000188820	1.42	0.000145	0.0133	CALHM6	7.48e-05	0.783
ENSG00000256870	0.47	0.000145	0.0133	SLC5A8	0.149	0.909
ENSG00000162688	-0.32	0.000146	0.0133	AGL	0.00311	0.639
ENSG00000160072	0.36	0.000147	0.0133	ATAD3B	2.95e-05	0.645
ENSG00000183020	0.23	0.000148	0.0133	AP2A2	0.000309	0.712
ENSG00000158258	-1.33	0.00015	0.0134	CLSTN2	0.6	0.783
ENSG00000198730	-0.21	0.000151	0.0134	CTR9	0.000401	0.639
ENSG00000198753	0.49	0.000151	0.0134	PLXNB3	0.00237	0.792
ENSG00000144028	-0.14	0.000156	0.0138	SNRNP200	0.0895	0.758
ENSG00000164484	0.95	0.000158	0.0139	TMEM200A	0.0343	0.792
ENSG00000224520	1.17	0.000168	0.0147	KRT8P45	0.0496	0.639
ENSG00000144045	0.72	0.00017	0.0148	DQX1	0.00254	0.881
ENSG00000144791	0.19	0.000171	0.0148	LIMD1	2.8e-05	0.712
ENSG00000121380	0.49	0.000175	0.015	BCL2L14	0.0401	0.916
ENSG00000159842	0.23	0.000175	0.015	ABR	0.00022	0.915
ENSG00000276168	0.47	0.000181	0.0154	RN7SL1	0.000234	0.858
ENSG00000135722	0.4	0.000182	0.0155	FBXL8	0.00188	0.639
ENSG00000169660	0.36	0.000193	0.0163	HEXD	0.000107	0.655
ENSG00000171451	-0.42	0.000196	0.0165	DSEL	0.00331	0.946
ENSG00000134762	-0.61	0.000199	0.0166	DSC3	0.0026	0.813

ENSG00000124574	0.24	0.000199	0.0166	ABCC10	0.000252	0.858
ENSG00000005007	0.22	0.0002	0.0166	UPF1	0.000295	0.639
ENSG00000262074	1.02	0.000201	0.0166	SNORD3B-2	0.00137	0.902
ENSG00000073849	0.38	0.000206	0.0169	ST6GAL1	0.0663	0.884
ENSG00000210107	0.78	0.000208	0.017	MT-TQ	0.194	0.758
ENSG00000072195	0.92	0.000211	0.0172	SPEG	0.00417	0.695
ENSG00000065485	0.35	0.000216	0.0175	PDIA5	0.000256	0.858
ENSG00000148229	-0.28	0.000217	0.0175	POLE3	0.0107	0.792
ENSG00000287839	0.71	0.000223	0.0179	nan	0.413	0.858
ENSG00000163191	-0.29	0.000226	0.018	S100A11	0.00105	0.639
ENSG00000176731	-0.32	0.000226	0.018	RBIS	4.26e-05	0.639
ENSG00000184164	0.34	0.000228	0.0181	CRELD2	0.000477	0.815
ENSG00000180902	0.42	0.000229	0.0181	D2HGDH	2.51e-05	0.639
ENSG00000105523	0.36	0.00023	0.0181	FAM83E	3.4e-05	0.669
ENSG00000117480	0.42	0.000234	0.0182	FAAH	4,00E-05	0.639
ENSG00000182544	-0.25	0.000234	0.0182	MFSD5	0.0292	0.647
ENSG00000151632	-0.54	0.000235	0.0182	AKR1C2	0.0181	0.973
ENSG00000169474	-1.19	0.000237	0.0183	SPRR1A	0.107	0.639
ENSG00000101452	0.33	0.000239	0.0184	DHX35	0.0546	0.722
ENSG00000140511	0.95	0.000241	0.0184	HAPLN3	0.000294	0.985
ENSG00000110719	0.43	0.000242	0.0184	TCIRG1	3.78e-05	0.639
ENSG00000149534	1.26	0.000244	0.0185	MS4A2	0.0228	0.902
ENSG00000175899	-0.68	0.000247	0.0187	A2M	0.000347	0.639
ENSG00000261123	0.73	0.00025	0.0187	nan	0.0291	0.669
ENSG00000143442	0.18	0.00025	0.0187	POGZ	0.334	0.997
ENSG00000166123	0.31	0.000251	0.0187	GPT2	0.00111	0.749
ENSG00000084234	-0.15	0.000253	0.0187	APLP2	0.0782	0.858
ENSG00000168062	0.86	0.000254	0.0187	BATF2	0.000482	0.915
ENSG00000179639	-0.84	0.000254	0.0187	FCER1A	0.00033	0.639
ENSG00000263528	0.35	0.000254	0.0187	IKBKE	0.0141	0.881
ENSG00000160271	0.37	0.000257	0.0188	RALGDS	0.00568	0.645

ENSG00000135272	-0.34	0.000258	0.0188	MDFIC	0.00523	0.645
ENSG00000114784	-0.36	0.000259	0.0188	EIF1B	0.00104	0.813
ENSG00000153147	-0.19	0.00026	0.0188	SMARCA5	0.000111	0.873
ENSG00000197253	1.34	0.000261	0.0188	TPSB2	0.0112	0.858
ENSG00000135637	0.36	0.000263	0.0188	CCDC142	0.000871	0.796
ENSG00000088812	-0.19	0.000264	0.0189	ATRN	0.046	0.954
ENSG00000163945	0.31	0.000269	0.0192	UVSSA	6.99e-07	0.639
ENSG00000259605	0.67	0.000272	0.0193	nan	0.021	0.924
ENSG00000140807	-0.46	0.000275	0.0194	NKD1	0.555	0.639
ENSG00000106624	-0.91	0.000276	0.0194	AEBP1	0.195	0.935
ENSG00000116667	-0.23	0.000278	0.0194	C1orf21	0.0697	0.639
ENSG00000012171	0.47	0.000278	0.0194	SEMA3B	0.000267	0.783
ENSG00000004455	0.18	0.000279	0.0194	AK2	0.148	0.805
ENSG00000182512	-0.32	0.000283	0.0196	GLRX5	0.0715	0.783
ENSG00000166025	-0.49	0.000287	0.0198	AMOTL1	0.0343	0.881
ENSG00000087008	0.26	0.000294	0.0201	ACOX3	0.000943	0.657
ENSG00000166987	0.34	0.000296	0.0201	MBD6	2.14e-05	0.911
ENSG00000143320	-0.75	0.000296	0.0201	CRABP2	0.072	0.722
ENSG00000189339	0.27	0.000297	0.0201	SLC35E2B	0.000966	0.639
ENSG00000111199	0.32	0.000297	0.0201	TRPV4	0.0145	0.647
ENSG00000117266	0.32	0.000301	0.0203	CDK18	5.76e-06	0.639
ENSG00000127511	0.27	0.000303	0.0203	SIN3B	6.84e-05	0.792
ENSG00000197070	0.31	0.000307	0.0206	ARRDC1	0.000527	0.725
ENSG00000099994	0.82	0.000311	0.0207	SUSD2	0.0656	0.858
ENSG00000160883	1.44	0.000313	0.0208	HK3	0.000481	0.999
ENSG00000162654	0.92	0.000315	0.0208	GBP4	0.044	0.748
ENSG00000067082	0.44	0.000318	0.0209	KLF6	0.0733	0.902
ENSG00000021645	0.86	0.000321	0.0211	NRXN3	0.179	0.85
ENSG00000204291	-1.37	0.000322	0.0211	COL15A1	0.157	0.882
ENSG00000197119	0.3	0.00033	0.0215	SLC25A29	0.000155	0.639
ENSG00000023734	-0.2	0.000334	0.0217	STRAP	0.000309	0.639

ENSG00000100836	0.27	0.000338	0.0219	PABPN1	2.92e-05	0.639
ENSG00000129657	0.34	0.000341	0.022	SEC14L1	0.000772	0.783
ENSG00000197530	0.37	0.000344	0.0221	MIB2	5.03e-06	0.639
ENSG00000146205	0.67	0.000349	0.0223	ANO7	0.0391	0.902
ENSG00000142961	0.37	0.00035	0.0223	MOB3C	8.73e-05	0.796
ENSG00000136235	-0.54	0.000354	0.0224	GPNMB	0.00628	0.641
ENSG00000149090	-1.27	0.000355	0.0224	PAMR1	0.0371	0.927
ENSG00000130529	0.34	0.000355	0.0224	TRPM4	0.000234	0.858
ENSG00000188290	0.55	0.000365	0.0228	HES4	0.00032	0.645
ENSG00000239437	0.82	0.000365	0.0228	RN7SL752P	0.00014	0.712
ENSG00000167280	0.36	0.000365	0.0228	ENGASE	4.29e-05	0.639
ENSG00000164050	0.32	0.00037	0.0231	PLXNB1	5.05e-05	0.639
ENSG00000183625	1.44	0.000375	0.0233	CCR3	NA	NA
ENSG00000112769	-0.61	0.000377	0.0233	LAMA4	0.0646	0.639
ENSG00000099326	0.31	0.000381	0.0235	MZF1	0.00664	0.722
ENSG00000234423	0.9	0.000384	0.0236	LINC01250	0.028	0.639
ENSG00000111817	0.32	0.000386	0.0236	DSE	0.59	0.813
ENSG00000167685	0.37	0.000387	0.0236	ZNF444	0.00632	0.813
ENSG00000134321	1.01	0.00039	0.0237	RSAD2	0.00348	0.975
ENSG00000164609	-0.25	0.000392	0.0237	SLU7	0.00137	0.966
ENSG00000138182	-0.43	0.000392	0.0237	KIF20B	2.39e-05	0.745
ENSG00000005206	0.36	0.000395	0.0238	SPPL2B	3.42e-05	0.641
ENSG00000100234	-0.62	0.000396	0.0238	TIMP3	0.145	0.813
ENSG00000164338	-0.32	0.000403	0.0241	UTP15	0.000612	0.646
ENSG00000187498	-1.09	0.000408	0.0243	COL4A1	0.925	0.975
ENSG00000109736	0.28	0.000415	0.0247	MFSD10	0.000143	0.639
ENSG00000179023	0.38	0.000421	0.0249	KLHDC7A	0.0115	0.849
ENSG00000253368	0.48	0.000428	0.0252	TRNP1	0.00347	0.983
ENSG00000228423	0.79	0.00043	0.0253	nan	0.00308	0.892
ENSG00000154451	1.04	0.000431	0.0253	GBP5	0.0759	0.796
ENSG00000181163	-0.23	0.000435	0.0254	NPM1	9.8e-05	0.639

ENSG00000196504	-0.2	0.000436	0.0254	PRPF40A	0.000103	0.641
ENSG00000114686	-0.25	0.000454	0.0263	MRPL3	0.00255	0.653
ENSG00000275215	0.95	0.000454	0.0263	RNA5-8SN3	1.28e-05	0.639
ENSG00000182389	0.44	0.000456	0.0263	CACNB4	0.962	0.881
ENSG00000170500	0.48	0.000461	0.0265	LONRF2	0.00967	0.639
ENSG00000262655	-1.16	0.000467	0.0267	SPON1	0.0371	0.639
ENSG00000112378	-0.23	0.000468	0.0267	PERP	0.00771	0.669
ENSG00000160753	0.3	0.000469	0.0267	RUSC1	0.0213	0.651
ENSG00000142798	0.31	0.000474	0.0269	HSPG2	0.00126	0.639
ENSG00000165119	-0.2	0.000476	0.0269	HNRNPK	0.000527	0.645
ENSG00000241973	0.15	0.000477	0.0269	PI4KA	0.0482	0.914
ENSG00000181523	0.3	0.000486	0.0273	SGSH	8.42e-05	0.858
ENSG00000071242	0.27	0.000488	0.0273	RPS6KA2	0.0458	0.749
ENSG00000088035	-0.35	0.000489	0.0273	ALG6	0.00535	0.85
ENSG00000134057	-0.42	0.00049	0.0273	CCNB1	0.00126	0.676
ENSG00000124155	-0.16	0.000502	0.0278	PIGT	0.0327	0.783
ENSG00000032444	0.25	0.000502	0.0278	PNPLA6	0.00113	0.805
ENSG00000178372	-0.98	0.000508	0.028	CALML5	0.203	0.641
ENSG00000107201	0.59	0.000509	0.028	DDX58	0.0322	0.783
ENSG00000140937	-0.71	0.000519	0.0285	CDH11	0.013	0.796
ENSG00000084093	-0.2	0.000521	0.0285	REST	9.36e-05	0.641
ENSG00000177548	0.35	0.000528	0.0287	RABEP2	7.3e-06	0.639
ENSG00000227036	0.32	0.000528	0.0287	LINC00511	0.000295	0.882
ENSG00000132646	-0.28	0.000529	0.0287	PCNA	0.00385	0.783
ENSG00000111665	-0.69	0.000539	0.0291	CDCA3	0.991	0.639
ENSG00000152661	-0.83	0.000544	0.0293	GJA1	0.0856	0.915
ENSG00000187608	1.08	0.000549	0.0295	ISG15	0.000636	0.87
ENSG00000089220	-0.23	0.000556	0.0298	PEBP1	0.00339	0.849
ENSG00000197635	0.91	0.000559	0.0299	DPP4	0.0558	0.741
ENSG00000134153	-0.29	0.000561	0.0299	EMC7	0.00568	0.639
ENSG00000179832	0.26	0.000563	0.0299	MROH1	0.000106	0.749

ENSG00000130158	0.27	0.000565	0.0299	DOCK6	2.92e-05	0.725
ENSG00000070610	0.18	0.000574	0.0302	GBA2	0.00276	0.749
ENSG00000268621	-1.24	0.000578	0.0302	IGFL2-AS1	0.0342	0.645
ENSG00000198700	-0.15	0.000583	0.0302	IPO9	0.0185	0.93
ENSG00000070759	0.33	0.000584	0.0302	TESK2	0.101	0.823
ENSG00000080986	-0.7	0.000585	0.0302	NDC80	0.0134	0.645
ENSG00000105221	0.19	0.000585	0.0302	AKT2	1.87e-05	0.783
ENSG00000146842	-0.34	0.000586	0.0302	TMEM209	0.000414	0.639
ENSG00000072071	0.55	0.000587	0.0302	ADGRL1	0.00087	0.876
ENSG00000175634	0.21	0.000589	0.0302	RPS6KB2	0.00967	0.858
ENSG00000078898	-1.11	0.000589	0.0302	BPIFB2	0.112	0.651
ENSG00000162069	0.42	0.000589	0.0302	BICDL2	3.32e-05	0.639
ENSG00000164284	-0.3	0.000598	0.0305	GRPEL2	0.0055	0.858
ENSG00000159216	0.15	0.000601	0.0306	RUNX1	0.0147	0.639
ENSG00000198488	0.84	0.00061	0.031	B3GNT6	0.0329	0.915
ENSG00000161057	-0.22	0.000611	0.031	PSMC2	0.000366	0.639
ENSG00000101246	0.32	0.000616	0.0311	ARFRP1	0.000295	0.645
ENSG00000207730	0.65	0.000617	0.0311	MIR200B	0.000361	0.783
ENSG00000104331	-0.18	0.000621	0.0312	BPNT2	0.0015	0.749
ENSG00000198934	-0.48	0.000623	0.0312	MAGEE1	0.0738	0.985
ENSG00000126524	-0.25	0.000624	0.0312	SBDS	0.0176	0.639
ENSG00000124207	-0.27	0.000627	0.0312	CSE1L	0.000143	0.639
ENSG00000105443	0.23	0.000629	0.0312	CYTH2	7,00E-05	0.639
ENSG00000091164	-0.21	0.000629	0.0312	TXNL1	0.00199	0.645
ENSG00000124664	0.4	0.000633	0.0313	SPDEF	0.175	0.858
ENSG00000177301	-1.04	0.000634	0.0313	KCNA2	0.0878	0.896
ENSG00000180104	0.22	0.000646	0.0317	EXOC3	0.000366	0.639
ENSG00000153933	-0.26	0.000648	0.0318	DGKE	0.0166	0.639
ENSG00000116031	-0.82	0.000656	0.0321	CD207	0.00113	0.639
ENSG00000135387	-0.18	0.000662	0.0322	CAPRIN1	3.9e-05	0.669
ENSG00000140297	0.64	0.000662	0.0322	GCNT3	0.273	0.881

ENSG00000184371	0.64	0.000664	0.0322	CSF1	0.0488	0.641
ENSG00000173209	0.26	0.000666	0.0322	AHSA2P	0.00391	0.783
ENSG00000183323	0.23	0.000669	0.0322	CCDC125	0.748	0.749
ENSG00000089685	-0.66	0.00067	0.0322	BIRC5	0.28	0.722
ENSG00000249780	1.88	0.000685	0.0329	nan	NA	NA
ENSG00000207414	0.68	0.000696	0.0333	RNU6-768P	0.992	0.755
ENSG00000171097	0.88	0.000698	0.0333	KYAT1	0.00674	0.858
ENSG00000196187	0.22	0.0007	0.0333	TMEM63A	0.00128	0.655
ENSG00000079785	-0.24	0.000713	0.0338	DDX1	0.00184	0.792
ENSG00000130595	0.92	0.000721	0.0341	TNNT3	0.0698	0.975
ENSG00000248587	0.97	0.000723	0.0341	GDNF-AS1	0.0247	0.639
ENSG00000067057	0.25	0.000724	0.0341	PFKP	0.0321	0.87
ENSG00000099949	0.39	0.00073	0.0341	LZTR1	0.00251	0.639
ENSG00000256751	-0.27	0.000732	0.0341	PLBD1-AS1	0.0957	0.645
ENSG00000099889	0.36	0.000732	0.0341	ARVCF	5.85e-05	0.641
ENSG00000127578	0.77	0.000737	0.0341	WFIKKN1	0.000233	0.639
ENSG00000176715	0.23	0.000741	0.0341	ACSF3	0.00109	0.639
ENSG00000154122	-0.48	0.000742	0.0341	ANKH	0.0176	0.999
ENSG00000172943	0.17	0.000744	0.0341	PHF8	0.000866	0.792
ENSG00000137672	-0.88	0.000745	0.0341	TRPC6	0.0289	0.813
ENSG00000185507	0.62	0.000745	0.0341	IRF7	0.000668	0.783
ENSG00000137496	0.51	0.000745	0.0341	IL18BP	0.00171	0.722
ENSG00000171456	0.15	0.000746	0.0341	ASXL1	0.0802	0.639
ENSG00000241529	0.93	0.000762	0.0348	RN7SL767P	0.000953	0.858
ENSG00000131620	0.7	0.000764	0.0348	ANO1	0.555	0.891
ENSG00000102317	-0.17	0.000766	0.0348	RBM3	0.00967	0.645
ENSG00000188312	-0.38	0.000772	0.0349	CENPP	0.0168	0.639
ENSG00000135451	-0.76	0.000773	0.0349	TROAP	0.59	0.796
ENSG00000151468	-0.81	0.000787	0.0355	CCDC3	0.0341	0.783
ENSG00000142235	0.62	0.0008	0.036	LMTK3	0.000309	0.858
ENSG00000172845	-0.18	0.000806	0.036	SP3	0.000641	0.712

ENSG00000131747	-0.52	0.000809	0.036	TOP2A	0.0199	0.858
ENSG00000116514	0.4	0.000809	0.036	RNF19B	0.00426	0.999
ENSG00000066926	-0.24	0.000809	0.036	FECH	0.00111	0.983
ENSG00000264063	0.92	0.000812	0.0361	nan	7.67e-05	0.641
ENSG00000278291	-0.63	0.000818	0.0363	nan	0.0546	0.995
ENSG00000100714	-0.21	0.000829	0.0366	MTHFD1	0.00141	0.639
ENSG00000100722	-0.22	0.000843	0.0372	ZC3H14	0.000894	0.858
ENSG00000138760	-0.19	0.000848	0.0373	SCARB2	0.00563	0.717
ENSG00000142002	0.26	0.00085	0.0373	DPP9	0.0147	0.995
ENSG00000169714	-0.16	0.000853	0.0373	CNBP	0.00156	0.783
ENSG00000165458	0.2	0.000854	0.0373	INPPL1	0.000361	0.758
ENSG00000178188	0.24	0.000857	0.0373	SH2B1	0.00017	0.858
ENSG00000099341	-0.2	0.000862	0.0374	PSMD8	4.72e-05	0.758
ENSG00000273748	-0.58	0.000863	0.0374	nan	0.0266	0.639
ENSG00000117748	-0.3	0.000866	0.0375	RPA2	0.00113	0.909
ENSG00000139926	-0.41	0.000876	0.0378	FRMD6	0.0129	0.783
ENSG00000287306	0.47	0.000878	0.0378	nan	0.167	0.995
ENSG00000076944	0.32	0.000888	0.0381	STXBP2	0.000106	0.936
ENSG00000104957	0.24	0.000896	0.0384	YJU2B	0.00967	0.639
ENSG00000287059	0.96	0.000904	0.0386	nan	0.586	0.722
ENSG00000096696	-0.21	0.000908	0.0386	DSP	0.00262	0.954
ENSG00000165092	-0.27	0.000908	0.0386	ALDH1A1	0.00967	0.641
ENSG00000198466	0.21	0.000916	0.0388	ZNF587	0.0212	0.858
ENSG00000169612	-0.32	0.000921	0.039	RAMAC	0.0146	0.874
ENSG00000125089	0.4	0.000928	0.0392	SH3TC1	0.00183	0.639
ENSG00000274487	0.43	0.000929	0.0392	NPEPPSP1	0.00188	0.99
ENSG00000180251	-0.57	0.000934	0.0393	SLC9A4	0.00132	0.639
ENSG00000126016	0.3	0.000941	0.0395	AMOT	0.701	0.749
ENSG00000140105	0.73	0.000945	0.0396	WARS1	0.0439	0.758
ENSG00000250320	-0.63	0.000952	0.0396	EDIL3-DT	0.0223	0.881
ENSG00000166260	-0.26	0.000954	0.0396	COX11	0.00339	0.792

ENSG00000266714	0.49	0.000954	0.0396	MYO15B	0.000945	0.639
ENSG00000120314	0.17	0.000956	0.0396	WDR55	0.00665	0.813
ENSG00000165629	-0.23	0.000959	0.0396	ATP5F1C	0.000878	0.783
ENSG00000110848	0.93	0.000959	0.0396	CD69	0.215	0.639
ENSG00000172172	-0.34	0.000965	0.0397	MRPL13	2.97e-05	0.973
ENSG00000179348	1.13	0.000966	0.0397	GATA2	0.0187	0.905
ENSG00000145014	0.32	0.00097	0.0397	TMEM44	0.000256	0.639
ENSG00000119917	0.9	0.000977	0.0399	IFIT3	0.0234	0.915
ENSG00000235776	0.72	0.000979	0.0399	RPL7AP70	0.409	0.985
ENSG00000145423	-1.11	0.000987	0.0402	SFRP2	0.178	0.944
ENSG00000164163	-0.26	0.000998	0.0404	ABCE1	0.000482	0.639
ENSG00000111615	-0.25	0.001	0.0404	KRR1	0.000778	0.645
ENSG00000280248	0.8	0.001	0.0404	nan	0.00137	0.884
ENSG00000111058	-0.39	0.001	0.0404	ACSS3	0.0177	0.915
ENSG00000133943	0.19	0.00101	0.0404	DGLUCY	0.00727	0.781
ENSG00000175309	0.28	0.00101	0.0404	PHYKPL	0.00434	0.884
ENSG00000149499	0.26	0.00101	0.0404	EML3	4.16e-05	0.639
ENSG00000213614	0.19	0.00101	0.0406	HEXA	0.162	0.975
ENSG00000181885	0.21	0.00102	0.0406	CLDN7	0.00146	0.954
ENSG00000162591	0.54	0.00102	0.0406	MEGF6	0.000156	0.639
ENSG00000092529	0.38	0.00103	0.0407	CAPN3	0.12	0.95
ENSG00000163629	-0.29	0.00103	0.041	PTPN13	0.0104	0.813
ENSG00000166888	0.18	0.00104	0.0411	STAT6	0.0011	0.792
ENSG00000188747	0.4	0.00104	0.0411	NOXA1	0.000256	0.639
ENSG00000049239	0.19	0.00104	0.0411	H6PD	0.00151	0.645
ENSG00000175221	0.27	0.00105	0.0411	MED16	0.000624	0.645
ENSG00000139579	-0.25	0.00105	0.0411	NABP2	0.000986	0.641
ENSG00000178927	0.37	0.00105	0.0411	CYBC1	0.000135	0.816
ENSG00000270792	0.61	0.00105	0.0411	nan	0.161	0.901
ENSG00000151239	-0.28	0.00106	0.0411	TWF1	0.000136	0.792
ENSG00000275708	0.96	0.00106	0.0411	MIR3648-1	0.000361	0.783

ENSG00000166428	-0.62	0.00106	0.0411	PLD4	0.00444	0.639
ENSG00000280138	0.41	0.00107	0.0413	nan	0.0055	0.95
ENSG00000157322	0.73	0.00108	0.0415	CLEC18A	0.0593	0.781
ENSG00000130164	0.43	0.00108	0.0416	LDLR	0.0199	0.712
ENSG00000187715	0.93	0.00109	0.0419	KBTBD12	0.0245	0.639
ENSG00000118200	-0.26	0.0011	0.0422	CAMSAP2	0.0045	0.641
ENSG00000076706	-0.53	0.0011	0.0422	MCAM	0.00914	0.994
ENSG00000149428	0.22	0.0011	0.0422	HYOU1	0.101	0.667
ENSG00000155256	0.21	0.00111	0.0422	ZFYVE27	0.00972	0.858
ENSG00000151065	-0.34	0.00112	0.0426	DCP1B	0.0421	0.983
ENSG00000174943	0.28	0.00113	0.0426	KCTD13	0.000496	0.712
ENSG00000177169	0.33	0.00113	0.0426	ULK1	5.56e-05	0.639
ENSG00000166147	-0.63	0.00113	0.0426	FBN1	0.0733	0.915
ENSG00000120049	0.62	0.00113	0.0426	KCNIP2	0.109	0.909
ENSG00000105492	0.99	0.00114	0.0426	SIGLEC6	0.132	0.758
ENSG00000129925	0.24	0.00114	0.0426	PGAP6	5.05e-05	0.669
ENSG00000170608	0.63	0.00114	0.0426	FOXA3	0.0618	0.915
ENSG00000237441	0.21	0.00114	0.0426	RGL2	0.000335	0.639
ENSG00000239899	0.77	0.00116	0.0431	RN7SL674P	0.000308	0.858
ENSG00000085978	0.19	0.00116	0.0431	ATG16L1	0.0213	0.854
ENSG00000182473	0.15	0.00116	0.0431	EXOC7	0.000704	0.783
ENSG00000259570	0.97	0.00116	0.0431	nan	0.192	0.994
ENSG00000198198	0.19	0.00118	0.0436	SZT2	0.000431	0.669
ENSG00000182087	0.26	0.00118	0.0436	TMEM259	5.76e-06	0.645
ENSG00000039650	0.28	0.00119	0.0437	PNKP	7.48e-05	0.641
ENSG00000185359	0.22	0.00119	0.0438	HGS	0.00346	0.639
ENSG00000228300	0.29	0.00119	0.0438	FAM174C	0.000953	0.639
ENSG00000103044	0.58	0.0012	0.0438	HAS3	0.0928	0.858
ENSG00000154263	0.43	0.00121	0.044	ABCA10	0.112	0.881
ENSG00000108669	0.2	0.00121	0.044	CYTH1	3.42e-05	0.881
ENSG00000101972	-0.21	0.00121	0.044	STAG2	0.00332	0.87

ENSG00000120215	-1.44	0.00121	0.044	MLANA	0.961	0.858
ENSG00000243679	0.56	0.00121	0.044	nan	5.03e-06	0.639
ENSG00000101773	-0.26	0.00122	0.044	RBBP8	0.000793	0.86
ENSG00000124491	0.81	0.00122	0.044	F13A1	0.495	0.997
ENSG00000160216	0.17	0.00122	0.0441	AGPAT3	0.00154	0.712
ENSG00000129197	0.2	0.00123	0.0442	RPAIN	0.542	0.858
ENSG00000025708	0.57	0.00123	0.0442	TYMP	0.000943	0.722
ENSG00000171853	0.21	0.00123	0.0442	TRAPPC12	3.52e-05	0.858
ENSG00000189043	-0.29	0.00124	0.0443	NDUFA4	0.000527	0.641
ENSG00000134046	-0.21	0.00124	0.0444	MBD2	0.00142	0.639
ENSG00000274917	1.04	0.00125	0.0445	RNA5-8SN5	0.000188	0.641
ENSG00000117450	-0.23	0.00125	0.0445	PRDX1	0.00669	0.691
ENSG00000183682	0.75	0.00126	0.0447	BMP8A	0.0371	0.725
ENSG00000173805	0.84	0.00126	0.0448	HAP1	0.104	0.639
ENSG00000280265	0.58	0.00127	0.0449	nan	0.0508	0.858
ENSG00000164828	0.15	0.00129	0.0453	SUN1	0.0115	0.645
ENSG00000223756	0.76	0.0013	0.0457	TSSC2	0.000943	0.639
ENSG00000111341	-0.8	0.0013	0.0457	MGP	0.00297	0.783
ENSG00000090615	0.17	0.00131	0.0457	GOLGA3	0.00193	0.909
ENSG00000164109	-0.47	0.00131	0.0457	MAD2L1	0.00553	0.639
ENSG00000150093	-0.19	0.00131	0.0457	ITGB1	0.000482	0.639
ENSG00000167703	0.35	0.00132	0.0459	SLC43A2	3.72e-05	0.639
ENSG00000004534	0.18	0.00132	0.0459	RBM6	0.0128	0.884
ENSG00000163959	0.32	0.00133	0.0461	SLC51A	0.00151	0.882
ENSG00000064687	0.37	0.00133	0.0461	ABCA7	3.4e-05	0.639
ENSG00000196230	-0.21	0.00134	0.0464	TUBB	0.00689	0.645
ENSG00000168395	0.25	0.00135	0.0465	ING5	1.28e-05	0.639
ENSG00000086289	-0.55	0.00135	0.0465	EPDR1	0.127	0.981
ENSG00000162378	-0.19	0.00136	0.0466	ZYG11B	0.0121	0.639
ENSG00000137100	-0.22	0.00137	0.0469	DCTN3	0.0496	0.653
ENSG00000178971	0.22	0.00137	0.047	CTC1	0.0291	0.641

ENSG00000185479	-0.88	0.00138	0.0471	KRT6B	0.186	0.655
ENSG00000148082	-0.43	0.00138	0.0471	SHC3	0.17	0.792
ENSG00000173540	0.23	0.00138	0.0471	GMPPB	0.0248	0.775
ENSG00000060138	0.15	0.00139	0.0471	YBX3	0.0145	0.873
ENSG00000265735	0.8	0.00139	0.0471	RN7SL5P	0.000121	0.805
ENSG00000239306	0.21	0.0014	0.0473	RBM14	0.171	0.792
ENSG00000283167	-1.36	0.0014	0.0474	nan	0.59	0.729
ENSG00000132793	0.28	0.00141	0.0477	LPIN3	3.23e-05	0.639
ENSG00000135709	0.28	0.00142	0.048	KIAA0513	0.000415	0.915
ENSG00000167702	0.38	0.00143	0.0481	KIFC2	0.000216	0.646
ENSG00000255847	0.75	0.00144	0.0482	nan	NA	NA
ENSG00000164418	-0.45	0.00144	0.0482	GRIK2	0.0364	0.973
ENSG00000127129	1.31	0.00145	0.0482	EDN2	0.00706	0.711
ENSG00000162613	-0.21	0.00145	0.0482	FUBP1	0.00054	0.639
ENSG00000272333	0.29	0.00145	0.0482	KMT2B	1.73e-05	0.655
ENSG00000184545	0.43	0.00145	0.0482	DUSP8	0.0531	0.641
ENSG00000167632	0.18	0.00145	0.0482	TRAPPC9	0.00794	0.639
ENSG00000081760	0.19	0.00145	0.0482	AACS	0.0733	0.937
ENSG00000092201	-0.18	0.00146	0.0483	SUPT16H	0.000556	0.796
ENSG00000279281	0.42	0.00146	0.0483	nan	0.000773	0.669
ENSG00000089472	-0.59	0.00147	0.0483	HEPH	0.589	0.792
ENSG00000156869	-0.3	0.00148	0.0487	FRRS1	0.0597	0.645
ENSG00000186806	-0.99	0.00148	0.0487	VSIG10L	0.435	0.639
ENSG00000130940	0.21	0.00149	0.0487	CASZ1	0.000966	0.823
ENSG00000063176	0.29	0.00149	0.0487	SPHK2	0.00689	0.749
ENSG00000141367	-0.16	0.0015	0.0489	CLTC	0.00043	0.748
ENSG00000056972	-0.2	0.0015	0.049	TRAF3IP2	0.00176	0.669
ENSG00000264546	0.6	0.00152	0.0494	nan	0.17	1
ENSG00000198924	-0.3	0.00152	0.0494	DCLRE1A	0.00348	0.858
ENSG00000181444	0.52	0.00153	0.0496	ZNF467	0.000986	0.783
ENSG00000174837	0.99	0.00154	0.0496	ADGRE1	0.00967	0.669

ENSG00000072571	-0.62	0.00154	0.0496	HMMR	0.0391	0.639
ENSG00000106460	-0.31	0.00154	0.0496	TMEM106B	0.00453	0.645
ENSG00000008196	0.97	0.00154	0.0496	TFAP2B	0.924	0.639
ENSG00000153815	0.2	0.00154	0.0496	CMIP	0.000309	0.858
ENSG00000051009	0.2	0.00156	0.0499	FHIP1B	0.000107	0.926
ENSG00000140464	0.39	0.00156	0.0499	PML	0.000204	0.973
ENSG00000079246	-0.16	0.00156	0.0499	XRCC5	0.000335	0.669

Supplementary Table 1. *Differentially expressed genes between T2-high asthma and non-asthmatic controls. High BEC asthma was defined as: subjects with high blood eosinophil counts (>0.3×10^9/L) and low FeNO (<25 ppb; n = 27), high FeNO asthma was defined as: subjects with low blood eosinophil counts (<0.3×10^9/L) and high FeNO (>25 ppb; n = 24).*

Gene	logFC	PValue	FDR	Gene name
ENSG00000276070	2.51	7.72e-15	1.4e-10	CCL4L2
ENSG00000276085	2.62	2.72e-13	2.47e-09	CCL3L3
ENSG00000275302	2.1	1.01e-12	6.12e-09	CCL4
ENSG00000277632	2.18	3.27e-12	1.48e-08	CCL3
ENSG00000110436	-1.89	1.8e-08	6.53e-05	SLC1A2
ENSG00000168542	-2.21	9.96e-08	0.000302	COL3A1
ENSG00000164692	-2.72	2.6e-07	0.000647	COL1A2
ENSG00000167779	-1.59	2.85e-07	0.000647	IGFBP6
ENSG00000087245	-1.68	5.65e-07	0.00114	MMP2
ENSG00000139329	-2.66	6.59e-07	0.0012	LUM
ENSG00000113140	-1.83	7.38e-07	0.00121	SPARC
ENSG00000065618	-1.45	7.97e-07	0.00121	COL17A1
ENSG00000187498	-1.76	9.37e-07	0.00122	COL4A1
ENSG00000113721	-1.99	9.6e-07	0.00122	PDGFRB
ENSG00000149090	-2.08	1.01e-06	0.00122	PAMR1
ENSG00000154175	-1.35	1.09e-06	0.00123	ABI3BP
ENSG00000163359	-1.79	2.09e-06	0.00223	COL6A3

ENSG00000279602	1.57	2.49e-06	0.00251	nan
ENSG00000142173	-1.85	3.32e-06	0.00317	COL6A2
ENSG00000170373	2.94	5.96e-06	0.00538	CST1
ENSG00000117616	0.34	6.22e-06	0.00538	RSRP1
ENSG00000145824	-1.7	8.56e-06	0.00707	CXCL14
ENSG00000086991	-1.29	9.41e-06	0.00743	NOX4
ENSG00000186847	-2.29	1.25e-05	0.00944	KRT14
ENSG00000106624	-1.21	1.37e-05	0.00998	AEBP1
ENSG00000123485	-1.1	1.62e-05	0.0104	HJURP
ENSG00000204539	-3.0	1.62e-05	0.0104	CDSN
ENSG00000178372	-1.5	1.63e-05	0.0104	CALML5
ENSG00000099953	-1.22	1.67e-05	0.0104	MMP11
ENSG00000130822	1.35	1.79e-05	0.0108	PNCK
ENSG00000078898	-1.69	1.96e-05	0.0109	BPIFB2
ENSG00000122786	-1.0	1.97e-05	0.0109	CALD1
ENSG00000177606	0.72	1.98e-05	0.0109	JUN
ENSG00000247095	0.74	2.11e-05	0.0113	MIR210HG
ENSG00000090512	2.34	2.21e-05	0.0115	FETUB
ENSG00000285492	1.99	2.33e-05	0.0116	nan
ENSG00000007171	1.26	2.37e-05	0.0116	NOS2
ENSG00000143320	-0.96	2.64e-05	0.0126	CRABP2
ENSG00000256612	0.54	2.71e-05	0.0126	CYP2B7P
ENSG00000130635	-1.18	2.78e-05	0.0126	COL5A1
ENSG00000134871	-1.09	2.97e-05	0.0132	COL4A2
ENSG00000120217	1.04	3.44e-05	0.0149	CD274
ENSG00000150471	-1.18	3.74e-05	0.0158	ADGRL3
ENSG00000186340	-2.14	3.85e-05	0.0159	THBS2
ENSG00000149021	-1.51	4.33e-05	0.0172	SCGB1A1
ENSG00000124215	0.87	4.35e-05	0.0172	CDH26
ENSG00000010438	-1.37	4.63e-05	0.0179	PRSS3
ENSG00000136938	-0.3	4.88e-05	0.0185	ANP32B

ENSG00000166428	-0.83	5.16e-05	0.0191	PLD4
ENSG00000172554	1.14	5.35e-05	0.0194	SNTG2
ENSG00000262074	1.17	7.17e-05	0.0255	SNORD3B-2
ENSG00000169245	2.32	7.44e-05	0.026	CXCL10
ENSG00000215853	-1.57	7.6e-05	0.0261	RPTN
ENSG00000283167	-2.01	8.01e-05	0.0269	nan
ENSG00000135919	-1.08	8.28e-05	0.0273	SERPINE2
ENSG00000111665	-0.85	8.7e-05	0.0282	CDCA3
ENSG00000274114	0.98	9.08e-05	0.0289	ALOX15P1
ENSG00000189001	-2.14	9.65e-05	0.0302	SBSN
ENSG00000162545	-1.67	0.000106	0.0327	CAMK2N1
ENSG00000169474	-1.5	0.000116	0.035	SPRR1A
ENSG00000129521	0.92	0.00012	0.0358	EGLN3
ENSG00000179639	-0.91	0.000126	0.0369	FCER1A
ENSG00000280274	0.89	0.000129	0.0371	nan
ENSG00000137968	-0.98	0.000132	0.0375	SLC44A5
ENSG00000185479	-1.29	0.000143	0.0398	KRT6B
ENSG00000089327	-0.78	0.000151	0.0414	FXYD5
ENSG00000102755	1.11	0.000154	0.0414	FLT1
ENSG00000187609	0.5	0.000155	0.0414	EXD3
ENSG00000184371	0.78	0.000169	0.0445	CSF1
ENSG00000207414	0.77	0.000173	0.0449	RNU6-768P
ENSG00000234423	0.98	0.000176	0.045	LINC01250
ENSG00000182512	-0.37	0.000189	0.0476	GLRX5
ENSG00000115705	0.9	0.0002	0.0497	TPO

Supplementary Table 2. *Differentially expressed genes between stable T2-high asthma and non-asthmatic controls.*

ONT	ID	Description	setSize	Enrichment Score	NES	pvalue	p.adjust	qvalue
CC	GO:0022626	cytosolic ribosome	115	-0.821	-3.06	1,00E-10	2.33e-08	2.09e-08
CC	GO:0022625	cytosolic large ribosomal subunit	56	-0.88	-2.94	1,00E-10	2.33e-08	2.09e-08
CC	GO:0044391	ribosomal subunit	184	-0.74	-2.92	1,00E-10	2.33e-08	2.09e-08
CC	GO:0005840	ribosome	226	-0.695	-2.84	1,00E-10	2.33e-08	2.09e-08
CC	GO:0015934	large ribosomal subunit	113	-0.761	-2.84	1,00E-10	2.33e-08	2.09e-08
CC	GO:0022627	cytosolic small ribosomal subunit	42	-0.822	-2.61	1,00E-10	2.33e-08	2.09e-08
CC	GO:0015935	small ribosomal subunit	75	-0.706	-2.48	1,00E-10	2.33e-08	2.09e-08
CC	GO:1990904	ribonucleoprotein complex	682	-0.465	-2.1	1,00E-10	2.33e-08	2.09e-08
CC	GO:0031012	extracellular matrix	340	-0.452	-1.92	1,00E-10	2.33e-08	2.09e-08
CC	GO:0005925	focal adhesion	390	-0.437	-1.88	1,00E-10	2.33e-08	2.09e-08
CC	GO:0005615	extracellular space	2338	-0.317	-1.52	1,00E-10	2.33e-08	2.09e-08
CC	GO:0030054	cell junction	1684	-0.317	-1.5	1,00E-10	2.33e-08	2.09e-08
CC	GO:0005576	extracellular region	2808	-0.31	-1.5	1,00E-10	2.33e-08	2.09e-08
CC	GO:1903561	extracellular vesicle	1782	-0.311	-1.48	1,00E-10	2.33e-08	2.09e-08
CC	GO:0043230	extracellular organelle	1783	-0.311	-1.48	1,00E-10	2.33e-08	2.09e-08
CC	GO:0065010	extracellular membrane-bounded organelle	1783	-0.311	-1.48	1,00E-10	2.33e-08	2.09e-08
CC	GO:0070062	extracellular exosome	1767	-0.31	-1.47	1,00E-10	2.33e-08	2.09e-08
CC	GO:0043232	intracellular non-membrane-bounded organelle	4356	-0.28	-1.38	1,00E-10	2.33e-08	2.09e-08
CC	GO:0043228	non-membrane-bounded organelle	4357	-0.28	-1.38	1,00E-10	2.33e-08	2.09e-08
CC	GO:0032991	protein-containing complex	5082	-0.278	-1.37	1,00E-10	2.33e-08	2.09e-08
BP	GO:0002181	cytoplasmic translation	154	-0.767	-2.99	1,00E-10	2.33e-08	2.09e-08
BP	GO:0043043	peptide biosynthetic process	632	-0.47	-2.11	1,00E-10	2.33e-08	2.09e-08
BP	GO:0006412	translation	612	-0.472	-2.11	1,00E-10	2.33e-08	2.09e-08
BP	GO:0042254	ribosome biogenesis	308	-0.477	-2.01	1,00E-10	2.33e-08	2.09e-08

BP	GO:0022613	ribonucleoprotein complex biogenesis	454	-0.455	-1.98	1,00E-10	2.33e-08	2.09e-08
BP	GO:0043604	amide biosynthetic process	737	-0.435	-1.98	1,00E-10	2.33e-08	2.09e-08
BP	GO:0006518	peptide metabolic process	763	-0.43	-1.96	1,00E-10	2.33e-08	2.09e-08
BP	GO:0043603	amide metabolic process	993	-0.376	-1.74	1,00E-10	2.33e-08	2.09e-08
BP	GO:0071840	cellular component organization or biogenesis	5531	-0.268	-1.32	1,00E-10	2.33e-08	2.09e-08
BP	GO:0010467	gene expression	4961	-0.265	-1.3	1,00E-10	2.33e-08	2.09e-08
MF	GO:0003735	structural constituent of ribosome	155	-0.76	-2.97	1,00E-10	2.33e-08	2.09e-08
MF	GO:0005198	structural molecule activity	570	-0.55	-2.45	1,00E-10	2.33e-08	2.09e-08
MF	GO:0003723	RNA binding	1542	-0.368	-1.74	1,00E-10	2.33e-08	2.09e-08
MF	GO:0003676	nucleic acid binding	3330	-0.287	-1.4	1,00E-10	2.33e-08	2.09e-08
CC	GO:0099080	supramolecular complex	1065	-0.339	-1.57	4.52e-10	9.45e-08	8.45e-08
BP	GO:1901566	organonitrogen compound biosynthetic process	1499	-0.316	-1.49	7.92e-10	1.57e-07	1.41e-07
BP	GO:0050789	regulation of biological process	8517	-0.247	-1.23	1.38e-09	2.55e-07	2.28e-07
BP	GO:0032502	developmental process	4696	-0.261	-1.28	1.68e-09	2.96e-07	2.65e-07
CC	GO:0005730	nucleolus	945	-0.343	-1.58	1.97e-09	3.4e-07	3.04e-07
BP	GO:0032501	multicellular organismal process	5175	-0.257	-1.27	1.39e-08	2.16e-06	1.93e-06
BP	GO:0030199	collagen fibril organization	58	-0.675	-2.27	1.91e-08	2.92e-06	2.61e-06
BP	GO:0140014	mitotic nuclear division	255	-0.456	-1.89	2.63e-08	3.79e-06	3.39e-06
BP	GO:0002250	adaptive immune response	384	-0.398	-1.71	4.9e-08	6.38e-06	5.7e-06
CC	GO:0001533	cornified envelope	44	-0.705	-2.24	1.08e-07	1.3e-05	1.17e-05
BP	GO:0006364	rRNA processing	216	-0.459	-1.87	1.08e-07	1.3e-05	1.17e-05
BP	GO:0000278	mitotic cell cycle	799	-0.337	-1.54	1.28e-07	1.5e-05	1.34e-05
BP	GO:0051301	cell division	560	-0.364	-1.62	1.32e-07	1.52e-05	1.36e-05
BP	GO:0098813	nuclear chromosome segregation	279	-0.424	-1.77	1.58e-07	1.76e-05	1.58e-05

BP	GO:0140694	non-membrane-bounded organelle assembly	358	-0.399	-1.7	1.79e-07	1.95e-05	1.74e-05
BP	GO:0007059	chromosome segregation	378	-0.39	-1.68	3.9e-07	3.97e-05	3.55e-05
BP	GO:0022618	protein-RNA complex assembly	196	-0.462	-1.84	4.6e-07	4.56e-05	4.08e-05
CC	GO:0016020	membrane	6829	-0.242	-1.2	7.51e-07	7.09e-05	6.34e-05
BP	GO:0034502	protein localization to chromosome	120	-0.518	-1.96	7.82e-07	7.3e-05	6.53e-05
BP	GO:0016072	rRNA metabolic process	256	-0.428	-1.78	8.61e-07	7.68e-05	6.87e-05
BP	GO:0030154	cell differentiation	3133	-0.263	-1.28	9.77e-07	8.34e-05	7.46e-05
BP	GO:0006629	lipid metabolic process	1086	0.292	1.4	1.04e-06	8.75e-05	7.83e-05
BP	GO:0044281	small molecule metabolic process	1473	0.277	1.35	1.22e-06	9.94e-05	8.89e-05
CC	GO:0031982	vesicle	3242	-0.263	-1.28	1.5e-06	0.000118	0.000105
BP	GO:0046649	lymphocyte activation	649	-0.336	-1.52	1.55e-06	0.00012	0.000108
MF	GO:0050839	cell adhesion molecule binding	477	-0.352	-1.54	1.95e-06	0.000147	0.000132
CC	GO:0009986	cell surface	619	-0.336	-1.51	3.22e-06	0.000236	0.000211
BP	GO:1904666	regulation of ubiquitin protein ligase activity	17	-0.82	-2.16	3.34e-06	0.000243	0.000218
BP	GO:0016054	organic acid catabolic process	202	0.419	1.74	3.46e-06	0.000247	0.000221
BP	GO:0046395	carboxylic acid catabolic process	202	0.419	1.74	3.46e-06	0.000247	0.000221
BP	GO:0009058	biosynthetic process	6208	-0.241	-1.19	3.96e-06	0.000281	0.000251
BP	GO:0051783	regulation of nuclear division	118	-0.499	-1.88	4.53e-06	0.000315	0.000282
BP	GO:0002376	immune system process	1966	-0.272	-1.3	5,00E-06	0.000339	0.000303
MF	GO:0005518	collagen binding	58	-0.608	-2.05	5.16e-06	0.000346	0.00031
MF	GO:0097159	organic cyclic compound binding	4979	-0.245	-1.21	5.19e-06	0.000346	0.00031
BP	GO:0007088	regulation of mitotic nuclear division	103	-0.516	-1.91	6.46e-06	0.000427	0.000382
CC	GO:0005829	cytosol	4758	-0.246	-1.21	6.57e-06	0.000431	0.000385
BP	GO:0044282	small molecule catabolic process	294	0.371	1.62	7.85e-06	0.000498	0.000446

BP	GO:2001251	negative regulation of chromosome organization	87	-0.535	-1.92	8.17e-06	0.000514	0.00046	
MF	GO:0015631	tubulin binding	336	-0.374	-1.6	1.1e-05	0.00067	0.000599	
BP	GO:0050851	antigen receptor-mediated signaling pathway	174	-0.443	-1.75	1.14e-05	0.000692	0.000619	
CC	GO:0031974	membrane-enclosed lumen	4931	-0.245	-1.21	1.28e-05	0.00075	0.000671	
CC	GO:0043233	organelle lumen	4931	-0.245	-1.21	1.28e-05	0.00075	0.000671	
CC	GO:0070013	intracellular organelle lumen	4931	-0.245	-1.21	1.28e-05	0.00075	0.000671	
BP	GO:0009056	catabolic process	2195	0.252	1.25	1.58e-05	0.000894		8,00E-04
BP	GO:0032787	monocarboxylic acid metabolic process	503	0.328	1.5	1.95e-05	0.00108	0.000967	
BP	GO:0050776	regulation of immune response	748	-0.314	-1.43	1.97e-05	0.00109	0.000973	
BP	GO:0051310	metaphase chromosome alignment	95	-0.505	-1.84	2.4e-05	0.0013	0.00117	
MF	GO:0005539	glycosaminoglycan binding	147	-0.452	-1.75	2.47e-05	0.00134	0.00119	
CC	GO:0000786	nucleosome	103	-0.495	-1.84	2.51e-05	0.00134	0.0012	
BP	GO:0001775	cell activation	883	-0.303	-1.39	2.65e-05	0.00141	0.00126	
BP	GO:1905818	regulation of chromosome separation	70	-0.553	-1.93	2.73e-05	0.00145	0.00129	
CC	GO:0097014	ciliary plasm	163	-0.442	-1.73	2.88e-05	0.0015	0.00134	
BP	GO:0007094	mitotic spindle assembly checkpoint signaling	43	-0.627	-1.99	3.3e-05	0.00167	0.00149	
BP	GO:0071173	spindle assembly checkpoint signaling	43	-0.627	-1.99	3.3e-05	0.00167	0.00149	
BP	GO:0071174	mitotic spindle checkpoint signaling	43	-0.627	-1.99	3.3e-05	0.00167	0.00149	
BP	GO:0044248	cellular catabolic process	1373	0.266	1.29	3.92e-05	0.00194	0.00174	
BP	GO:0006807	nitrogen compound metabolic process	7681	-0.232	-1.16	3.95e-05	0.00195	0.00174	
BP	GO:0032094	response to food	23	0.717	2.02	4.47e-05	0.00218	0.00195	

BP	GO:1903131	mononuclear cell differentiation	395	-0.346	-1.49	4.53e-05	0.00219	0.00196
BP	GO:0019730	antimicrobial humoral response	74	-0.541	-1.9	4.59e-05	0.00221	0.00197
BP	GO:1902100	negative regulation of metaphase/anaphase transition of cell cycle	46	-0.61	-1.95	5.02e-05	0.00232	0.00208
BP	GO:0046320	regulation of fatty acid oxidation	28	0.684	2	5.69e-05	0.00248	0.00222
BP	GO:0070542	response to fatty acid	45	0.61	1.98	6.18e-05	0.00267	0.00239
BP	GO:0010628	positive regulation of gene expression	913	-0.293	-1.35	6.7e-05	0.00283	0.00253
BP	GO:0071398	cellular response to fatty acid	28	0.681	1.99	6.93e-05	0.00288	0.00258
BP	GO:0000226	microtubule cytoskeleton organization	575	-0.323	-1.44	7.47e-05	0.00306	0.00273
BP	GO:0051443	positive regulation of ubiquitin-protein transferase activity	18	-0.769	-2.04	7.88e-05	0.00321	0.00287
BP	GO:0005996	monosaccharide metabolic process	207	0.389	1.62	8.07e-05	0.00323	0.00289
BP	GO:0007017	microtubule-based process	808	-0.299	-1.36	8.11e-05	0.00323	0.00289
BP	GO:0034248	regulation of amide metabolic process	394	-0.343	-1.48	8.65e-05	0.00339	0.00303
BP	GO:0042100	B cell proliferation	69	-0.541	-1.88	8.77e-05	0.00341	0.00305
CC	GO:0014069	postsynaptic density	230	-0.388	-1.59	9.35e-05	0.00361	0.00323
BP	GO:0044419	biological process involved in interspecies interaction between organisms	1292	-0.279	-1.3	1,00E-04	0.00377	0.00337
BP	GO:0006334	nucleosome assembly	94	-0.491	-1.79	0.000114	0.00418	0.00374
BP	GO:0019953	sexual reproduction	711	-0.306	-1.39	0.000119	0.0043	0.00384
MF	GO:0030695	GTPase regulator activity	436	0.319	1.44	0.000119	0.0043	0.00384
MF	GO:0060589	nucleoside-triphosphatase regulator activity	436	0.319	1.44	0.000119	0.0043	0.00384

BP	GO:0033631	cell-cell adhesion mediated by integrin	17	-0.755	-1.99	0.000157	0.00546	0.00488
MF	GO:0005178	integrin binding	118	-0.448	-1.68	0.000218	0.00718	0.00642
BP	GO:0007275	multicellular organism development	3436	-0.244	-1.19	0.000305	0.00961	0.00859
BP	GO:0001906	cell killing	146	-0.413	-1.59	0.000322	0.0101	0.009
BP	GO:0045834	positive regulation of lipid metabolic process	83	0.477	1.73	0.000374	0.0113	0.0101
BP	GO:0006613	cotranslational protein targeting to membrane	26	-0.677	-2	0.000395	0.0119	0.0106
BP	GO:0019637	organophosphate metabolic process	882	0.272	1.29	0.000397	0.0119	0.0106
BP	GO:0009607	response to biotic stimulus	1177	-0.274	-1.28	0.00041	0.0122	0.0109
BP	GO:0035987	endodermal cell differentiation	41	-0.592	-1.87	0.000439	0.0128	0.0114
MF	GO:0016491	oxidoreductase activity	610	0.289	1.34	0.000442	0.0128	0.0114
CC	GO:0002116	semaphorin receptor complex	12	0.787	1.89	0.000458	0.0131	0.0117
BP	GO:0031294	lymphocyte costimulation	40	-0.597	-1.88	0.000461	0.0131	0.0117
MF	GO:0055105	ubiquitin-protein transferase inhibitor activity	11	-0.812	-1.9	0.000485	0.0137	0.0123
BP	GO:0010565	regulation of cellular ketone metabolic process	99	0.44	1.64	0.000495	0.014	0.0125
BP	GO:0034728	nucleosome organization	113	-0.439	-1.64	0.000512	0.0143	0.0128
CC	GO:0030496	midbody	197	-0.388	-1.55	0.000537	0.0149	0.0133
BP	GO:0050000	chromosome localization	111	-0.445	-1.65	0.000543	0.015	0.0134
BP	GO:1990173	protein localization to nucleoplasm	14	-0.761	-1.91	0.00059	0.0159	0.0142
CC	GO:0031090	organelle membrane	3174	0.231	1.16	0.000603	0.0161	0.0144
MF	GO:0043178	alcohol binding	64	-0.517	-1.78	0.00061	0.0162	0.0145
BP	GO:0010556	regulation of macromolecule biosynthetic process	3938	-0.238	-1.17	0.000638	0.0169	0.0151

BP	GO:0015718	monocarboxylic acid transport	117	0.407	1.57	0.000676	0.0177	0.0158
MF	GO:0045182	translation regulator activity	126	-0.426	-1.61	0.000698	0.0181	0.0162
BP	GO:0001910	regulation of leukocyte mediated cytotoxicity	73	-0.48	-1.68	0.000726	0.0185	0.0166
MF	GO:0043022	ribosome binding	106	-0.443	-1.64	0.000757	0.0193	0.0172
MF	GO:0015026	coreceptor activity	41	-0.577	-1.82	0.000847	0.0213	0.019
BP	GO:0002285	lymphocyte activation involved in immune response	170	-0.394	-1.55	0.000923	0.0226	0.0202
BP	GO:0008284	positive regulation of cell population proliferation	693	-0.29	-1.31	0.000921	0.0226	0.0202
BP	GO:0051172	negative regulation of nitrogen compound metabolic process	1752	-0.254	-1.21	0.000932	0.0228	0.0204
BP	GO:0007004	telomere maintenance via telomerase	68	-0.482	-1.67	0.000967	0.0233	0.0209
BP	GO:0031424	keratinization	34	-0.598	-1.83	0.00106	0.025	0.0223
MF	GO:0032266	phosphatidylinositol-3-phosphate binding	44	0.548	1.77	0.00107	0.0251	0.0225
BP	GO:0032731	positive regulation of interleukin-1 beta production	53	0.512	1.7	0.00109	0.0255	0.0228
MF	GO:0003823	antigen binding	58	-0.51	-1.71	0.00112	0.0258	0.0231
MF	GO:0140678	molecular function inhibitor activity	366	-0.327	-1.4	0.00117	0.0269	0.0241
BP	GO:0045589	regulation of regulatory T cell differentiation	26	0.633	1.81	0.0013	0.0293	0.0262
BP	GO:2000573	positive regulation of DNA biosynthetic process	65	-0.499	-1.72	0.00132	0.0296	0.0265
BP	GO:0030029	actin filament-based process	658	0.281	1.3	0.00132	0.0296	0.0265
BP	GO:1902001	fatty acid transmembrane transport	18	0.69	1.84	0.00133	0.0296	0.0265

BP	GO:1904816	positive regulation of protein localization to chromosome telomeric region	12	-0.766	-1.83	0.00133	0.0296	0.0265
BP	GO:0002449	lymphocyte mediated immunity	241	-0.357	-1.48	0.00135	0.0298	0.0267
BP	GO:0072079	nephron tubule formation	11	0.789	1.85	0.00137	0.03	0.0269
MF	GO:0016788	hydrolase activity acting on ester bonds	615	0.28	1.3	0.00136	0.03	0.0269
MF	GO:0005048	signal sequence binding	41	-0.564	-1.78	0.00142	0.0309	0.0277
BP	GO:0009889	regulation of biosynthetic process	4061	-0.233	-1.15	0.00155	0.0333	0.0298
BP	GO:0031326	regulation of cellular biosynthetic process	4034	-0.233	-1.14	0.00158	0.0337	0.0302
CC	GO:0030137	COPI-coated vesicle	24	-0.644	-1.85	0.00164	0.0346	0.031
BP	GO:0007346	regulation of mitotic cell cycle	443	-0.303	-1.32	0.00165	0.0347	0.0311
BP	GO:0014911	positive regulation of smooth muscle cell migration	36	-0.573	-1.78	0.00167	0.035	0.0313
BP	GO:0032943	mononuclear cell proliferation	237	-0.351	-1.45	0.00173	0.0361	0.0323
CC	GO:0042101	T cell receptor complex	19	-0.688	-1.86	0.00176	0.0367	0.0328
MF	GO:0031492	nucleosomal DNA binding	51	-0.52	-1.71	0.00179	0.037	0.0331
BP	GO:0007339	binding of sperm to zona pellucida	22	-0.659	-1.83	0.0018	0.037	0.0331
BP	GO:0006952	defense response	1329	-0.261	-1.22	0.0018	0.037	0.0331
BP	GO:0019400	alditol metabolic process	17	0.701	1.84	0.00182	0.0372	0.0333
BP	GO:0032703	negative regulation of interleukin-2 production	22	0.639	1.77	0.002	0.0403	0.036
CC	GO:0005689	U12-type spliceosomal complex	26	-0.629	-1.85	0.00203	0.0406	0.0363
BP	GO:0044283	small molecule biosynthetic process	464	0.295	1.34	0.00214	0.0425	0.038
BP	GO:0050810	regulation of steroid biosynthetic process	49	0.502	1.63	0.00235	0.0456	0.0408

BP	GO:0032963	collagen metabolic process	79	-0.451	-1.6	0.00236	0.0457	0.0409
MF	GO:1901681	sulfur compound binding	184	-0.368	-1.45	0.00237	0.0457	0.0409
BP	GO:0018149	peptide cross-linking	19	-0.679	-1.84	0.00244	0.0468	0.0419
BP	GO:0030036	actin cytoskeleton organization	595	0.282	1.3	0.0025	0.0476	0.0426
BP	GO:0008283	cell population proliferation	1499	-0.254	-1.2	0.00251	0.0477	0.0427
BP	GO:1904869	regulation of protein localization to Cajal body	11	-0.769	-1.8	0.00252	0.0478	0.0428
BP	GO:1904871	positive regulation of protein localization to Cajal body	11	-0.769	-1.8	0.00252	0.0478	0.0428
BP	GO:1900152	negative regulation of nuclear-transcribed mRNA catabolic process deadenylation-dependent decay	11	-0.768	-1.8	0.00256	0.0484	0.0433
BP	GO:0033632	regulation of cell-cell adhesion mediated by integrin	11	-0.768	-1.79	0.00259	0.0488	0.0436
BP	GO:0090232	positive regulation of spindle checkpoint	13	-0.744	-1.83	0.00261	0.0489	0.0437

Supplementary Table 3. *Gene Ontology pathways identified by gene set enrichment using ranked gene list from the differential gene expression analysis between T2-low asthma and non-asthmatic controls. Genes were ranked by (-log10Pvalue)*± 1 (depending on the direction of logFC).*

ID	Description	setSize	Enrichment Score	NES	pvalue	p.adjust	qvalue
R-HSA-156827	L13a-mediated translational silencing of Ceruloplasmin expression	109	-0.833	-3.15	1,00E-10	4.35e-09	3.54e-09
R-HSA-72706	GTP hydrolysis and joining of the 60S ribosomal subunit	110	-0.83	-3.15	1,00E-10	4.35e-09	3.54e-09

R-HSA-72613	Eukaryotic Translation Initiation	117	-0.823	-3.14	1,00E-10	4.35e-09	3.54e-09
R-HSA-72737	Cap-dependent Translation Initiation	117	-0.823	-3.14	1,00E-10	4.35e-09	3.54e-09
R-HSA-156842	Eukaryotic Translation Elongation	89	-0.857	-3.13	1,00E-10	4.35e-09	3.54e-09
R-HSA-156902	Peptide chain elongation	87	-0.856	-3.11	1,00E-10	4.35e-09	3.54e-09
R-HSA-72689	Formation of a pool of free 40S subunits	99	-0.837	-3.11	1,00E-10	4.35e-09	3.54e-09
R-HSA-1799339	SRP-dependent cotranslational protein targeting to membrane	110	-0.819	-3.11	1,00E-10	4.35e-09	3.54e-09
R-HSA-2408557	Selenocysteine synthesis	91	-0.847	-3.1	1,00E-10	4.35e-09	3.54e-09
R-HSA-192823	Viral mRNA Translation	87	-0.85	-3.09	1,00E-10	4.35e-09	3.54e-09
R-HSA-975956	Nonsense Mediated Decay (NMD) independent of the Exon Junction Complex (EJC)	93	-0.837	-3.08	1,00E-10	4.35e-09	3.54e-09
R-HSA-2408522	Selenoamino acid metabolism	113	-0.805	-3.06	1,00E-10	4.35e-09	3.54e-09
R-HSA-72764	Eukaryotic Translation Termination	91	-0.833	-3.05	1,00E-10	4.35e-09	3.54e-09
R-HSA-927802	Nonsense-Mediated Decay (NMD)	113	-0.792	-3.02	1,00E-10	4.35e-09	3.54e-09
R-HSA-975957	Nonsense Mediated Decay (NMD) enhanced by the Exon Junction Complex (EJC)	113	-0.792	-3.02	1,00E-10	4.35e-09	3.54e-09
R-HSA-9633012	Response of EIF2AK4 (GCN2) to amino acid deficiency	98	-0.811	-3.01	1,00E-10	4.35e-09	3.54e-09
R-HSA-168273	Influenza Viral RNA Transcription and Replication	133	-0.773	-2.99	1,00E-10	4.35e-09	3.54e-09
R-HSA-6791226	Major pathway of rRNA processing in the nucleolus and cytosol	178	-0.73	-2.96	1,00E-10	4.35e-09	3.54e-09
R-HSA-8868773	rRNA processing in the nucleus and cytosol	188	-0.717	-2.93	1,00E-10	4.35e-09	3.54e-09
R-HSA-168255	Influenza Infection	154	-0.737	-2.92	1,00E-10	4.35e-09	3.54e-09
R-HSA-9010553	Regulation of expression of SLITs and ROBOs	161	-0.725	-2.9	1,00E-10	4.35e-09	3.54e-09
R-HSA-72312	rRNA processing	197	-0.693	-2.84	1,00E-10	4.35e-09	3.54e-09
R-HSA-376176	Signaling by ROBO receptors	201	-0.686	-2.82	1,00E-10	4.35e-09	3.54e-09

R-HSA-72766	Translation	283	-0.654	-2.77	1,00E-10	4.35e-09	3.54e-09
R-HSA-9711097	Cellular response to starvation	148	-0.687	-2.71	1,00E-10	4.35e-09	3.54e-09
R-HSA-72649	Translation initiation complex formation	58	-0.78	-2.64	1,00E-10	4.35e-09	3.54e-09
R-HSA-72662	Activation of the mRNA upon binding of the cap-binding complex and eIFs and subsequent binding to 43S	59	-0.775	-2.64	1,00E-10	4.35e-09	3.54e-09
R-HSA-72695	Formation of the ternary complex and subsequently the 43S complex	51	-0.783	-2.61	1,00E-10	4.35e-09	3.54e-09
R-HSA-72702	Ribosomal scanning and start codon recognition	58	-0.768	-2.6	1,00E-10	4.35e-09	3.54e-09
R-HSA-9754678	SARS-CoV-2 modulates host translation machinery	50	-0.776	-2.58	1,00E-10	4.35e-09	3.54e-09
R-HSA-9735869	SARS-CoV-1 modulates host translation machinery	36	-0.825	-2.53	1,00E-10	4.35e-09	3.54e-09
R-HSA-71291	Metabolism of amino acids and derivatives	308	-0.549	-2.35	1,00E-10	4.35e-09	3.54e-09
R-HSA-422475	Axon guidance	463	-0.52	-2.3	1,00E-10	4.35e-09	3.54e-09
R-HSA-9675108	Nervous system development	484	-0.511	-2.27	1,00E-10	4.35e-09	3.54e-09
R-HSA-9692914	SARS-CoV-1-host interactions	93	-0.635	-2.34	2.12e-10	8.95e-09	7.28e-09
R-HSA-9678108	SARS-CoV-1 Infection	136	-0.527	-2.05	4.71e-08	1.94e-06	1.58e-06
R-HSA-3000178	ECM proteoglycans	52	-0.666	-2.22	2.17e-07	8.68e-06	7.06e-06
R-HSA-1474244	Extracellular matrix organization	219	-0.453	-1.88	2.33e-07	9.09e-06	7.4e-06
R-HSA-69620	Cell Cycle Checkpoints	278	-0.429	-1.81	3.09e-07	1.17e-05	9.55e-06
R-HSA-1650814	Collagen biosynthesis and modifying enzymes	50	-0.664	-2.2	3.38e-07	1.25e-05	1.02e-05
R-HSA-216083	Integrin cell surface interactions	64	-0.635	-2.21	3.87e-07	1.4e-05	1.14e-05
R-HSA-68886	M Phase	385	-0.387	-1.69	4.97e-07	1.75e-05	1.42e-05
R-HSA-9705683	SARS-CoV-2-host interactions	180	-0.463	-1.88	1.39e-06	4.77e-05	3.88e-05
R-HSA-8948216	Collagen chain trimerization	30	-0.743	-2.19	1.57e-06	5.29e-05	4.3e-05
R-HSA-1442490	Collagen degradation	46	-0.655	-2.12	3.04e-06	1,00E-04	8.13e-05

R-HSA-606279	Deposition of new CENPA-containing nucleosomes at the centromere	63	-0.602	-2.09	3.44e-06	0.000108	8.8e-05
R-HSA-774815	Nucleosome assembly	63	-0.602	-2.09	3.44e-06	0.000108	8.8e-05
R-HSA-9694516	SARS-CoV-2 Infection	266	-0.404	-1.7	4.5e-06	0.000139	0.000113
R-HSA-2555396	Mitotic Metaphase and Anaphase	216	-0.429	-1.78	5.82e-06	0.000176	0.000143
R-HSA-9679506	SARS-CoV Infections	368	-0.376	-1.63	6.26e-06	0.000185	0.000151
R-HSA-68882	Mitotic Anaphase	215	-0.425	-1.76	9.35e-06	0.000271	0.000221
R-HSA-6803157	Antimicrobial peptides	33	-0.691	-2.1	9.74e-06	0.000277	0.000225
R-HSA-3000170	Syndecan interactions	24	-0.752	-2.14	1.3e-05	0.000357	0.00029
R-HSA-1474290	Collagen formation	71	-0.556	-1.97	1.29e-05	0.000357	0.00029
R-HSA-2500257	Resolution of Sister Chromatid Cohesion	110	-0.497	-1.88	1.57e-05	0.000422	0.000343
R-HSA-2467813	Separation of Sister Chromatids	171	-0.44	-1.77	2.67e-05	0.000705	0.000573
R-HSA-3000480	Scavenging by Class A Receptors	12	-0.846	-2.02	3.19e-05	0.000829	0.000675
R-HSA-198933	Immunoregulatory interactions between a Lymphoid and a non-Lymphoid cell	93	-0.508	-1.87	3.89e-05	0.000994	0.000808
R-HSA-1474228	Degradation of the extracellular matrix	100	-0.497	-1.85	4.12e-05	0.00103	0.00084
R-HSA-6806834	Signaling by MET	72	-0.537	-1.9	5.08e-05	0.00125	0.00102
R-HSA-2299718	Condensation of Prophase Chromosomes	63	-0.56	-1.94	5.18e-05	0.00126	0.00102
R-HSA-6805567	Keratinization	61	-0.575	-1.97	5.43e-05	0.00128	0.00104
R-HSA-6809371	Formation of the cornified envelope	61	-0.575	-1.97	5.43e-05	0.00128	0.00104
R-HSA-69306	DNA Replication	174	-0.43	-1.73	5.59e-05	0.00129	0.00105
R-HSA-195258	RHO GTPase Effectors	290	-0.373	-1.58	6.39e-05	0.00146	0.00118
R-HSA-69481	G2/M Checkpoints	160	-0.432	-1.73	7.89e-05	0.00177	0.00144
R-HSA-9609690	HCMV Early Events	115	-0.467	-1.78	8.39e-05	0.00185	0.00151
R-HSA-3000171	Non-integrin membrane-ECM interactions	49	-0.582	-1.92	0.000111	0.00242	0.00197
R-HSA-68867	Assembly of the pre-replicative complex	132	-0.451	-1.74	0.000118	0.00254	0.00207
R-HSA-2022090	Assembly of collagen fibrils and other multimeric structures	50	-0.581	-1.93	0.000123	0.0026	0.00212

R-HSA-8875878	MET promotes cell motility	37	-0.635	-1.96	0.000131	0.0027	0.0022
R-HSA-2559586	DNA Damage/Telomere Stress Induced Senescence	72	-0.524	-1.85	0.00013	0.0027	0.0022
R-HSA-109582	Hemostasis	466	-0.326	-1.44	0.000152	0.00308	0.00251
R-HSA-8874081	MET activates PTK2 signaling	26	-0.687	-1.96	0.000175	0.00334	0.00272
R-HSA-5633008	TP53 Regulates Transcription of Cell Death Genes	43	0.578	1.88	0.000175	0.00334	0.00272
R-HSA-141424	Amplification of signal from the kinetochores	89	-0.492	-1.8	0.000173	0.00334	0.00272
R-HSA-141444	Amplification of signal from unattached kinetochores via a MAD2 inhibitory signal	89	-0.492	-1.8	0.000173	0.00334	0.00272
R-HSA-68877	Mitotic Prometaphase	186	-0.41	-1.68	0.000176	0.00334	0.00272
R-HSA-171306	Packaging Of Telomere Ends	44	-0.606	-1.94	0.000184	0.00345	0.00281
R-HSA-110330	Recognition and association of DNA glycosylase with site containing an affected purine	48	-0.579	-1.9	0.000205	0.0037	0.00301
R-HSA-110331	Cleavage of the damaged purine	48	-0.579	-1.9	0.000205	0.0037	0.00301
R-HSA-73927	Depurination	48	-0.579	-1.9	0.000205	0.0037	0.00301
R-HSA-68616	Assembly of the ORC complex at the origin of replication	59	-0.55	-1.87	0.000228	0.00407	0.00331
R-HSA-69618	Mitotic Spindle Checkpoint	105	-0.472	-1.78	0.000256	0.00441	0.00359
R-HSA-9648025	EML4 and NUDC in mitotic spindle formation	100	-0.476	-1.77	0.000254	0.00441	0.00359
R-HSA-5663220	RHO GTPases Activate Formins	123	-0.447	-1.72	0.000256	0.00441	0.00359
R-HSA-202733	Cell surface interactions at the vascular wall	99	-0.474	-1.76	0.000284	0.00483	0.00393
R-HSA-211000	Gene Silencing by RNA	120	-0.438	-1.68	0.000297	0.005	0.00407
R-HSA-5578749	Transcriptional regulation by small RNAs	96	-0.478	-1.77	0.00033	0.0055	0.00447
R-HSA-200425	Carnitine metabolism	13	0.78	1.88	0.000384	0.00632	0.00514
R-HSA-73728	RNA Polymerase I Promoter Opening	53	-0.549	-1.83	0.000394	0.00632	0.00514

R-HSA-912446	Meiotic recombination	71	-0.5	-1.77	0.00041	0.00632	0.00514
R-HSA-174143	APC/C-mediated degradation of cell cycle proteins	85	-0.48	-1.74	0.000409	0.00632	0.00514
R-HSA-453276	Regulation of mitotic cell cycle	85	-0.48	-1.74	0.000409	0.00632	0.00514
R-HSA-202403	TCR signaling	111	-0.451	-1.71	0.000405	0.00632	0.00514
R-HSA-69002	DNA Replication Pre-Initiation	148	-0.414	-1.63	0.000406	0.00632	0.00514
R-HSA-73772	RNA Polymerase I Promoter Escape	80	-0.483	-1.73	0.000419	0.00639	0.0052
R-HSA-162906	HIV Infection	221	-0.378	-1.57	0.000426	0.00644	0.00524
R-HSA-5693532	DNA Double-Strand Break Repair	164	-0.404	-1.63	0.000443	0.00662	0.00538
R-HSA-162909	Host Interactions of HIV factors	127	-0.434	-1.67	0.000519	0.00768	0.00625
R-HSA-73886	Chromosome Maintenance	125	-0.427	-1.64	0.000532	0.0078	0.00635
R-HSA-5693538	Homology Directed Repair	134	-0.427	-1.66	0.00056	0.00812	0.00661
R-HSA-977225	Amyloid fiber formation	88	-0.476	-1.73	0.000584	0.00839	0.00683
R-HSA-5625886	Activated PKN1 stimulates transcription of AR (androgen receptor) regulated genes KLK2 and KLK3	55	-0.532	-1.79	0.000651	0.00926	0.00754
R-HSA-110328	Recognition and association of DNA glycosylase with site containing an affected pyrimidine	53	-0.535	-1.78	0.000731	0.0101	0.00822
R-HSA-110329	Cleavage of the damaged pyrimidine	53	-0.535	-1.78	0.000731	0.0101	0.00822
R-HSA-73928	Depyrimidination	53	-0.535	-1.78	0.000731	0.0101	0.00822
R-HSA-5620924	Intraflagellar transport	48	-0.546	-1.79	0.000784	0.0107	0.00874
R-HSA-5693606	DNA Double Strand Break Response	75	-0.487	-1.75	0.000886	0.012	0.00979
R-HSA-73929	Base-Excision Repair AP Site Formation	55	-0.523	-1.76	0.00094	0.0125	0.0102
R-HSA-176408	Regulation of APC/C activators between G1/S and early anaphase	78	-0.479	-1.71	0.000932	0.0125	0.0102
R-HSA-9820965	Respiratory syncytial virus (RSV) genome replication transcription and translation	16	-0.735	-1.89	0.00108	0.0143	0.0117

R-HSA-6803205	TP53 regulates transcription of several additional cell death genes whose specific roles in p53-dependent apoptosis remain uncertain	14	0.745	1.85	0.00111	0.0146	0.0119
R-HSA-5334118	DNA methylation	54	-0.528	-1.77	0.00119	0.0155	0.0126
R-HSA-9725554	Differentiation of keratinocytes in interfollicular epidermis in mammalian skin	34	-0.591	-1.79	0.00133	0.0168	0.0137
R-HSA-9734767	Developmental Cell Lineages	34	-0.591	-1.79	0.00133	0.0168	0.0137
R-HSA-373760	L1CAM interactions	88	-0.461	-1.68	0.00134	0.0168	0.0137
R-HSA-5693567	HDR through Homologous Recombination (HRR) or Single Strand Annealing (SSA)	128	-0.421	-1.62	0.00132	0.0168	0.0137
R-HSA-8939236	RUNX1 regulates transcription of genes involved in differentiation of HSCs	115	-0.421	-1.6	0.00137	0.017	0.0139
R-HSA-186797	Signaling by PDGF	48	-0.534	-1.75	0.00141	0.0174	0.0142
R-HSA-69473	G2/M DNA damage checkpoint	89	-0.455	-1.67	0.00147	0.018	0.0146
R-HSA-5693565	Recruitment and ATM-mediated phosphorylation of repair and signaling proteins at DNA double strand breaks	74	-0.473	-1.69	0.00159	0.0193	0.0157
R-HSA-5693607	Processing of DNA double-strand break ends	93	-0.446	-1.64	0.00163	0.0196	0.016
R-HSA-69275	G2/M Transition	182	-0.383	-1.55	0.00168	0.02	0.0163
R-HSA-453274	Mitotic G2-G2/M phases	184	-0.38	-1.55	0.00172	0.0203	0.0165
R-HSA-202424	Downstream TCR signaling	91	-0.446	-1.63	0.00197	0.0232	0.0189
R-HSA-75892	Platelet Adhesion to exposed collagen	13	-0.742	-1.82	0.00201	0.0233	0.0189
R-HSA-2173782	Binding and Uptake of Ligands by Scavenger Receptors	31	-0.609	-1.81	0.00201	0.0233	0.0189
R-HSA-2426168	Activation of gene expression by SREBF (SREBP)	42	0.533	1.74	0.00223	0.0256	0.0208
R-HSA-427389	ERCC6 (CSB) and EHMT2 (G9a) positively regulate rRNA expression	66	-0.466	-1.64	0.00237	0.0269	0.0219

R-HSA-3214858	RMTs methylate histone arginines	68	-0.471	-1.67	0.00244	0.0273	0.0222
R-HSA-9610379	HCMV Late Events	105	-0.434	-1.63	0.00245	0.0273	0.0222
R-HSA-8957322	Metabolism of steroids	122	0.401	1.58	0.00242	0.0273	0.0222
R-HSA-5250924	B-WICH complex positively regulates rRNA expression	81	-0.456	-1.64	0.00253	0.0279	0.0227
R-HSA-73884	Base Excision Repair	84	-0.452	-1.64	0.00262	0.0287	0.0233
R-HSA-212300	PRC2 methylates histones and DNA	63	-0.486	-1.69	0.00275	0.0297	0.0242
R-HSA-73864	RNA Polymerase I Transcription	100	-0.432	-1.61	0.00275	0.0297	0.0242
R-HSA-5625740	RHO GTPases activate PKNs	80	-0.448	-1.6	0.00298	0.0319	0.026
R-HSA-389960	Formation of tubulin folding intermediates by CCT/TriC	16	-0.698	-1.79	0.00301	0.0321	0.0261
R-HSA-8852135	Protein ubiquitination	78	-0.453	-1.62	0.00328	0.0346	0.0282
R-HSA-5617833	Cilium Assembly	184	-0.369	-1.5	0.00351	0.0369	0.03
R-HSA-8866910	TFAP2 (AP-2) family regulates transcription of growth factors and their receptors	11	0.754	1.74	0.0036	0.0374	0.0305
R-HSA-9670095	Inhibition of DNA recombination at telomere	60	-0.492	-1.68	0.00362	0.0374	0.0305
R-HSA-983705	Signaling by the B Cell Receptor (BCR)	109	-0.422	-1.6	0.00372	0.0382	0.0311
R-HSA-427359	SIRT1 negatively regulates rRNA expression	58	-0.501	-1.7	0.00379	0.0387	0.0315
R-HSA-8852276	The role of GTSE1 in G2/M progression after G2 checkpoint	65	-0.466	-1.63	0.00402	0.0407	0.0331
R-HSA-73854	RNA Polymerase I Promoter Clearance	99	-0.426	-1.58	0.00429	0.0432	0.0352
R-HSA-9845614	Sphingolipid catabolism	10	0.761	1.69	0.00434	0.0434	0.0353
R-HSA-176814	Activation of APC/C and APC/C:Cdc20 mediated degradation of mitotic proteins	74	-0.45	-1.61	0.00443	0.0439	0.0357
R-HSA-1500620	Meiosis	95	-0.425	-1.57	0.00445	0.0439	0.0357
R-HSA-8866654	E3 ubiquitin ligases ubiquitinate target proteins	58	-0.494	-1.67	0.00457	0.0448	0.0364

R-HSA-69231	Cyclin D associated events in G1	47	-0.496	-1.61	0.00468	0.0453	0.0369
R-HSA-69236	G1 Phase	47	-0.496	-1.61	0.00468	0.0453	0.0369
R-HSA-9609646	HCMV Infection	138	-0.383	-1.5	0.00472	0.0454	0.0369
R-HSA-390450	Folding of actin by CCT/TriC	10	-0.781	-1.8	0.00518	0.0495	0.0403

Supplementary Table 4. *Reactome pathways identified by gene set enrichment using ranked gene list from the differential gene expression analysis between T2-low asthma and non-asthmatic controls. Genes were ranked by (-log10Pvalue)*± 1 (depending on the direction of logFC).*

Gene	Correlation MEblack	p_val MEblack	p_val_adjusted MEblack	Gene name
ENSG00000231389	0.876714682	2.21e-45	1.75e-43	HLA-DPA1
ENSG00000223865	0.876024203	3.16e-45	1.75e-43	HLA-DPB1
ENSG00000161405	0.872864747	1.58e-44	5.86e-43	IKZF3
ENSG00000167984	0.842874002	1.08e-38	3.01e-37	NLRC3
ENSG00000135905	0.836607713	1.26e-37	2.8e-36	DOCK10
ENSG00000161791	0.833928164	3.49e-37	6.45e-36	FMNL3
ENSG00000197471	0.833493703	4.1e-37	6.51e-36	SPN
ENSG00000167895	0.81122273	9.9e-34	1.37e-32	TMC8
ENSG00000182866	0.804750687	7.86e-33	9.69e-32	LCK
ENSG00000185477	0.803930839	1.02e-32	1.13e-31	GPRIN3
ENSG00000172215	0.797912786	6.46e-32	6.24e-31	CXCR6
ENSG00000204252	0.797762593	6.76e-32	6.24e-31	HLA-DOA
ENSG00000083457	0.797501011	7.31e-32	6.24e-31	ITGAE
ENSG00000123338	0.783352847	4.42e-30	3.51e-29	NCKAP1L
ENSG00000071246	0.781085713	8.29e-30	6.14e-29	VASH1
ENSG00000133561	0.780267264	1.04e-29	7.21e-29	GIMAP6
ENSG00000185811	0.779789505	1.18e-29	7.73e-29	IKZF1
ENSG00000113263	0.777050245	2.49e-29	1.54e-28	ITK

ENSG00000115085	0.769891071	1.66e-28	9.72e-28	ZAP70
ENSG00000198879	0.766516033	3.97e-28	2.21e-27	SFMBT2
ENSG00000271503	0.764445735	6.73e-28	3.56e-27	CCL5
ENSG00000160791	0.760657256	1.74e-27	8.79e-27	CCR5
ENSG00000116824	0.760245906	1.93e-27	9.31e-27	CD2
ENSG00000072818	0.756682408	4.63e-27	2.14e-26	ACAP1
ENSG00000103522	0.75265252	1.23e-26	5.44e-26	IL21R
ENSG00000153283	0.752391443	1.31e-26	5.57e-26	CD96
ENSG00000198851	0.747905799	3.77e-26	1.55e-25	CD3E
ENSG00000105122	0.746299012	5.48e-26	2.17e-25	RASAL3
ENSG00000197043	0.746109686	5.72e-26	2.19e-25	ANXA6
ENSG00000124203	0.741081112	1.81e-25	6.71e-25	ZNF831
ENSG00000159189	0.737697139	3.88e-25	1.39e-24	C1QC
ENSG00000180644	0.736652374	4.89e-25	1.7e-24	PRF1
ENSG00000118971	0.735705914	6.03e-25	2.03e-24	CCND2
ENSG00000004399	0.731125553	1.64e-24	5.37e-24	PLXND1
ENSG00000182578	0.729689954	2.24e-24	7.11e-24	CSF1R
ENSG00000146192	0.727712958	3.42e-24	1.06e-23	FGD2
ENSG00000184371	0.723133549	9e-24	2.7e-23	CSF1
ENSG00000136286	0.719776804	1.81e-23	5.27e-23	MYO1G
ENSG00000167286	0.717575582	2.84e-23	8.07e-23	CD3D
ENSG00000066294	0.717327428	2.98e-23	8.28e-23	CD84
ENSG00000140968	0.715435159	4.38e-23	1.19e-22	IRF8
ENSG00000107742	0.714194688	5.62e-23	1.49e-22	SPOCK2
ENSG00000146285	0.71222406	8.34e-23	2.15e-22	SCML4
ENSG00000158714	0.711073286	1.05e-22	2.65e-22	SLAMF8
ENSG00000155657	0.709700021	1.38e-22	3.39e-22	TTN
ENSG00000211772	0.708713921	1.67e-22	4.03e-22	TRBC2
ENSG00000173372	0.708232049	1.84e-22	4.34e-22	C1QA
ENSG00000213949	0.707653226	2.06e-22	4.76e-22	ITGA1
ENSG00000131401	0.705968918	2.86e-22	6.47e-22	NAPSB

ENSG00000115232	0.704736587	3.63e-22	8.05e-22	ITGA4
ENSG00000173369	0.704112528	4.09e-22	8.9e-22	C1QB
ENSG00000153563	0.699821741	9.28e-22	1.98e-21	CD8A
ENSG00000108798	0.698226172	1.25e-21	2.63e-21	ABI3
ENSG00000121281	0.695005216	2.29e-21	4.71e-21	ADCY7
ENSG00000175463	0.693132062	3.24e-21	6.53e-21	TBC1D10C
ENSG00000010610	0.691212331	4.6e-21	9.13e-21	CD4
ENSG00000005844	0.689567463	6.21e-21	1.21e-20	ITGAL
ENSG00000096996	0.688488444	7.55e-21	1.45e-20	IL12RB1
ENSG00000162676	0.684031545	1.68e-20	3.16e-20	GFI1
ENSG00000277734	0.680410827	3.18e-20	5.88e-20	TRAC
ENSG00000236790	0.678206781	4.66e-20	8.49e-20	LINC00299
ENSG00000196405	0.67466421	8.59e-20	1.54e-19	EVL
ENSG00000013725	0.674424493	8.95e-20	1.58e-19	CD6
ENSG00000137491	0.674088978	9.47e-20	1.64e-19	SLCO2B1
ENSG00000152495	0.668887501	2.28e-19	3.9e-19	CAMK4
ENSG00000147138	0.668361719	2.49e-19	4.19e-19	GPR174
ENSG00000165168	0.667780543	2.74e-19	4.55e-19	CYBB
ENSG00000125354	0.666673232	3.3e-19	5.39e-19	SEPTIN6
ENSG00000166428	0.665078889	4.29e-19	6.91e-19	PLD4
ENSG00000172673	0.664572833	4.67e-19	7.4e-19	THEMIS
ENSG00000110448	0.662261639	6.82e-19	1.07e-18	CD5
ENSG00000010810	0.658043917	1.35e-18	2.08e-18	FYN
ENSG00000159753	0.65451911	2.37e-18	3.6e-18	CARMIL2
ENSG00000100628	0.650331499	4.57e-18	6.85e-18	ASB2
ENSG00000104972	0.646692348	8.02e-18	1.19e-17	LILRB1
ENSG00000101082	0.646420169	8.36e-18	1.22e-17	SLA2
ENSG00000160185	0.644068971	1.2e-17	1.73e-17	UBASH3A
ENSG00000160654	0.635535849	4.31e-17	6.13e-17	CD3G
ENSG00000186818	0.634368977	5.12e-17	7.19e-17	LILRB4
ENSG00000177374	0.631774805	7.48e-17	1.04e-16	HIC1

ENSG00000134242	0.629724113	1.01e-16	1.38e-16	PTPN22
ENSG00000104814	0.627044658	1.48e-16	2e-16	MAP4K1
ENSG00000100385	0.625116318	1.95e-16	2.61e-16	IL2RB
ENSG00000173200	0.624427458	2.15e-16	2.84e-16	PARP15
ENSG00000281103	0.621677526	3.17e-16	4.13e-16	TRG-AS1
ENSG00000126882	0.618703326	4.8e-16	6.16e-16	FAM78A
ENSG00000180096	0.618649415	4.83e-16	6.16e-16	SEPTIN1
ENSG00000254838	0.617345266	5.79e-16	7.3e-16	GVINP1
ENSG00000104951	0.616323612	6.67e-16	8.31e-16	IL4I1
ENSG00000106565	0.615719648	7.24e-16	8.88e-16	TMEM176B
ENSG00000100351	0.615678731	7.28e-16	8.88e-16	GRAP2
ENSG00000159618	0.609868112	1.61e-15	1.94e-15	ADGRG5
ENSG00000160219	0.608714159	1.87e-15	2.24e-15	GAB3
ENSG00000115523	0.608474967	1.94e-15	2.29e-15	GNLY
ENSG00000088827	0.607869042	2.1e-15	2.45e-15	SIGLEC1
ENSG00000167613	0.607410114	2.23e-15	2.58e-15	LAIR1
ENSG00000168685	0.597721867	7.96e-15	9.1e-15	IL7R
ENSG00000138378	0.595590952	1.05e-14	1.19e-14	STAT4
ENSG00000175899	0.594779219	1.16e-14	1.3e-14	A2M
ENSG00000010327	0.59447657	1.21e-14	1.34e-14	STAB1
ENSG00000184060	0.585452769	3.75e-14	4.12e-14	ADAP2
ENSG00000110077	0.582415789	5.45e-14	5.93e-14	MS4A6A
ENSG00000106066	0.57555621	1.25e-13	1.35e-13	CPVL
ENSG00000107551	0.574149604	1.48e-13	1.58e-13	RASSF4
ENSG00000150637	0.565164985	4.23e-13	4.48e-13	CD226
ENSG00000285744	0.551077426	2.07e-12	2.17e-12	nan
ENSG00000137265	0.547342074	3.12e-12	3.24e-12	IRF4
ENSG00000038945	0.529578628	2.04e-11	2.1e-11	MSR1
ENSG00000177575	0.495718844	5.48e-10	5.58e-10	CD163
ENSG00000134028	0.473623916	3.89e-09	3.93e-09	ADAMDEC1
ENSG00000145416	0.448093455	3.18e-08	3.18e-08	MARCHF1

Supplementary Table 5. *Module membership of the genes in gene module "black".*

Gene	Correlation MEpurple	p_val MEpurple	p_val_adjusted MEpurple	Gene name
ENSG00000125691	0,909203273	5,49E-54	4,83E-52	RPL23
ENSG00000174444	0,885373746	2,04E-47	8,99E-46	RPL4
ENSG00000198755	0,883999445	4,41E-47	1,29E-45	RPL10A
ENSG00000071082	0,876529597	2,43E-45	5,35E-44	RPL31
ENSG00000231500	0,862721349	2,12E-42	3,74E-41	RPS18
ENSG00000142937	0,86223451	2,66E-42	3,90E-41	RPS8
ENSG00000171863	0,85649642	3,54E-41	4,46E-40	RPS7
ENSG00000114391	0,850819677	4,13E-40	4,54E-39	RPL24
ENSG00000144713	0,8490429	8,71E-40	8,52E-39	RPL32
ENSG00000186468	0,847550848	1,62E-39	1,42E-38	RPS23
ENSG00000197958	0,843861689	7,30E-39	5,84E-38	RPL12
ENSG00000142534	0,843112827	9,86E-39	7,23E-38	RPS11
ENSG00000198242	0,838280146	6,62E-38	4,48E-37	RPL23A
ENSG00000156482	0,832795635	5,33E-37	3,35E-36	RPL30
ENSG00000089009	0,831629922	8,22E-37	4,82E-36	RPL6
ENSG00000147604	0,825757751	6,96E-36	3,83E-35	RPL7
ENSG00000137154	0,818540467	8,62E-35	4,46E-34	RPS6
ENSG00000182899	0,815595922	2,33E-34	1,14E-33	RPL35A
ENSG00000145592	0,815383324	2,50E-34	1,16E-33	RPL37
ENSG00000114942	0,814474595	3,39E-34	1,49E-33	EEF1B2
ENSG00000083845	0,813326083	4,96E-34	2,08E-33	RPS5
ENSG00000109475	0,804135278	9,53E-33	3,81E-32	RPL34
ENSG00000108298	0,802577462	1,55E-32	5,93E-32	RPL19

ENSG00000174748	0,802328462	1,67E-32	6,13E-32	RPL15
ENSG00000164587	0,798056707	6,18E-32	2,18E-31	RPS14
ENSG00000147403	0,797835521	6,61E-32	2,24E-31	RPL10
ENSG00000108107	0,796330003	1,04E-31	3,39E-31	RPL28
ENSG00000188846	0,795442621	1,36E-31	4,26E-31	RPL14
ENSG00000110700	0,794729591	1,68E-31	5,08E-31	RPS13
ENSG00000197756	0,794463286	1,81E-31	5,32E-31	RPL37A
ENSG00000145741	0,786177717	2,00E-30	5,67E-30	BTF3
ENSG00000122026	0,783397062	4,37E-30	1,20E-29	RPL21
ENSG00000140988	0,775218194	4,08E-29	1,09E-28	RPS2
ENSG00000204628	0,77413474	5,45E-29	1,41E-28	RACK1
ENSG00000112306	0,773889389	5,81E-29	1,46E-28	RPS12
ENSG00000136942	0,770765606	1,32E-28	3,24E-28	RPL35
ENSG00000122406	0,763466083	8,62E-28	2,05E-27	RPL5
ENSG00000118181	0,760326081	1,89E-27	4,38E-27	RPS25
ENSG00000145425	0,755242996	6,57E-27	1,48E-26	RPS3A
ENSG00000138326	0,754154375	8,55E-27	1,88E-26	RPS24
ENSG00000221983	0,753349883	1,04E-26	2,23E-26	UBA52
ENSG00000229117	0,751831612	1,49E-26	3,13E-26	RPL41
ENSG00000105193	0,750205262	2,19E-26	4,49E-26	RPS16
ENSG00000148303	0,747712939	3,94E-26	7,88E-26	RPL7A
ENSG00000008988	0,743189583	1,12E-25	2,19E-25	RPS20
ENSG00000166441	0,742827847	1,22E-25	2,33E-25	RPL27A
ENSG00000143947	0,742019942	1,46E-25	2,74E-25	RPS27A
ENSG00000177954	0,738936784	2,94E-25	5,39E-25	RPS27
ENSG00000171858	0,737622757	3,94E-25	7,08E-25	RPS21
ENSG00000156508	0,735545578	6,25E-25	1,10E-24	EEF1A1

ENSG00000133112	0,733301943	1,02E-24	1,77E-24	TPT1
ENSG00000175061	0,73012899	2,04E-24	3,45E-24	SNHG29
ENSG00000198918	0,728234366	3,06E-24	5,08E-24	RPL39
ENSG00000065978	0,726045346	4,88E-24	7,95E-24	YBX1
ENSG00000161016	0,723696332	8,00E-24	1,28E-23	RPL8
ENSG00000142541	0,714104277	5,73E-23	9,00E-23	RPL13A
ENSG00000142676	0,709881666	1,33E-22	2,05E-22	RPL11
ENSG00000131469	0,705530087	3,11E-22	4,72E-22	RPL27
ENSG00000162244	0,691465765	4,40E-21	6,56E-21	RPL29
ENSG00000130255	0,690798206	4,97E-21	7,28E-21	RPL36
ENSG00000196531	0,687925468	8,36E-21	1,21E-20	NACA
ENSG00000181163	0,683804627	1,75E-20	2,48E-20	NPM1
ENSG00000137818	0,678333257	4,56E-20	6,37E-20	RPLP1
ENSG00000089157	0,673141069	1,11E-19	1,53E-19	RPLP0
ENSG00000173064	-0,669274434	2,14E-19	2,90E-19	HECTD4
ENSG00000125743	0,659134399	1,13E-18	1,51E-18	SNRPD2
ENSG00000127184	0,658337076	1,29E-18	1,69E-18	COX7C
ENSG00000070756	0,652543753	3,23E-18	4,18E-18	PABPC1
ENSG00000178035	0,648450103	6,12E-18	7,80E-18	IMPDH2
ENSG00000233927	0,648320903	6,24E-18	7,84E-18	RPS28
ENSG00000163682	0,641151856	1,86E-17	2,31E-17	RPL9
ENSG00000171490	0,639901248	2,25E-17	2,75E-17	RSL1D1
ENSG00000135486	0,631136575	8,20E-17	9,89E-17	HNRNPA1
ENSG00000127481	-0,627684476	1,35E-16	1,61E-16	UBR4
ENSG00000169567	0,627058817	1,48E-16	1,73E-16	HINT1
ENSG00000213741	0,620138333	3,93E-16	4,55E-16	RPS29
ENSG00000105640	0,609181283	1,76E-15	2,01E-15	RPL18A

ENSG00000196683	0,600842003	5,31E-15	5,99E-15	TOMM7
ENSG00000165502	0,593770054	1,32E-14	1,47E-14	RPL36AL
ENSG00000102007	0,587196655	3,02E-14	3,32E-14	PLP2
ENSG00000172809	0,555427247	1,28E-12	1,39E-12	RPL38
ENSG00000154723	0,547476764	3,08E-12	3,30E-12	ATP5PF
ENSG00000175390	0,542020026	5,54E-12	5,88E-12	EIF3F
ENSG00000169714	0,530101981	1,93E-11	2,03E-11	CNBP
ENSG00000215021	0,509153149	1,55E-10	1,61E-10	PHB2
ENSG00000126457	0,506368533	2,02E-10	2,07E-10	PRMT1
ENSG00000147684	0,493683748	6,60E-10	6,68E-10	NDUFB9
ENSG00000126267	0,461332689	1,09E-08	1,09E-08	COX6B1

Supplementary Table 6. *Module membership of the genes in gene module "purple".*

Gene	Correlation MEgreenyellow	p_val MEgreenyellow	p_val_adjusted MEgreenyellow	Gene name
ENSG00000012817	0,9910498	1,06E-121	2,86E-120	KDM5D
ENSG00000067048	0,98999516	2,10E-118	2,84E-117	DDX3Y
ENSG00000183878	0,98973685	1,20E-117	1,08E-116	UTY
ENSG00000229807	-0,9892751	2,40E-116	1,62E-115	XIST
ENSG00000114374	0,98790349	8,72E-113	4,71E-112	USP9Y
ENSG00000231535	0,98693343	1,66E-110	7,49E-110	LINC00278
ENSG00000131002	0,98631795	3,81E-109	1,47E-108	TXLNGY
ENSG00000099725	0,98343308	1,70E-103	5,75E-103	PRKY
ENSG00000165246	0,98305714	7,82E-103	2,35E-102	NLGN4Y
ENSG00000129824	0,98230855	1,47E-101	3,98E-101	RPS4Y1
ENSG00000198692	0,98189088	7,19E-101	1,76E-100	EIF1AY
ENSG00000067646	0,98112449	1,20E-99	2,70E-99	ZFY

ENSG00000176728	0,98020645	3,01E-98	6,24E-98	TTTY14
ENSG00000092377	0,9694922	1,55E-85	3,00E-85	TBL1Y
ENSG00000005889	-0,8225108	2,19E-35	3,94E-35	ZFX
ENSG00000130021	-0,8128808	5,75E-34	9,70E-34	PUDP
ENSG00000006756	-0,7991318	4,46E-32	7,09E-32	ARSD
ENSG00000147050	-0,7963724	1,03E-31	1,54E-31	KDM6A
ENSG00000072501	-0,7175297	2,86E-23	4,07E-23	SMC1A
ENSG00000198034	-0,6766	6,16E-20	8,31E-20	RPS4X
ENSG00000225470	-0,6621703	6,92E-19	8,89E-19	JPX
ENSG00000126012	-0,6411904	1,85E-17	2,27E-17	KDM5C
ENSG00000046651	-0,6299054	9,80E-17	1,15E-16	OFD1
ENSG00000006757	-0,5984099	7,28E-15	8,19E-15	PNPLA4
ENSG00000196459	-0,5203495	5,19E-11	5,60E-11	TRAPPC2
ENSG00000285679	-0,4515401	2,42E-08	2,51E-08	
ENSG00000285756	-0,378724	4,28E-06	4,28E-06	

Supplementary Table 7. *Module membership of the genes in gene module "greenyellow".*

ONT	ID	Description	GeneRatio	BgRatio	pvalue	p.adjust	qvalue
BP	GO:0007159	leukocyte cell-cell adhesion	22/103	296/11893	5.9e-15	4e-13	3e-13
CC	GO:0098802	plasma membrane signaling receptor complex	15/104	106/12371	9e-15	4e-13	3.3e-13
BP	GO:0002764	immune response-regulating signaling pathway	23/103	393/11893	2.2e-13	1e-11	7.6e-12
BP	GO:0045087	innate immune response	26/103	655/11893	3.4e-11	1e-09	7.6e-10
MF	GO:0004888	transmembrane signaling receptor activity	20/103	387/12230	5.8e-11	6e-09	5e-09
BP	GO:0001819	positive regulation of cytokine production	17/103	355/11893	9e-09	2e-07	1.5e-07

BP	GO:0002707	negative regulation of lymphocyte mediated immunity	7/103	40/11893	4.4e-08	9e-07	6.7e-07
BP	GO:0045580	regulation of T cell differentiation	10/103	138/11893	3e-07	6e-06	4.5e-06
MF	GO:0023023	MHC protein complex binding	6/103	30/12230	1.6e-07	8e-06	6.8e-06
BP	GO:0002703	regulation of leukocyte mediated immunity	10/103	173/11893	2.4e-06	4e-05	3e-05
BP	GO:0010628	positive regulation of gene expression	21/103	830/11893	6.9e-06	0.0001	7.8e-05
BP	GO:0051241	negative regulation of multicellular organismal process	19/103	734/11893	1.5e-05	0.0002	0.00015
BP	GO:0001909	leukocyte mediated cytotoxicity	7/103	92/11893	1.4e-05	0.0002	0.00015
BP	GO:0001913	T cell mediated cytotoxicity	5/103	40/11893	2.3e-05	0.0003	0.00023
BP	GO:0042127	regulation of cell population proliferation	24/103	1142/11893	3.1e-05	0.0004	0.0003
BP	GO:0016477	cell migration	23/103	1066/11893	3.2e-05	0.0004	0.0003
MF	GO:0003823	antigen binding	5/103	41/12230	2.3e-05	0.0007	0.00059
BP	GO:0050858	negative regulation of antigen receptor-mediated signaling pathway	4/103	25/11893	5.8e-05	0.0007	0.00053
BP	GO:0032715	negative regulation of interleukin-6 production	5/103	32/11893	7.4e-06	0.00011	8.2e-05
CC	GO:0031982	vesicle	46/104	3004/12371	6.2e-06	0.00012	9.6e-05
CC	GO:0031410	cytoplasmic vesicle	34/104	1902/12371	7.7e-06	0.00013	0.00011
CC	GO:0097708	intracellular vesicle	34/104	1907/12371	8.2e-06	0.00013	0.00011
BP	GO:0042113	B cell activation	10/103	200/11893	8.8e-06	0.00013	9.7e-05
BP	GO:0001906	cell killing	8/103	120/11893	9.1e-06	0.00013	9.9e-05
CC	GO:0030669	clathrin-coated endocytic vesicle membrane	6/104	58/12371	8.7e-06	0.00013	0.00011
BP	GO:0002831	regulation of response to biotic stimulus	14/103	406/11893	1e-05	0.00015	0.00011
BP	GO:0002274	myeloid leukocyte activation	9/103	163/11893	1.1e-05	0.00016	0.00012
BP	GO:0032640	tumor necrosis factor production	8/103	123/11893	1.1e-05	0.00016	0.00012
BP	GO:0032680	regulation of tumor necrosis factor production	8/103	123/11893	1.1e-05	0.00016	0.00012
BP	GO:0002366	leukocyte activation involved in immune response	10/103	209/11893	1.3e-05	0.00018	0.00014

BP	GO:0071706	tumor necrosis factor superfamily cytokine production	8/103	127/11893	1.4e-05	0.00019	0.00014
BP	GO:1903555	regulation of tumor necrosis factor superfamily cytokine production	8/103	127/11893	1.4e-05	0.00019	0.00014
BP	GO:0002263	cell activation involved in immune response	10/103	213/11893	1.5e-05	0.00021	0.00016
BP	GO:0002573	myeloid leukocyte differentiation	9/103	172/11893	1.8e-05	0.00024	0.00018
BP	GO:0019724	B cell mediated immunity	7/103	96/11893	1.9e-05	0.00025	0.00019
BP	GO:0034097	response to cytokine	18/103	681/11893	2e-05	0.00026	0.0002
BP	GO:0002532	production of molecular mediator involved in inflammatory response	6/103	67/11893	2.4e-05	0.00031	0.00023
CC	GO:0045334	clathrin-coated endocytic vesicle	6/104	68/12371	2.2e-05	0.00032	0.00027
BP	GO:0030183	B cell differentiation	7/103	101/11893	2.6e-05	0.00034	0.00026
BP	GO:0031341	regulation of cell killing	6/103	68/11893	2.6e-05	0.00034	0.00025
CC	GO:0008305	integrin complex	4/104	21/12371	2.5e-05	0.00035	0.00029
BP	GO:0050860	negative regulation of T cell receptor signaling pathway	4/103	21/11893	2.8e-05	0.00037	0.00028
BP	GO:0030099	myeloid cell differentiation	12/103	334/11893	3.1e-05	0.00039	0.00029
BP	GO:0072676	lymphocyte migration	6/103	71/11893	3.3e-05	0.00042	0.00031
BP	GO:0010758	regulation of macrophage chemotaxis	4/103	22/11893	3.4e-05	0.00043	0.00032
BP	GO:0036336	dendritic cell migration	4/103	22/11893	3.4e-05	0.00043	0.00032
CC	GO:0098796	membrane protein complex	20/104	876/12371	3.5e-05	0.00046	0.00038
MF	GO:0019956	chemokine binding	4/103	18/12230	1.3e-05	0.00046	0.00039
BP	GO:1902107	positive regulation of leukocyte differentiation	8/103	149/11893	4.4e-05	0.00054	0.0004
BP	GO:1903708	positive regulation of hemopoiesis	8/103	149/11893	4.4e-05	0.00054	0.0004
BP	GO:0002534	cytokine production involved in inflammatory response	5/103	47/11893	5.1e-05	0.00062	0.00046
BP	GO:0030225	macrophage differentiation	5/103	47/11893	5.1e-05	0.00062	0.00046
BP	GO:1900015	regulation of cytokine production involved in inflammatory response	5/103	47/11893	5.1e-05	0.00062	0.00046

BP	GO:0045577	regulation of B cell differentiation	4/103	26/11893	6.9e-05	0.00082	0.00061
BP	GO:0008283	cell population proliferation	26/103	1364/11893	7.3e-05	0.00086	0.00065
BP	GO:0048856	anatomical structure development	53/103	3917/11893	7.3e-05	0.00087	0.00065
BP	GO:0048246	macrophage chemotaxis	4/103	27/11893	8e-05	0.00094	0.0007
BP	GO:0048870	cell motility	24/103	1214/11893	8.4e-05	0.00098	0.00073
BP	GO:0002695	negative regulation of leukocyte activation	7/103	122/11893	8.8e-05	0.001	0.00077
MF	GO:0023026	MHC class II protein complex binding	4/103	23/12230	3.7e-05	0.001	0.00089
BP	GO:2000026	regulation of multicellular organismal development	21/103	988/11893	9.3e-05	0.0011	0.0008
BP	GO:0001818	negative regulation of cytokine production	9/103	214/11893	9.7e-05	0.0011	0.00083
BP	GO:0046637	regulation of alpha-beta T cell differentiation	5/103	54/11893	0.0001	0.0011	0.00085
BP	GO:0006956	complement activation	4/103	28/11893	9.3e-05	0.0011	0.0008
BP	GO:0006909	phagocytosis	8/103	168/11893	0.0001	0.0012	0.00086
BP	GO:0002761	regulation of myeloid leukocyte differentiation	6/103	87/11893	0.0001	0.0012	0.00088
BP	GO:0035710	CD4-positive, alpha-beta T cell activation	6/103	88/11893	0.00011	0.0012	0.00092
BP	GO:0071675	regulation of mononuclear cell migration	6/103	88/11893	0.00011	0.0012	0.00092
BP	GO:0002495	antigen processing and presentation of peptide antigen via MHC class II	4/103	29/11893	0.00011	0.0012	0.00089
BP	GO:0031589	cell-substrate adhesion	10/103	272/11893	0.00012	0.0013	0.00099
BP	GO:0050848	regulation of calcium-mediated signaling	5/103	57/11893	0.00013	0.0014	0.0011
BP	GO:0098883	synapse pruning	3/103	12/11893	0.00013	0.0014	0.0011
MF	GO:0005102	signaling receptor binding	19/103	838/12230	6.3e-05	0.0015	0.0013
BP	GO:0002504	antigen processing and presentation of peptide or polysaccharide antigen via MHC class II	4/103	31/11893	0.00014	0.0015	0.0011
MF	GO:0042287	MHC protein binding	4/103	26/12230	6.2e-05	0.0015	0.0013
BP	GO:0045582	positive regulation of T cell differentiation	6/103	93/11893	0.00015	0.0016	0.0012
BP	GO:0007160	cell-matrix adhesion	8/103	180/11893	0.00016	0.0017	0.0013
BP	GO:0050866	negative regulation of cell activation	7/103	135/11893	0.00017	0.0017	0.0013

BP	GO:0016064	immunoglobulin mediated immune response	6/103	94/11893	0.00016	0.0017	0.0013
BP	GO:0097530	granulocyte migration	6/103	95/11893	0.00017	0.0017	0.0013
BP	GO:0001910	regulation of leukocyte mediated cytotoxicity	5/103	60/11893	0.00017	0.0017	0.0013
BP	GO:1905521	regulation of macrophage migration	4/103	32/11893	0.00016	0.0017	0.0013
BP	GO:0002295	T-helper cell lineage commitment	3/103	13/11893	0.00017	0.0017	0.0013
BP	GO:0002399	MHC class II protein complex assembly	3/103	13/11893	0.00017	0.0017	0.0013
BP	GO:0002503	peptide antigen assembly with MHC class II protein complex	3/103	13/11893	0.00017	0.0017	0.0013
CC	GO:0030659	cytoplasmic vesicle membrane	20/104	970/12371	0.00014	0.0018	0.0015
BP	GO:0002832	negative regulation of response to biotic stimulus	6/103	96/11893	0.00018	0.0018	0.0014
BP	GO:0001914	regulation of T cell mediated cytotoxicity	4/103	33/11893	0.00018	0.0018	0.0014
BP	GO:0002347	response to tumor cell	4/103	33/11893	0.00018	0.0018	0.0014
BP	GO:0019722	calcium-mediated signaling	7/103	138/11893	0.00019	0.0019	0.0014
BP	GO:0032502	developmental process	55/103	4272/11893	0.0002	0.002	0.0015
BP	GO:0006897	endocytosis	14/103	536/11893	0.00021	0.002	0.0015
BP	GO:0045088	regulation of innate immune response	11/103	346/11893	0.0002	0.002	0.0015
BP	GO:0018108	peptidyl-tyrosine phosphorylation	8/103	186/11893	0.00021	0.002	0.0015
MF	GO:0042608	T cell receptor binding	3/103	11/12230	9.1e-05	0.002	0.0017
CC	GO:0012506	vesicle membrane	20/104	984/12371	0.00017	0.0021	0.0017
BP	GO:0033630	positive regulation of cell adhesion mediated by integrin	3/103	14/11893	0.00021	0.0021	0.0016
BP	GO:0033631	cell-cell adhesion mediated by integrin	3/103	14/11893	0.00021	0.0021	0.0016
BP	GO:0050790	regulation of catalytic activity	21/103	1051/11893	0.00022	0.0022	0.0016
BP	GO:0018212	peptidyl-tyrosine modification	8/103	188/11893	0.00022	0.0022	0.0016
BP	GO:0002687	positive regulation of leukocyte migration	6/103	100/11893	0.00023	0.0022	0.0016
CC	GO:0098636	protein complex involved in cell adhesion	4/104	35/12371	0.0002	0.0022	0.0018
CC	GO:0042613	MHC class II protein complex	3/104	14/12371	0.0002	0.0022	0.0018
BP	GO:0060326	cell chemotaxis	8/103	192/11893	0.00026	0.0025	0.0018

BP	GO:0043373	CD4-positive, alpha-beta T cell lineage commitment	3/103	15/11893	0.00027	0.0025	0.0019
BP	GO:0030595	leukocyte chemotaxis	7/103	147/11893	0.00028	0.0027	0.002
CC	GO:0005581	collagen trimer	4/104	37/12371	0.00025	0.0027	0.0022
BP	GO:0071310	cellular response to organic substance	25/103	1403/11893	0.0003	0.0028	0.0021
BP	GO:0008284	positive regulation of cell population proliferation	15/103	623/11893	0.00029	0.0028	0.0021
BP	GO:0045621	positive regulation of lymphocyte differentiation	6/103	106/11893	0.00031	0.0029	0.0022
BP	GO:0071622	regulation of granulocyte chemotaxis	4/103	38/11893	0.00031	0.0029	0.0022
BP	GO:0002685	regulation of leukocyte migration	7/103	150/11893	0.00032	0.003	0.0022
CC	GO:0030665	clathrin-coated vesicle membrane	6/104	108/12371	0.00029	0.003	0.0025
BP	GO:0002407	dendritic cell chemotaxis	3/103	16/11893	0.00033	0.003	0.0022
BP	GO:0010759	positive regulation of macrophage chemotaxis	3/103	16/11893	0.00033	0.003	0.0022
BP	GO:0046641	positive regulation of alpha-beta T cell proliferation	3/103	16/11893	0.00033	0.003	0.0022
BP	GO:0051250	negative regulation of lymphocyte activation	6/103	108/11893	0.00034	0.0031	0.0023
BP	GO:0043367	CD4-positive, alpha-beta T cell differentiation	5/103	70/11893	0.00034	0.0031	0.0023
BP	GO:2000106	regulation of leukocyte apoptotic process	5/103	70/11893	0.00034	0.0031	0.0023
BP	GO:0046638	positive regulation of alpha-beta T cell differentiation	4/103	39/11893	0.00035	0.0031	0.0023
CC	GO:0005576	extracellular region	36/104	2480/12371	0.00035	0.0034	0.0029
BP	GO:1905517	macrophage migration	4/103	40/11893	0.00038	0.0034	0.0026
BP	GO:0042742	defense response to bacterium	7/103	156/11893	0.0004	0.0035	0.0027
BP	GO:0045824	negative regulation of innate immune response	5/103	72/11893	0.00039	0.0035	0.0026
BP	GO:0002363	alpha-beta T cell lineage commitment	3/103	17/11893	0.00039	0.0035	0.0026

BP	GO:0002438	acute inflammatory response to antigenic stimulus	3/103	17/11893	0.00039	0.0035	0.0026
BP	GO:2001185	regulation of CD8-positive, alpha-beta T cell activation	3/103	17/11893	0.00039	0.0035	0.0026
BP	GO:0032720	negative regulation of tumor necrosis factor production	4/103	41/11893	0.00042	0.0037	0.0028
MF	GO:0044877	protein-containing complex binding	25/103	1399/12230	0.00019	0.0038	0.0032
MF	GO:0016004	phospholipase activator activity	3/103	14/12230	0.0002	0.0038	0.0032
BP	GO:0030316	osteoclast differentiation	5/103	74/11893	0.00044	0.0039	0.0029
BP	GO:0048519	negative regulation of biological process	50/103	3855/11893	0.00046	0.004	0.003
BP	GO:2000107	negative regulation of leukocyte apoptotic process	4/103	42/11893	0.00046	0.004	0.003
BP	GO:0001773	myeloid dendritic cell activation	3/103	18/11893	0.00047	0.004	0.003
BP	GO:0002396	MHC protein complex assembly	3/103	18/11893	0.00047	0.004	0.003
BP	GO:0002501	peptide antigen assembly with MHC protein complex	3/103	18/11893	0.00047	0.004	0.003
BP	GO:0043369	CD4-positive or CD8-positive, alpha-beta T cell lineage commitment	3/103	18/11893	0.00047	0.004	0.003
BP	GO:0008037	cell recognition	5/103	76/11893	0.0005	0.0042	0.0032
BP	GO:0045637	regulation of myeloid cell differentiation	7/103	162/11893	0.00051	0.0043	0.0032
BP	GO:1903556	negative regulation of tumor necrosis factor superfamily cytokine production	4/103	43/11893	0.00051	0.0043	0.0032
MF	GO:0060229	lipase activator activity	3/103	15/12230	0.00025	0.0045	0.0038
BP	GO:0002418	immune response to tumor cell	3/103	19/11893	0.00055	0.0046	0.0035
BP	GO:0050793	regulation of developmental process	28/103	1725/11893	0.00056	0.0047	0.0035
BP	GO:0071621	granulocyte chemotaxis	5/103	78/11893	0.00057	0.0047	0.0035
MF	GO:0004713	protein tyrosine kinase activity	6/103	109/12230	0.00031	0.0051	0.0043
MF	GO:0004715	non-membrane spanning protein tyrosine kinase activity	4/103	39/12230	0.00031	0.0051	0.0043
BP	GO:0001562	response to protozoan	3/103	20/11893	0.00065	0.0053	0.004

BP	GO:0002719	negative regulation of cytokine production involved in immune response	3/103	20/11893	0.00065	0.0053	0.004
BP	GO:0036037	CD8-positive, alpha-beta T cell activation	3/103	20/11893	0.00065	0.0053	0.004
CC	GO:0042611	MHC protein complex	3/104	20/12371	0.00059	0.0056	0.0046
BP	GO:0002688	regulation of leukocyte chemotaxis	5/103	82/11893	0.00071	0.0058	0.0043
BP	GO:0045670	regulation of osteoclast differentiation	4/103	47/11893	0.00071	0.0058	0.0043
BP	GO:0050868	negative regulation of T cell activation	5/103	83/11893	0.00075	0.006	0.0045
BP	GO:0006958	complement activation, classical pathway	3/103	21/11893	0.00075	0.006	0.0045
BP	GO:1905523	positive regulation of macrophage migration	3/103	21/11893	0.00075	0.006	0.0045
BP	GO:0032623	interleukin-2 production	4/103	48/11893	0.00077	0.0061	0.0046
BP	GO:0032663	regulation of interleukin-2 production	4/103	48/11893	0.00077	0.0061	0.0046
BP	GO:0050853	B cell receptor signaling pathway	4/103	48/11893	0.00077	0.0061	0.0046
BP	GO:0032101	regulation of response to external stimulus	16/103	761/11893	0.0008	0.0063	0.0047
BP	GO:0006935	chemotaxis	9/103	285/11893	0.00081	0.0063	0.0047
BP	GO:0042330	taxis	9/103	285/11893	0.00081	0.0063	0.0047
BP	GO:0031529	ruffle organization	4/103	49/11893	0.00083	0.0065	0.0049
BP	GO:0042130	negative regulation of T cell proliferation	4/103	49/11893	0.00083	0.0065	0.0049
BP	GO:0001911	negative regulation of leukocyte mediated cytotoxicity	3/103	22/11893	0.00086	0.0067	0.005
BP	GO:1900016	negative regulation of cytokine production involved in inflammatory response	3/103	22/11893	0.00086	0.0067	0.005
MF	GO:0005044	scavenger receptor activity	3/103	18/12230	0.00043	0.0067	0.0057
BP	GO:0002437	inflammatory response to antigenic stimulus	4/103	50/11893	0.0009	0.0069	0.0052
BP	GO:0001776	leukocyte homeostasis	5/103	87/11893	0.00093	0.0071	0.0053
BP	GO:0002228	natural killer cell mediated immunity	4/103	51/11893	0.00097	0.0074	0.0055
BP	GO:0045123	cellular extravasation	4/103	51/11893	0.00097	0.0074	0.0055
BP	GO:0002360	T cell lineage commitment	3/103	23/11893	0.00099	0.0074	0.0056
BP	GO:0031342	negative regulation of cell killing	3/103	23/11893	0.00099	0.0074	0.0056
BP	GO:0150146	cell junction disassembly	3/103	23/11893	0.00099	0.0074	0.0056

BP	GO:0019882	antigen processing and presentation	5/103	89/11893	0.001	0.0077	0.0058
CC	GO:0005768	endosome	17/104	867/12371	0.00086	0.0078	0.0065
BP	GO:0045595	regulation of cell differentiation	20/103	1102/11893	0.0011	0.0082	0.0061
BP	GO:0006816	calcium ion transport	8/103	240/11893	0.0011	0.0083	0.0062
BP	GO:0038094	Fc-gamma receptor signaling pathway	3/103	24/11893	0.0011	0.0083	0.0062
BP	GO:0070887	cellular response to chemical stimulus	29/103	1894/11893	0.0011	0.0084	0.0063
BP	GO:1903038	negative regulation of leukocyte cell-cell adhesion	5/103	91/11893	0.0011	0.0084	0.0063
BP	GO:0040011	locomotion	17/103	865/11893	0.0012	0.0085	0.0063
BP	GO:0031348	negative regulation of defense response	7/103	187/11893	0.0012	0.0086	0.0065
BP	GO:0031347	regulation of defense response	13/103	567/11893	0.0012	0.0087	0.0065
BP	GO:0065009	regulation of molecular function	24/103	1454/11893	0.0012	0.0089	0.0067
BP	GO:0071887	leukocyte apoptotic process	5/103	93/11893	0.0013	0.0091	0.0068
BP	GO:0019886	antigen processing and presentation of exogenous peptide antigen via MHC class II	3/103	25/11893	0.0013	0.0091	0.0068
BP	GO:2000514	regulation of CD4-positive, alpha-beta T cell activation	4/103	55/11893	0.0013	0.0093	0.0069
BP	GO:0006915	apoptotic process	24/103	1465/11893	0.0014	0.0098	0.0073
BP	GO:0042093	T-helper cell differentiation	4/103	56/11893	0.0014	0.0098	0.0074
BP	GO:0002287	alpha-beta T cell activation involved in immune response	4/103	57/11893	0.0015	0.01	0.0077
BP	GO:0002293	alpha-beta T cell differentiation involved in immune response	4/103	57/11893	0.0015	0.01	0.0077
BP	GO:0002294	CD4-positive, alpha-beta T cell differentiation involved in immune response	4/103	57/11893	0.0015	0.01	0.0077
BP	GO:0048002	antigen processing and presentation of peptide antigen	4/103	57/11893	0.0015	0.01	0.0077
BP	GO:0002335	mature B cell differentiation	3/103	26/11893	0.0014	0.01	0.0076
MF	GO:0042802	identical protein binding	25/103	1532/12230	0.00075	0.011	0.0093

BP	GO:0098657	import into cell	14/103	658/11893	0.0016	0.011	0.0081
BP	GO:0071222	cellular response to lipopolysaccharide	6/103	145/11893	0.0016	0.011	0.0082
BP	GO:0002701	negative regulation of production of molecular mediator of immune response	3/103	27/11893	0.0016	0.011	0.0081
BP	GO:0030224	monocyte differentiation	3/103	27/11893	0.0016	0.011	0.0081
BP	GO:0032689	negative regulation of type II interferon production	3/103	27/11893	0.0016	0.011	0.0081
BP	GO:0050798	activated T cell proliferation	3/103	27/11893	0.0016	0.011	0.0081
BP	GO:0050850	positive regulation of calcium-mediated signaling	3/103	27/11893	0.0016	0.011	0.0081
BP	GO:1900027	regulation of ruffle assembly	3/103	27/11893	0.0016	0.011	0.0081
BP	GO:2000406	positive regulation of T cell migration	3/103	27/11893	0.0016	0.011	0.0081
BP	GO:0048585	negative regulation of response to stimulus	21/103	1233/11893	0.0018	0.012	0.009
BP	GO:0097529	myeloid leukocyte migration	6/103	148/11893	0.0018	0.012	0.009
BP	GO:0002292	T cell differentiation involved in immune response	4/103	60/11893	0.0018	0.012	0.009
BP	GO:0032945	negative regulation of mononuclear cell proliferation	4/103	60/11893	0.0018	0.012	0.009
BP	GO:0050672	negative regulation of lymphocyte proliferation	4/103	60/11893	0.0018	0.012	0.009
BP	GO:0007259	receptor signaling pathway via JAK-STAT	5/103	102/11893	0.0019	0.013	0.0094
BP	GO:0050864	regulation of B cell activation	5/103	102/11893	0.0019	0.013	0.0094
BP	GO:0002260	lymphocyte homeostasis	4/103	61/11893	0.0019	0.013	0.0094
BP	GO:0002431	Fc receptor mediated stimulatory signaling pathway	3/103	29/11893	0.002	0.013	0.0096
BP	GO:0032743	positive regulation of interleukin-2 production	3/103	29/11893	0.002	0.013	0.0096
BP	GO:2000403	positive regulation of lymphocyte migration	3/103	29/11893	0.002	0.013	0.0096
BP	GO:2000516	positive regulation of CD4-positive, alpha-beta T cell activation	3/103	29/11893	0.002	0.013	0.0096

BP	GO:0012501	programmed cell death	24/103	1518/11893	0.0022	0.014	0.011
BP	GO:0071219	cellular response to molecule of bacterial origin	6/103	153/11893	0.0021	0.014	0.01
BP	GO:0050730	regulation of peptidyl-tyrosine phosphorylation	6/103	155/11893	0.0023	0.014	0.011
CC	GO:0098562	cytoplasmic side of membrane	6/104	150/12371	0.0016	0.014	0.012
BP	GO:0050731	positive regulation of peptidyl-tyrosine phosphorylation	5/103	105/11893	0.0022	0.014	0.01
BP	GO:0002526	acute inflammatory response	4/103	63/11893	0.0021	0.014	0.01
BP	GO:0002690	positive regulation of leukocyte chemotaxis	4/103	64/11893	0.0023	0.014	0.011
BP	GO:0001782	B cell homeostasis	3/103	30/11893	0.0022	0.014	0.01
BP	GO:0008219	cell death	24/103	1521/11893	0.0023	0.015	0.011
BP	GO:0051336	regulation of hydrolase activity	12/103	541/11893	0.0024	0.015	0.011
BP	GO:0097696	receptor signaling pathway via STAT	5/103	107/11893	0.0023	0.015	0.011
BP	GO:0032479	regulation of type I interferon production	5/103	108/11893	0.0024	0.015	0.011
BP	GO:0032606	type I interferon production	5/103	108/11893	0.0024	0.015	0.011
BP	GO:0002455	humoral immune response mediated by circulating immunoglobulin	3/103	31/11893	0.0024	0.015	0.011
BP	GO:0046640	regulation of alpha-beta T cell proliferation	3/103	31/11893	0.0024	0.015	0.011
BP	GO:0010720	positive regulation of cell development	9/103	336/11893	0.0025	0.016	0.012
BP	GO:0070664	negative regulation of leukocyte proliferation	4/103	66/11893	0.0025	0.016	0.012
BP	GO:0002715	regulation of natural killer cell mediated immunity	3/103	32/11893	0.0026	0.016	0.012
BP	GO:0007162	negative regulation of cell adhesion	7/103	217/11893	0.0028	0.017	0.013
BP	GO:1902106	negative regulation of leukocyte differentiation	4/103	68/11893	0.0028	0.017	0.013
BP	GO:0002478	antigen processing and presentation of exogenous peptide antigen	3/103	33/11893	0.0029	0.017	0.013
BP	GO:0042088	T-helper 1 type immune response	3/103	33/11893	0.0029	0.017	0.013

BP	GO:0042221	response to chemical	34/103	2493/11893	0.0029	0.018	0.013
BP	GO:0006959	humoral immune response	5/103	113/11893	0.003	0.018	0.014
BP	GO:0048523	negative regulation of cellular process	45/103	3636/11893	0.0032	0.019	0.014
CC	GO:0031012	extracellular matrix	8/104	275/12371	0.0022	0.019	0.015
CC	GO:0030312	external encapsulating structure	8/104	276/12371	0.0022	0.019	0.015
BP	GO:0032635	interleukin-6 production	5/103	114/11893	0.0031	0.019	0.014
BP	GO:0032675	regulation of interleukin-6 production	5/103	114/11893	0.0031	0.019	0.014
BP	GO:0002762	negative regulation of myeloid leukocyte differentiation	3/103	34/11893	0.0031	0.019	0.014
BP	GO:0042531	positive regulation of tyrosine phosphorylation of STAT protein	3/103	34/11893	0.0031	0.019	0.014
BP	GO:0046633	alpha-beta T cell proliferation	3/103	34/11893	0.0031	0.019	0.014
BP	GO:0002291	T cell activation via T cell receptor contact with antigen bound to MHC molecule on antigen presenting cell	2/103	10/11893	0.0032	0.019	0.014
BP	GO:0002645	positive regulation of tolerance induction	2/103	10/11893	0.0032	0.019	0.014
BP	GO:0035723	interleukin-15-mediated signaling pathway	2/103	10/11893	0.0032	0.019	0.014
BP	GO:0036005	response to macrophage colony-stimulating factor	2/103	10/11893	0.0032	0.019	0.014
BP	GO:0043301	negative regulation of leukocyte degranulation	2/103	10/11893	0.0032	0.019	0.014
BP	GO:0043374	CD8-positive, alpha-beta T cell differentiation	2/103	10/11893	0.0032	0.019	0.014
BP	GO:0071350	cellular response to interleukin-15	2/103	10/11893	0.0032	0.019	0.014
BP	GO:2001198	regulation of dendritic cell differentiation	2/103	10/11893	0.0032	0.019	0.014
CC	GO:0062023	collagen-containing extracellular matrix	7/104	220/12371	0.0025	0.02	0.017
BP	GO:1903707	negative regulation of hemopoiesis	4/103	72/11893	0.0035	0.02	0.015
BP	GO:0035556	intracellular signal transduction	29/103	2045/11893	0.0037	0.021	0.016
BP	GO:0043067	regulation of programmed cell death	19/103	1134/11893	0.0037	0.021	0.016
BP	GO:0032496	response to lipopolysaccharide	7/103	229/11893	0.0037	0.021	0.016
BP	GO:0008360	regulation of cell shape	5/103	118/11893	0.0036	0.021	0.016

MF	GO:0042605	peptide antigen binding	3/103	27/12230	0.0015	0.021	0.017
BP	GO:0010562	positive regulation of phosphorus metabolic process	11/103	498/11893	0.0038	0.022	0.016
BP	GO:0045937	positive regulation of phosphate metabolic process	11/103	498/11893	0.0038	0.022	0.016
BP	GO:0097305	response to alcohol	6/103	173/11893	0.0039	0.022	0.016
CC	GO:0030136	clathrin-coated vesicle	6/104	169/12371	0.003	0.022	0.018
CC	GO:0030662	coated vesicle membrane	6/104	169/12371	0.003	0.022	0.018
CC	GO:0009898	cytoplasmic side of plasma membrane	5/104	117/12371	0.003	0.022	0.018
MF	GO:0035591	signaling adaptor activity	5/103	101/12230	0.0016	0.022	0.018
BP	GO:0019884	antigen processing and presentation of exogenous antigen	3/103	37/11893	0.004	0.022	0.017
BP	GO:0033628	regulation of cell adhesion mediated by integrin	3/103	37/11893	0.004	0.022	0.017
BP	GO:0043029	T cell homeostasis	3/103	37/11893	0.004	0.022	0.017
BP	GO:2000404	regulation of T cell migration	3/103	37/11893	0.004	0.022	0.017
BP	GO:0046643	regulation of gamma-delta T cell activation	2/103	11/11893	0.0039	0.022	0.016
BP	GO:0061517	macrophage proliferation	2/103	11/11893	0.0039	0.022	0.016
BP	GO:0070672	response to interleukin-15	2/103	11/11893	0.0039	0.022	0.016
BP	GO:0006468	protein phosphorylation	17/103	974/11893	0.004	0.023	0.017
BP	GO:2000145	regulation of cell motility	14/103	736/11893	0.0043	0.024	0.018
BP	GO:0071216	cellular response to biotic stimulus	6/103	177/11893	0.0043	0.024	0.018
CC	GO:0030027	lamellipodium	6/104	174/12371	0.0035	0.024	0.02
BP	GO:1990266	neutrophil migration	4/103	76/11893	0.0042	0.024	0.018
BP	GO:0032480	negative regulation of type I interferon production	3/103	38/11893	0.0043	0.024	0.018
CC	GO:0031209	SCAR complex	2/104	11/12371	0.0037	0.024	0.02
CC	GO:0032156	septin cytoskeleton	2/104	11/12371	0.0037	0.024	0.02
CC	GO:0044194	cytolytic granule	2/104	11/12371	0.0037	0.024	0.02
BP	GO:0032103	positive regulation of response to external stimulus	10/103	438/11893	0.0046	0.025	0.019

MF	GO:0005178	integrin binding	5/103	105/12230	0.0019	0.025	0.021
BP	GO:0097178	ruffle assembly	3/103	39/11893	0.0046	0.025	0.019
BP	GO:0002357	defense response to tumor cell	2/103	12/11893	0.0046	0.025	0.019
BP	GO:0071888	macrophage apoptotic process	2/103	12/11893	0.0046	0.025	0.019
BP	GO:0010033	response to organic substance	27/103	1889/11893	0.0048	0.026	0.019
BP	GO:0042325	regulation of phosphorylation	14/103	744/11893	0.0048	0.026	0.019
BP	GO:0032760	positive regulation of tumor necrosis factor production	4/103	79/11893	0.0048	0.026	0.02
BP	GO:0042327	positive regulation of phosphorylation	10/103	444/11893	0.0051	0.027	0.02
BP	GO:0002237	response to molecule of bacterial origin	7/103	242/11893	0.005	0.027	0.02
BP	GO:0043370	regulation of CD4-positive, alpha-beta T cell differentiation	3/103	40/11893	0.0049	0.027	0.02
BP	GO:0070098	chemokine-mediated signaling pathway	3/103	40/11893	0.0049	0.027	0.02
BP	GO:0071346	cellular response to type II interferon	4/103	81/11893	0.0053	0.028	0.021
BP	GO:0030838	positive regulation of actin filament polymerization	3/103	41/11893	0.0053	0.028	0.021
BP	GO:0032613	interleukin-10 production	3/103	41/11893	0.0053	0.028	0.021
BP	GO:0032653	regulation of interleukin-10 production	3/103	41/11893	0.0053	0.028	0.021
BP	GO:0040012	regulation of locomotion	14/103	755/11893	0.0054	0.029	0.021
BP	GO:0050808	synapse organization	8/103	310/11893	0.0054	0.029	0.021
BP	GO:0006898	receptor-mediated endocytosis	6/103	186/11893	0.0055	0.029	0.022
BP	GO:1903557	positive regulation of tumor necrosis factor superfamily cytokine production	4/103	82/11893	0.0055	0.029	0.022
BP	GO:0001771	immunological synapse formation	2/103	13/11893	0.0054	0.029	0.021
BP	GO:0032693	negative regulation of interleukin-10 production	2/103	13/11893	0.0054	0.029	0.021
BP	GO:0043011	myeloid dendritic cell differentiation	2/103	13/11893	0.0054	0.029	0.021
BP	GO:0050665	hydrogen peroxide biosynthetic process	2/103	13/11893	0.0054	0.029	0.021
BP	GO:1902531	regulation of intracellular signal transduction	21/103	1366/11893	0.0061	0.031	0.023
BP	GO:0042981	regulation of apoptotic process	18/103	1098/11893	0.0059	0.031	0.023

BP	GO:0022408	negative regulation of cell-cell adhesion	5/103	134/11893	0.0061	0.031	0.023
BP	GO:0031343	positive regulation of cell killing	3/103	43/11893	0.0061	0.031	0.023
BP	GO:0034113	heterotypic cell-cell adhesion	3/103	43/11893	0.0061	0.031	0.023
BP	GO:0042509	regulation of tyrosine phosphorylation of STAT protein	3/103	43/11893	0.0061	0.031	0.023
BP	GO:0016310	phosphorylation	19/103	1191/11893	0.0062	0.032	0.024
BP	GO:0032688	negative regulation of interferon-beta production	2/103	14/11893	0.0063	0.032	0.024
BP	GO:0033003	regulation of mast cell activation	2/103	14/11893	0.0063	0.032	0.024
BP	GO:0050862	positive regulation of T cell receptor signaling pathway	2/103	14/11893	0.0063	0.032	0.024
BP	GO:0038093	Fc receptor signaling pathway	3/103	44/11893	0.0065	0.033	0.025
BP	GO:0030334	regulation of cell migration	13/103	693/11893	0.0067	0.034	0.025
BP	GO:0019220	regulation of phosphate metabolic process	15/103	865/11893	0.0074	0.036	0.027
BP	GO:0051924	regulation of calcium ion transport	5/103	140/11893	0.0073	0.036	0.027
BP	GO:0007229	integrin-mediated signaling pathway	4/103	88/11893	0.0071	0.036	0.027
BP	GO:0007260	tyrosine phosphorylation of STAT protein	3/103	46/11893	0.0073	0.036	0.027
BP	GO:0070228	regulation of lymphocyte apoptotic process	3/103	46/11893	0.0073	0.036	0.027
BP	GO:0002643	regulation of tolerance induction	2/103	15/11893	0.0072	0.036	0.027
BP	GO:0002837	regulation of immune response to tumor cell	2/103	15/11893	0.0072	0.036	0.027
BP	GO:0033033	negative regulation of myeloid cell apoptotic process	2/103	15/11893	0.0072	0.036	0.027
BP	GO:0050849	negative regulation of calcium-mediated signaling	2/103	15/11893	0.0072	0.036	0.027
BP	GO:0051174	regulation of phosphorus metabolic process	15/103	866/11893	0.0074	0.037	0.027
BP	GO:0050920	regulation of chemotaxis	5/103	141/11893	0.0075	0.037	0.028
BP	GO:2000401	regulation of lymphocyte migration	3/103	47/11893	0.0078	0.038	0.029
BP	GO:0048522	positive regulation of cellular process	48/103	4122/11893	0.0079	0.039	0.029

BP	GO:0002718	regulation of cytokine production involved in immune response	4/103	91/11893	0.008	0.039	0.029
BP	GO:0002367	cytokine production involved in immune response	4/103	92/11893	0.0083	0.04	0.03
BP	GO:0050921	positive regulation of chemotaxis	4/103	92/11893	0.0083	0.04	0.03
BP	GO:1990868	response to chemokine	3/103	48/11893	0.0082	0.04	0.03
BP	GO:1990869	cellular response to chemokine	3/103	48/11893	0.0082	0.04	0.03
BP	GO:0002834	regulation of response to tumor cell	2/103	16/11893	0.0082	0.04	0.03
BP	GO:0045655	regulation of monocyte differentiation	2/103	16/11893	0.0082	0.04	0.03
MF	GO:0019903	protein phosphatase binding	5/103	119/12230	0.0033	0.041	0.034
BP	GO:0042267	natural killer cell mediated cytotoxicity	3/103	49/11893	0.0087	0.042	0.031
MF	GO:0016493	C-C chemokine receptor activity	2/103	11/12230	0.0037	0.042	0.036
MF	GO:0019957	C-C chemokine binding	2/103	11/12230	0.0037	0.042	0.036
BP	GO:0001934	positive regulation of protein phosphorylation	9/103	409/11893	0.009	0.043	0.032
CC	GO:0005770	late endosome	7/104	263/12371	0.0067	0.043	0.036
BP	GO:0071677	positive regulation of mononuclear cell migration	3/103	50/11893	0.0092	0.044	0.033
BP	GO:0002716	negative regulation of natural killer cell mediated immunity	2/103	17/11893	0.0093	0.044	0.033
BP	GO:0032753	positive regulation of interleukin-4 production	2/103	17/11893	0.0093	0.044	0.033
BP	GO:0042832	defense response to protozoan	2/103	17/11893	0.0093	0.044	0.033
BP	GO:0045063	T-helper 1 cell differentiation	2/103	17/11893	0.0093	0.044	0.033
BP	GO:0045591	positive regulation of regulatory T cell differentiation	2/103	17/11893	0.0093	0.044	0.033
BP	GO:0031401	positive regulation of protein modification process	11/103	568/11893	0.01	0.047	0.035
BP	GO:0034341	response to type II interferon	4/103	98/11893	0.01	0.048	0.036
BP	GO:0050688	regulation of defense response to virus	3/103	52/11893	0.01	0.048	0.036

BP	GO:0010743	regulation of macrophage derived foam cell differentiation	2/103	18/11893	0.01	0.048	0.036
BP	GO:0045649	regulation of macrophage differentiation	2/103	18/11893	0.01	0.048	0.036
BP	GO:0070233	negative regulation of T cell apoptotic process	2/103	18/11893	0.01	0.048	0.036
CC	GO:0045121	membrane raft	6/104	207/12371	0.008	0.049	0.041
CC	GO:0098857	membrane microdomain	6/104	208/12371	0.0082	0.049	0.041
CC	GO:0032588	trans-Golgi network membrane	4/104	93/12371	0.0078	0.049	0.04
MF	GO:0030169	low-density lipoprotein particle binding	2/103	12/12230	0.0044	0.049	0.041
BP	GO:0043087	regulation of GTPase activity	6/103	214/11893	0.011	0.05	0.037
BP	GO:0031294	lymphocyte costimulation	6/103	36/11893	5.7e-07	1.1e-05	8.2e-06
BP	GO:0032946	positive regulation of mononuclear cell proliferation	11/103	94/11893	4.5e-10	1.1e-08	8.6e-09
BP	GO:0032729	positive regulation of type II interferon production	9/103	51/11893	4.3e-10	1.1e-08	8.3e-09
BP	GO:0044419	biological process involved in interspecies interaction between organisms	36/103	1163/11893	3e-12	1.1e-10	8e-11
BP	GO:0140546	defense response to symbiont	29/103	747/11893	3.2e-12	1.1e-10	8.6e-11
BP	GO:0098609	cell-cell adhesion	27/103	638/11893	2.9e-12	1.1e-10	8e-11
BP	GO:0002682	regulation of immune system process	50/103	1099/11893	4.2e-25	1.1e-22	8.1e-23
BP	GO:0002819	regulation of adaptive immune response	10/103	150/11893	6.5e-07	1.2e-05	9.1e-06
BP	GO:0045061	thymic T cell selection	5/103	20/11893	6.2e-07	1.2e-05	8.7e-06
BP	GO:0046629	gamma-delta T cell activation	5/103	20/11893	6.2e-07	1.2e-05	8.7e-06
BP	GO:0050671	positive regulation of lymphocyte proliferation	10/103	91/11893	5.5e-09	1.2e-07	9.3e-08
BP	GO:0007155	cell adhesion	35/103	1011/11893	2.6e-13	1.2e-11	8.9e-12
BP	GO:0051251	positive regulation of lymphocyte activation	19/103	247/11893	2.8e-13	1.2e-11	9.3e-12
BP	GO:0050870	positive regulation of T cell activation	19/103	187/11893	1.7e-15	1.2e-13	9.2e-14

BP	GO:0050776	regulation of immune response	38/103	665/11893	6.700000000000001e-22	1.2e-19	8.6e-20
BP	GO:0002823	negative regulation of adaptive immune response based on somatic recombination of immune receptors built from immunoglobulin superfamily domains	6/103	37/11893	6.8e-07	1.3e-05	9.4e-06
CC	GO:0005886	plasma membrane	64/104	3249/12371	3.4e-14	1.3e-12	1.1e-12
CC	GO:0009897	external side of plasma membrane	22/104	203/12371	1e-18	1.3e-16	1.1e-16
BP	GO:0042110	T cell activation	36/103	410/11893	4.4e-27	1.3 e-24	9.6e-25
BP	GO:0002710	negative regulation of T cell mediated immunity	6/103	18/11893	6.2e-09	1.4e-07	1e-07
BP	GO:0007154	cell communication	66/103	4076/11893	5.7e-10	1.4e-08	1.1e-08
MF	GO:0140375	immune receptor activity	11/103	89/12230	1.9e-10	1.4e-08	1.2e-08
BP	GO:0032609	type II interferon production	11/103	77/11893	5e-11	1.4e-09	1.1e-09
BP	GO:0032649	regulation of type II interferon production	11/103	77/11893	5e-11	1.4e-09	1.1e-09
BP	GO:0050670	regulation of lymphocyte proliferation	15/103	155/11893	4.1e-12	1.4e-10	1.1e-10
BP	GO:0045785	positive regulation of cell adhesion	22/103	360/11893	3.3e-13	1.4e-11	1.1e-11
BP	GO:0046631	alpha-beta T cell activation	15/103	131/11893	3.4e-13	1.4e-11	1.1e-11
BP	GO:0009607	response to biotic stimulus	37/103	1052/11893	2.5e-14	1.4e-12	1.1e-12
BP	GO:0002376	immune system process	69/103	1750/11893	1.4e-33	1.4e-30	1e-30
BP	GO:0050856	regulation of T cell receptor signaling pathway	6/103	38/11893	8e-07	1.5e-05	1.1e-05
CC	GO:0001772	immunological synapse	6/104	37/12371	5.7e-07	1.5e-05	1.3e-05
BP	GO:0050789	regulation of biological process	93/103	7818/11893	6.6e-09	1.5e-07	1.1e-07
BP	GO:0002704	negative regulation of leukocyte mediated immunity	8/103	48/11893	6.8e-09	1.5e-07	1.1e-07
BP	GO:0043207	response to external biotic stimulus	35/103	1023/11893	3.7e-13	1.5e-11	1.1e-11
BP	GO:0051707	response to other organism	35/103	1023/11893	3.7e-13	1.5e-11	1.1e-11
BP	GO:0032944	regulation of mononuclear cell proliferation	16/103	158/11893	3.6e-13	1.5e-11	1.1e-11

BP	GO:0002250	adaptive immune response	37/103	318/11893	2.1e-32	1.5e-29	1.1e-29
MF	GO:0038023	signaling receptor activity	25/103	528/12230	1e-12	1.6e-10	1.4e-10
MF	GO:0060089	molecular transducer activity	25/103	528/12230	1e-12	1.6e-10	1.4e-10
BP	GO:0006952	defense response	43/103	1170/11893	1.4e-17	1.6e-15	1.2e-15
BP	GO:0002252	immune effector process	28/103	425/11893	1.4e-17	1.6e-15	1.2e-15
BP	GO:0050778	positive regulation of immune response	32/103	543/11893	1.1e-18	1.6e-16	1.2e-16
BP	GO:0051716	cellular response to stimulus	73/103	4872/11893	6.8e-10	1.7e-08	1.2e-08
BP	GO:0045321	leukocyte activation	47/103	686/11893	3.3e-31	1.7000000000000003e-28	1.3000000000000002e-28
BP	GO:0030154	cell differentiation	47/103	2843/11893	1e-06	1.9e-05	1.4e-05
BP	GO:0048869	cellular developmental process	47/103	2844/11893	1.1e-06	1.9e-05	1.4e-05
BP	GO:0045058	T cell selection	9/103	42/11893	6.8e-11	1.9e-09	1.4e-09
BP	GO:0002768	immune response-regulating cell surface receptor signaling pathway	20/103	252/11893	3.4e-14	1.9e-12	1.4e-12
CC	GO:0042101	T cell receptor complex	9/104	14/12371	2.8e-16	1.9e-14	1.6e-14
BP	GO:0002706	regulation of lymphocyte mediated immunity	9/103	124/11893	1.2e-06	2.1e-05	1.6e-05
BP	GO:0051240	positive regulation of multicellular organismal process	30/103	1117/11893	1e-08	2.2e-07	1.7e-07
CC	GO:0030666	endocytic vesicle membrane	10/104	161/12371	9.6e-07	2.3e-05	1.9e-05
BP	GO:0046632	alpha-beta T cell differentiation	8/103	93/11893	1.3e-06	2.3e-05	1.7e-05
MF	GO:0004896	cytokine receptor activity	7/103	59/12230	5.8e-07	2.3e-05	1.9e-05
BP	GO:0002820	negative regulation of adaptive immune response	6/103	41/11893	1.3e-06	2.3e-05	1.7e-05
MF	GO:0015026	coreceptor activity	6/103	37/12230	5.8e-07	2.3e-05	1.9e-05
CC	GO:0043235	receptor complex	17/104	230/12371	6.9e-12	2.3e-10	1.9e-10
BP	GO:0050865	regulation of cell activation	28/103	432/11893	2.2e-17	2.3e-15	1.8e-15
BP	GO:0019221	cytokine-mediated signaling pathway	14/103	342/11893	1.4e-06	2.4e-05	1.8e-05
BP	GO:0048584	positive regulation of response to stimulus	39/103	1650/11893	1e-09	2.5e-08	1.9e-08
BP	GO:0002460	adaptive immune response based on somatic recombination of immune	20/103	224/11893	3.5e-15	2.5e-13	1.9e-13

		receptors built from immunoglobulin superfamily domains					
BP	GO:0002709	regulation of T cell mediated immunity	7/103	66/11893	1.5e-06	2.6e-05	1.9e-05
BP	GO:0033627	cell adhesion mediated by integrin	7/103	66/11893	1.5e-06	2.6e-05	1.9e-05
MF	GO:0019955	cytokine binding	9/103	87/12230	4.5e-08	2.8e-06	2.3e-06
BP	GO:0045619	regulation of lymphocyte differentiation	12/103	161/11893	1.3e-08	2.8e-07	2.1e-07
CC	GO:0009986	cell surface	28/104	503/12371	5.5e-16	2.9e-14	2.4e-14
BP	GO:0050851	antigen receptor-mediated signaling pathway	18/103	147/11893	3.5e-16	2.9e-14	2.2e-14
BP	GO:0071674	mononuclear cell migration	11/103	130/11893	1.5e-08	3.1e-07	2.3e-07
BP	GO:0002757	immune response-activating signaling pathway	22/103	376/11893	7.9e-13	3.1e-11	2.3e-11
BP	GO:0007165	signal transduction	64/103	3736/11893	1.2e-10	3.2e-09	2.4e-09
BP	GO:0070665	positive regulation of leukocyte proliferation	12/103	107/11893	1.2e-10	3.2e-09	2.4e-09
BP	GO:0032501	multicellular organismal process	76/103	4656/11893	8.3e-13	3.2e-11	2.4e-11
BP	GO:0001775	cell activation	47/103	778/11893	9.4e-29	3.2e-26	2.4e-26
BP	GO:0007166	cell surface receptor signaling pathway	45/103	1858/11893	9.7e-12	3.3e-10	2.5e-10
BP	GO:0046649	lymphocyte activation	44/103	582/11893	8.3e-31	3.4e-28	2.5e-28
BP	GO:0002698	negative regulation of immune effector process	10/103	80/11893	1.5e-09	3.7e-08	2.7e-08
BP	GO:0048583	regulation of response to stimulus	49/103	2877/11893	1.9e-07	3.8e-06	2.9e-06
BP	GO:0033077	T cell differentiation in thymus	7/103	70/11893	2.3e-06	3.9e-05	2.9e-05
BP	GO:0065007	biological regulation	94/103	8092/11893	1.9e-08	3.9e-07	2.9e-07
BP	GO:0001817	regulation of cytokine production	25/103	573/11893	1.2e-11	3.9e-10	2.9e-10
BP	GO:0046651	lymphocyte proliferation	17/103	199/11893	1e-12	3.9e-11	2.9e-11
BP	GO:1903706	regulation of hemopoiesis	17/103	318/11893	1.7e-09	4.1e-08	3e-08
BP	GO:0051249	regulation of lymphocyte activation	26/103	366/11893	4e-17	4.1e-15	3.1e-15
BP	GO:0002696	positive regulation of leukocyte activation	21/103	263/11893	6.3e-15	4.2e-13	3.1e-13
CC	GO:0071944	cell periphery	72/104	3507/12371	4.8e-18	4.2e-16	3.5e-16

BP	GO:0002684	positive regulation of immune system process	42/103	777/11893	1.8e-23	4.2e-21	3.1e-21
BP	GO:0048518	positive regulation of biological process	62/103	4478/11893	2.7e-06	4.5e-05	3.3e-05
BP	GO:0002822	regulation of adaptive immune response based on somatic recombination of immune receptors built from immunoglobulin superfamily domains	9/103	137/11893	2.8e-06	4.5e-05	3.4e-05
BP	GO:0070661	leukocyte proliferation	19/103	231/11893	8.3e-14	4.5e-12	3.3e-12
BP	GO:1903039	positive regulation of leukocyte cell-cell adhesion	20/103	204/11893	5.7e-16	4.5e-14	3.3e-14
BP	GO:0098542	defense response to other organism	32/103	805/11893	9e-14	4.7e-12	3.5e-12
BP	GO:0030217	T cell differentiation	22/103	236/11893	4.9e-17	4.7e-15	3.5e-15
BP	GO:0006954	inflammatory response	23/103	545/11893	1.7e-10	4.8e-09	3.6e-09
BP	GO:0009617	response to bacterium	15/103	419/11893	3e-06	4.9e-05	3.6e-05
BP	GO:0042102	positive regulation of T cell proliferation	10/103	65/11893	1.8e-10	4.9e-09	3.7e-09
BP	GO:0006950	response to stress	46/103	2853/11893	3.1e-06	5.1e-05	3.8e-05
BP	GO:1903037	regulation of leukocyte cell-cell adhesion	21/103	266/11893	7.9e-15	5.1e-13	3.8e-13
BP	GO:0002694	regulation of leukocyte activation	28/103	404/11893	3.7e-18	5.1e-16	3.8e-16
BP	GO:0030097	hemopoiesis	34/103	716/11893	5.6e-17	5.2e-15	3.9e-15
BP	GO:0001816	cytokine production	25/103	582/11893	1.6e-11	5.3e-10	4e-10
BP	GO:0002683	negative regulation of immune system process	20/103	351/11893	1.6e-11	5.3e-10	4e-10
BP	GO:0002449	lymphocyte mediated immunity	19/103	204/11893	8.5e-15	5.3e-13	3.9e-13
BP	GO:1903131	mononuclear cell differentiation	31/103	351/11893	2.9e-23	5.9e-21	4.4e-21
BP	GO:0022407	regulation of cell-cell adhesion	21/103	349/11893	1.6e-12	6.1e-11	4.6e-11
BP	GO:0050863	regulation of T cell activation	23/103	271/11893	7e-17	6.2e-15	4.6e-15
BP	GO:0002429	immune response-activating cell surface receptor signaling pathway	19/103	236/11893	1.2e-13	6.3e-12	4.7e-12
BP	GO:0022409	positive regulation of cell-cell adhesion	20/103	237/11893	1e-14	6.3e-13	4.7e-13
BP	GO:0071345	cellular response to cytokine stimulus	18/103	606/11893	3.9e-06	6.4e-05	4.8e-05
BP	GO:0050794	regulation of cellular process	88/103	7474/11893	3.2e-07	6.4e-06	4.8e-06

BP	GO:0002443	leukocyte mediated immunity	20/103	271/11893	1.3e-13	6.5e-12	4.9e-12
BP	GO:0032943	mononuclear cell proliferation	18/103	205/11893	1.3e-13	6.5e-12	4.8e-12
BP	GO:0070663	regulation of leukocyte proliferation	17/103	176/11893	1.4e-13	6.5e-12	4.9e-12
BP	GO:0002253	activation of immune response	26/103	415/11893	8.5e-16	6.5e-14	4.8e-14
BP	GO:0060284	regulation of cell development	18/103	608/11893	4.1e-06	6.6e-05	4.9e-05
BP	GO:0046635	positive regulation of alpha-beta T cell activation	6/103	50/11893	4.3e-06	6.7e-05	5e-05
BP	GO:0072678	T cell migration	6/103	50/11893	4.3e-06	6.7e-05	5e-05
BP	GO:0050867	positive regulation of cell activation	21/103	271/11893	1.1e-14	6.7e-13	5e-13
BP	GO:1902105	regulation of leukocyte differentiation	17/103	240/11893	2.1e-11	6.8e-10	5.1e-10
BP	GO:0048468	cell development	43/103	1885/11893	2.7e-10	7.1e-09	5.3e-09
BP	GO:0050852	T cell receptor signaling pathway	17/103	114/11893	8.3e-17	7.1e-15	5.3e-15
BP	GO:0002521	leukocyte differentiation	33/103	453/11893	3.8e-22	7.1e-20	5.3e-20
BP	GO:0006955	immune response	60/103	1174/11893	3.500000 0000000 003e-34	7.1e-31	5.3e-31
CC	GO:0098552	side of membrane	29/104	413/12371	2.7e-19	7.2e-17	6e-17
BP	GO:0043368	positive T cell selection	8/103	33/11893	2.8e-10	7.3e-09	5.5e-09
BP	GO:0030155	regulation of cell adhesion	25/103	592/11893	2.4e-11	7.3e-10	5.5e-10
BP	GO:0042129	regulation of T cell proliferation	13/103	118/11893	2.3e-11	7.3e-10	5.5e-10
BP	GO:0050854	regulation of antigen receptor-mediated signaling pathway	6/103	51/11893	4.8e-06	7.5e-05	5.6e-05
BP	GO:0009605	response to external stimulus	40/103	1794/11893	3.2e-09	7.5e-08	5.6e-08
BP	GO:0023052	signaling	66/103	4019/11893	2.9e-10	7.5e-09	5.6e-09
BP	GO:0050896	response to stimulus	91/103	5720/11893	5.9e-18	7.5e-16	5.6e-16
BP	GO:0002285	lymphocyte activation involved in immune response	9/103	147/11893	4.9e-06	7.6e-05	5.7e-05
BP	GO:0045059	positive thymic T cell selection	4/103	14/11893	5e-06	7.7e-05	5.7e-05
BP	GO:0002697	regulation of immune effector process	13/103	262/11893	3.9e-07	7.7e-06	5.8e-06
CC	GO:0016020	membrane	75/104	6197/12371	3.7e-06	7.9e-05	6.5e-05
CC	GO:0030139	endocytic vesicle	12/104	280/12371	3.9e-06	7.9e-05	6.5e-05

BP	GO:0006968	cellular defense response	5/103	30/11893	5.3e-06	8.1e-05	6.1e-05
BP	GO:0051239	regulation of multicellular organismal process	46/103	1992/11893	2.7e-11	8.1e-10	6.1e-10
BP	GO:0002456	T cell mediated immunity	10/103	87/11893	3.5e-09	8.2e-08	6.1e-08
CC	GO:0098797	plasma membrane protein complex	19/104	330/12371	2.9e-11	8.4e-10	7e-10
BP	GO:0042098	T cell proliferation	14/103	147/11893	2.8e-11	8.4e-10	6.3e-10
BP	GO:0002286	T cell activation involved in immune response	7/103	80/11893	5.6e-06	8.5e-05	6.4e-05
BP	GO:0050777	negative regulation of immune response	11/103	144/11893	4.3e-08	8.9e-07	6.7e-07
BP	GO:0030098	lymphocyte differentiation	27/103	313/11893	5.8e-20	9.1e-18	6.8e-18
BP	GO:0031295	T cell costimulation	6/103	35/11893	4.8e-07	9.4e-06	7e-06
BP	GO:0097028	dendritic cell differentiation	5/103	31/11893	6.3e-06	9.5e-05	7.1e-05
BP	GO:0050900	leukocyte migration	15/103	253/11893	4.2e-09	9.7e-08	7.2e-08
BP	GO:0046634	regulation of alpha-beta T cell activation	8/103	82/11893	5.1e-07	9.8e-06	7.3e-06

Supplementary Table 8. *Gene ontology pathways enriched in "black" gene module.*

ONT	ID	Description	GeneRatio	BgRatio	pvalue	p.adjust	qvalue
BP	GO:0002181	cytoplasmic translation	65/88	151/11893	4e-110	4.8e-107	4.6e-107
BP	GO:0006412	translation	70/88	593/11893	1.3e-75	8e-73	7.6e-73
BP	GO:0043043	peptide biosynthetic process	70/88	610/11893	1.1e-74	4.2e-72	4.1e-72
BP	GO:0043604	amide biosynthetic process	70/88	710/11893	6.8e-70	2e-67	1.9e-67
BP	GO:0006518	peptide metabolic process	70/88	732/11893	6.2e-69	1.5e-66	1.4e-66
BP	GO:0043603	amide metabolic process	70/88	949/11893	7.6e-61	1.5e-58	1.5e-58
BP	GO:1901566	organonitrogen compound biosynthetic process	73/88	1426/11893	5.7 e-53	9.8e-51	9.4e-51
BP	GO:0042254	ribosome biogenesis	32/88	305/11893	6.6e-29	9.9e-27	9.5e-27
BP	GO:0044271	cellular nitrogen compound biosynthetic process	76/88	3650/11893	1.7e-27	2.3e-25	2.2 e-25
BP	GO:0022613	ribonucleoprotein complex biogenesis	35/88	449/11893	2.8e-27	3.4e-25	3.2 e-25

BP	GO:0019538	protein metabolic process	75/88	3759/11893	1.8e-25	2e-23	1.9e-23
BP	GO:1901564	organonitrogen compound metabolic process	78/88	4540/11893	6.7e-23	6.7e-21	6.4e-21
BP	GO:0034641	cellular nitrogen compound metabolic process	79/88	4870/11893	9.1e-22	8.4e-20	8.1e-20
BP	GO:0010467	gene expression	76/88	4683/11893	7.2e-20	6.2e-18	5.9e-18
BP	GO:0042274	ribosomal small subunit biogenesis	17/88	100/11893	4.8e-19	3.8e-17	3.7e-17
BP	GO:0044249	cellular biosynthetic process	80/88	5751/11893	1.6e-17	1.2e-15	1.2e-15
BP	GO:0009059	macromolecule biosynthetic process	76/88	5078/11893	1.8e-17	1.3e-15	1.2e-15
BP	GO:1901576	organic substance biosynthetic process	80/88	5804/11893	3.2e-17	2.1e-15	2e-15
BP	GO:0006364	rRNA processing	20/88	213/11893	5e-17	3.1e-15	3e-15
BP	GO:0009058	biosynthetic process	80/88	5846/11893	5.3e-17	3.2e-15	3.1e-15
BP	GO:0042273	ribosomal large subunit biogenesis	14/88	70/11893	7e-17	4e-15	3.8e-15
BP	GO:0016072	rRNA metabolic process	20/88	253/11893	1.4e-15	7.8e-14	7.5e-14
BP	GO:0042255	ribosome assembly	12/88	55/11893	4.2e-15	2.2e-13	2.1e-13
BP	GO:0044237	cellular metabolic process	83/88	7477/11893	5e-12	2.5e-10	2.4e-10
BP	GO:0034470	ncRNA processing	20/88	394/11893	6.1e-12	2.9e-10	2.8e-10
BP	GO:0006396	RNA processing	28/88	881/11893	1.5e-11	6.8e-10	6.5e-10
BP	GO:0006807	nitrogen compound metabolic process	81/88	7244/11893	3.2e-11	1.4e-09	1.4e-09
BP	GO:0022618	protein-RNA complex assembly	14/88	192/11893	1.1e-10	4.9e-09	4.7e-09
BP	GO:0043170	macromolecule metabolic process	78/88	6861/11893	2e-10	8.4e-09	8.1e-09
BP	GO:0071826	protein-RNA complex organization	14/88	202/11893	2.3e-10	9e-09	8.6e-09
BP	GO:0044238	primary metabolic process	81/88	7579/11893	7.8e-10	3e-08	2.9e-08
BP	GO:0008152	metabolic process	84/88	8393/11893	3.4e-09	1.3e-07	1.2e-07
BP	GO:0034660	ncRNA metabolic process	20/88	569/11893	4.1e-09	1.5e-07	1.4e-07
BP	GO:0000027	ribosomal large subunit assembly	6/88	21/11893	6.9e-09	2.4e-07	2.3e-07
BP	GO:0071704	organic substance metabolic process	81/88	8001/11893	3.2e-08	1.1e-06	1e-06
BP	GO:1901798	positive regulation of signal transduction by p53 class mediator	5/88	25/11893	9.3e-07	3.1e-05	3e-05
BP	GO:0140694	non-membrane-bounded organelle assembly	13/88	339/11893	1.1e-06	3.6e-05	3.5e-05
BP	GO:0000470	maturation of LSU-rRNA	5/88	26/11893	1.2e-06	3.6e-05	3.5e-05
BP	GO:0051444	negative regulation of ubiquitin-protein transferase activity	4/88	14/11893	2.6e-06	8.1e-05	7.8e-05

BP	GO:0000028	ribosomal small subunit assembly	4/88	15/11893	3.6e-06	0.00011	0.0001
BP	GO:0051438	regulation of ubiquitin-protein transferase activity	5/88	33/11893	4e-06	0.00012	0.00011
BP	GO:1904666	regulation of ubiquitin protein ligase activity	4/88	16/11893	4.8e-06	0.00014	0.00013
BP	GO:1901796	regulation of signal transduction by p53 class mediator	7/88	95/11893	6.2e-06	0.00017	0.00017
BP	GO:0044085	cellular component biogenesis	38/88	2698/11893	1.5e-05	0.0004	0.00038
BP	GO:0072331	signal transduction by p53 class mediator	8/88	156/11893	1.9e-05	0.00051	0.00049
BP	GO:0006417	regulation of translation	11/88	327/11893	2.8e-05	0.00072	0.00069
BP	GO:0030490	maturation of SSU-rRNA	5/88	53/11893	4.3e-05	0.0011	0.001
BP	GO:2000765	regulation of cytoplasmic translation	4/88	28/11893	5e-05	0.0012	0.0012
BP	GO:2000767	positive regulation of cytoplasmic translation	3/88	12/11893	8.2e-05	0.002	0.0019
BP	GO:0034248	regulation of amide metabolic process	11/88	384/11893	0.00012	0.0028	0.0027
BP	GO:0045727	positive regulation of translation	6/88	114/11893	0.00019	0.0046	0.0044
BP	GO:2000059	negative regulation of ubiquitin-dependent protein catabolic process	4/88	47/11893	0.00039	0.009	0.0086
BP	GO:0034250	positive regulation of amide metabolic process	6/88	139/11893	0.00057	0.013	0.012
BP	GO:0010608	post-transcriptional regulation of gene expression	11/88	472/11893	0.00069	0.015	0.015
BP	GO:1903051	negative regulation of proteolysis involved in protein catabolic process	4/88	59/11893	0.00093	0.02	0.019
BP	GO:0006414	translational elongation	4/88	63/11893	0.0012	0.026	0.024
BP	GO:0006413	translational initiation	5/88	112/11893	0.0014	0.03	0.029
BP	GO:0031397	negative regulation of protein ubiquitination	4/88	68/11893	0.0016	0.033	0.031
BP	GO:0031396	regulation of protein ubiquitination	6/88	176/11893	0.0019	0.039	0.037
BP	GO:0000462	maturation of SSU-rRNA from tricistronic rRNA transcript (SSU-rRNA, 5.8S rRNA, LSU-rRNA)	3/88	35/11893	0.0022	0.043	0.041
BP	GO:0050821	protein stabilization	6/88	184/11893	0.0024	0.047	0.045

BP	GO:1903321	negative regulation of protein modification by small protein conjugation or removal	4/88	77/11893	0.0025	0.048	0.046
CC	GO:0022626	cytosolic ribosome	64/88	112/12371	4.9e-120	1e-117	7.7e-118
CC	GO:0044391	ribosomal subunit	64/88	178/12371	6.7e-103	7e-101	5.3e-101
CC	GO:0005840	ribosome	65/88	220/12371	3.6e-98	2.6 e-96	1.9e-96
CC	GO:0022625	cytosolic large ribosomal subunit	41/88	54/12371	1.3e-81	6.9e-80	5.2e-80
CC	GO:1990904	ribonucleoprotein complex	69/88	669/12371	4.2e-71	1.8 8e-69	1.3 8e-69
CC	GO:0015934	large ribosomal subunit	42/88	108/12371	6.5e-66	2.3e-64	1.7e-64
CC	GO:0022627	cytosolic small ribosomal subunit	23/88	41/12371	3.2e-40	9.6 e-39	7.2e-39
CC	GO:0015935	small ribosomal subunit	24/88	74/12371	1.2e-34	3.3e-33	2.4e-33
CC	GO:0005925	focal adhesion	32/88	377/12371	1.7e-26	4.1e-25	3.1e-25
CC	GO:0030055	cell-substrate junction	32/88	385/12371	3.4e-26	7.2e-25	5.4 e-25
CC	GO:0032991	protein-containing complex	81/88	4764/12371	4.1e-26	7.9e-25	5.9e-25
CC	GO:0030054	cell junction	51/88	1523/12371	1.6 e-24	2.9e-23	2.2e-23
CC	GO:0005829	cytosol	78/88	4547/12371	4.3e-24	7e-23	5.3e-23
CC	GO:0045202	synapse	40/88	922/12371	2e-22	3e-21	2.3e-21
CC	GO:0043232	intracellular non-membrane-bounded organelle	72/88	4139/12371	1e-20	1.3e-19	1e-19
CC	GO:0043228	non-membrane-bounded organelle	72/88	4140/12371	1e-20	1.3e-19	1e-19
CC	GO:0032040	small-subunit processome	15/88	71/12371	1.2e-18	1.4e-17	1.1e-17
CC	GO:0070161	anchoring junction	32/88	727/12371	9.1e-18	1.1e-16	8e-17
CC	GO:0030684	preribosome	16/88	105/12371	1.9e-17	2.2e-16	1.6e-16
CC	GO:0005730	nucleolus	34/88	910/12371	9.4e-17	9.9e-16	7.4e-16
CC	GO:0070062	extracellular exosome	41/88	1662/12371	4e-14	4e-13	3e-13
CC	GO:1903561	extracellular vesicle	41/88	1673/12371	5e-14	4.5e-13	3.3e-13
CC	GO:0043230	extracellular organelle	41/88	1674/12371	5.1e-14	4.5e-13	3.3e-13
CC	GO:0065010	extracellular membrane-bounded organelle	41/88	1674/12371	5.1e-14	4.5e-13	3.3e-13
CC	GO:0005615	extracellular space	42/88	2092/12371	1.9e-11	1.6e-10	1.2e-10
CC	GO:0014069	postsynaptic density	14/88	204/12371	1.5e-10	1.3e-09	9.4e-10
CC	GO:0032279	asymmetric synapse	14/88	213/12371	2.7e-10	2.1e-09	1.6e-09
CC	GO:0099572	postsynaptic specialization	14/88	216/12371	3.3e-10	2.5e-09	1.9e-09

CC	GO:0098984	neuron to neuron synapse	14/88	231/12371	8e-10	5.8e-09	4.4e-09
CC	GO:0005737	cytoplasm	86/88	9197/12371	2.1e-09	1.5e-08	1.1e-08
CC	GO:0005576	extracellular region	42/88	2480/12371	4.6e-09	3.1e-08	2.4e-08
CC	GO:0005791	rough endoplasmic reticulum	7/88	45/12371	2.6e-08	1.7e-07	1.3e-07
CC	GO:0031981	nuclear lumen	51/88	3877/12371	2.1e-07	1.3e-06	1e-06
CC	GO:0016020	membrane	67/88	6197/12371	4.6e-07	2.9e-06	2.1e-06
CC	GO:0098794	postsynapse	14/88	397/12371	7.2e-07	4.4e-06	3.3e-06
CC	GO:0031982	vesicle	42/88	3004/12371	1.4e-06	8.1e-06	6.1e-06
CC	GO:0005783	endoplasmic reticulum	27/88	1514/12371	3.5e-06	2e-05	1.5e-05
CC	GO:0005634	nucleus	64/88	6056/12371	4.6e-06	2.5e-05	1.9e-05
CC	GO:0031974	membrane-enclosed lumen	52/88	4714/12371	5e-05	0.00026	0.00019
CC	GO:0043233	organelle lumen	52/88	4714/12371	5e-05	0.00026	0.00019
CC	GO:0070013	intracellular organelle lumen	52/88	4714/12371	5e-05	0.00026	0.00019
CC	GO:0098554	cytoplasmic side of endoplasmic reticulum membrane	3/88	15/12371	0.00015	0.00075	0.00056
CC	GO:0030867	rough endoplasmic reticulum membrane	3/88	24/12371	0.00063	0.0031	0.0023
CC	GO:0036464	cytoplasmic ribonucleoprotein granule	6/88	208/12371	0.0036	0.017	0.013
CC	GO:0034709	methylosome	2/88	13/12371	0.0037	0.017	0.013
CC	GO:0098800	inner mitochondrial membrane protein complex	5/88	148/12371	0.004	0.019	0.014
CC	GO:0035770	ribonucleoprotein granule	6/88	225/12371	0.0053	0.024	0.018
CC	GO:0005654	nucleoplasm	37/88	3584/12371	0.0058	0.026	0.019
CC	GO:0005751	mitochondrial respiratory chain complex IV	2/88	19/12371	0.0079	0.034	0.025
CC	GO:0045277	respiratory chain complex IV	2/88	20/12371	0.0087	0.037	0.028
MF	GO:0003735	structural constituent of ribosome	61/87	150/12230	1.6e-101	3.4e-99	3.1e-99
MF	GO:0005198	structural molecule activity	61/87	523/12230	3.6e-64	3.9e-62	3.5e-62
MF	GO:0003723	RNA binding	70/87	1504/12230	2.8e-48	2e-46	1.8e-46
MF	GO:0003676	nucleic acid binding	73/87	3192/12230	1.3e-29	6.8e-28	6.2e-28
MF	GO:0097159	organic cyclic compound binding	74/87	4727/12230	4.3e-19	1.8e-17	1.7e-17
MF	GO:0019843	rRNA binding	14/87	64/12230	1e-17	3.6e-16	3.3e-16
MF	GO:0003729	mRNA binding	17/87	285/12230	1.3e-11	3.8e-10	3.5e-10

MF	GO:0055105	ubiquitin-protein transferase inhibitor activity	6/87	11/12230	4.9e-11	1.3e-09	1.2e-09
MF	GO:0048027	mRNA 5'-UTR binding	7/87	22/12230	1.1e-10	2.7e-09	2.4e-09
MF	GO:0055106	ubiquitin-protein transferase regulator activity	6/87	29/12230	4.5e-08	9.7e-07	8.8e-07
MF	GO:0045182	translation regulator activity	8/87	118/12230	1.8e-06	3.5e-05	3.2e-05
MF	GO:0045296	cadherin binding	11/87	306/12230	1e-05	0.00018	0.00017
MF	GO:0003730	mRNA 3'-UTR binding	6/87	87/12230	3.5e-05	0.00057	0.00052
MF	GO:0050839	cell adhesion molecule binding	11/87	446/12230	0.00031	0.0047	0.0043
MF	GO:0004857	enzyme inhibitor activity	8/87	256/12230	0.00046	0.0066	0.006
MF	GO:0097371	MDM2/MDM4 family protein binding	2/87	11/12230	0.0026	0.034	0.031
MF	GO:0140678	molecular function inhibitor activity	8/87	337/12230	0.0027	0.034	0.031
MF	GO:0031386	protein tag activity	2/87	12/12230	0.0032	0.035	0.032
MF	GO:0141047	molecular tag activity	2/87	12/12230	0.0032	0.035	0.032
MF	GO:0090079	translation regulator activity, nucleic acid binding	4/87	92/12230	0.0041	0.044	0.04
MF	GO:0003746	translation elongation factor activity	2/87	14/12230	0.0043	0.044	0.04

Supplementary Table 9. *Gene ontology pathways enriched in "purple" gene module.*

ONT	ID	Description	GeneRatio	BgRatio	pvalue	p.adjust	qvalue
CC	GO:0044666	MLL3/4 complex	2/20	12/12371	0.00016	0.018	0.014
MF	GO:0141052	histone H3 demethylase activity	4/20	25/12230	6.4e-08	3.5e-06	2.8e-06
MF	GO:0032452	histone demethylase activity	4/20	27/12230	8.9e-08	3.5e-06	2.8e-06
MF	GO:0140457	protein demethylase activity	4/20	27/12230	8.9e-08	3.5e-06	2.8e-06
MF	GO:0032451	demethylase activity	4/20	34/12230	2.3e-07	6.9e-06	5.6e-06
MF	GO:0016706	2-oxoglutarate-dependent dioxygenase activity	4/20	57/12230	1.9e-06	4.6e-05	3.7e-05
MF	GO:0051213	dioxygenase activity	4/20	79/12230	7.2e-06	0.00014	0.00012
MF	GO:0016705	oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen	4/20	126/12230	4.6e-05	0.00077	0.00063

MF	GO:0140993	histone modifying activity	4/20	186/12230	0.00021	0.0031	0.0025
MF	GO:0031490	chromatin DNA binding	3/20	112/12230	0.00076	0.01	0.0081
MF	GO:0003676	nucleic acid binding	12/20	3192/12230	0.0014	0.017	0.013

Supplementary Table 10. *Gene ontology pathways enriched in "greenyellow" gene module.*

	p-value	FDR
Sex, Male	0.55	0.92
Age, years	0.028	0.33
Ex-smoker	0.082	0.51
Non-smoker	0.04	0.33
PACKNO	0.13	0.55
BMI, kg/m2	0.1	0.52
Phadiatop test, Positive,	0.8	0.95
Disease duration, years	0.97	0.99
Age of diagnosis, years	0.3	0.75
GINA	0.16	0.56
Daily ICS dose (beclomethasone equivalent)*	0.55	0.92
Acq6 score	0.68	0.95
Blood eosinophils, 10^9/L	0.03	0.33
Blood neutrophils, 10^9/L	0.21	0.58
Blood monocytes, 10^9/L	0.39	0.85
Sputum eosinophils, % of non-squamous cells	0.76	0.95
Sputum neutrophils, % of non-squamous cells	0.84	0.95
FEV1 % predicted	0.19	0.58

FeNO, ppb	0.59	0.92
RV/TLC % predicted	0.41	0.85
R5-R20, % predicted	0.8	0.95
S_{cond}*VT, % predicted	0.59	0.92
S_{acin}*VT, % predicted	0.99	0.99
FEF_{50}, % predicted	0.83	0.95
1+ exacerbation last year	0.95	0.99

Supplementary Table 11. *Statistical significance of the correlation analysis between first black module eigengene with available clinical data.*

	Healthy	**T2-low**	**p**	**T2-high**	**p**	**p high vs low**
N	282	24		30		
Age, years (mean (SD))	22.28 (0.66)	22.28 (0.56)	0.988	22.33 (0.70)	0.682	0.776
Male n (%)	**110 (39.0)**	7 (29.2)	0.463	**18 (60.0)**	**0.043**	**0.047**
BMI, kg/m2 (mean (SD))	22.80 (3.54)	**26.85 (7.71)**	**<0.001**	24.18 (4.81)	0.052	0.126
Smoking, yes n (%)	65 (23.0)	3 (12.5)	0.348	8 (26.7)	0.827	0.345
Emergency visit, n/N (%)	3/43 (7.0)	5/24 (20.8)	0.199	5/28 (17.9)	0.302	1.000
Hospitalization, n/N (%)	0/42 (0.0)	1/23 (4.3)	0.758	3/27 (11.1)	0.109	0.722
ICS prescription, n/N (%)	**1/282 (0.4)**	**13/23 (56.5)**	**<0.001**	**17/29 (58.6)**	**<0.001**	1.000
Phadiatop test, positive n (%)	**84 (29.8)**	**14 (58.3)**	**0.008**	**28 (93.3)**	**<0.001**	**0.006**
Nasal steroid use, yes n/N (%)	**12/248 (4.8)**	**6/17 (35.3)**	**<0.001**	**11/18 (61.1)**	**<0.001**	0.234
Allergic rhinitis, n (%)	**29 (10.3)**	**10 (41.7)**	**<0.001**	**17 (56.7)**	**<0.001**	0.411

Blood eosinophils, 10^9/L (median [IQR])	**0.00 [0.00, 0.10]**	**0.00 [0.00, 0.10]**	0.153	**0.20 [0.20, 0.30]**	**<0.001**	**<0.001**
Blood monocytes, 10^9/L (mean (SD))	**0.49 (0.13)**	0.49 (0.12)	0.952	**0.55 (0.16)**	**0.014**	0.123
FeNO, ppb (median [IQR])	**12.00 [9.00, 17.00]**	**14.00 [10.50, 20.00]**	0.072	**31.50 [20.25, 49.25]**	**<0.001**	**<0.001**
FEV_1, % predicted (mean (SD))	97.82 (9.50)	96.41 (9.83)	0.499	**93.72 (9.68)**	**0.034**	0.336

Supplementary Table 12. *Clinical characteristics of the participants of BAMSE cohort. BMI: Body mass index, ppb: parts per billion, FEV1: Forced expiratory volume in 1 second; mean and standard deviation (SD) were calculated for normally distributed data, median and quartile 1 and quartile 3 for non-normally distributed data. T-test for normally distributed variables, Mann-Whitney U test for non-normally distributed variables and chi-square test for categorical variables was used. A comparison was performed between T2-low and non-asthmatics controls and T2-high and non-asthmatic controls.*

	Healthy	T2-low	p	T2-high	p	**p high vs low**
N	23	16		9		
Male, n (%)	**16 (69.6)**	**3 (18.8)**	**0.005**	4 (44.4)	0.361	0.363
Age, years (mean (SD))	**34.35 (11.20)**	**45.50 (12.81)**	**0.007**	**46.67 (22.07)**	**0.044**	0.868
Smoking status			0.219		0.919	0.479
Current-smoker, n (%)	0 (0.0)	2 (12.5)		0 (0.0)		
Ex-smoker, n (%)	3 (13.0)	2 (12.5)		2 (22.2)		
Non-smoker, n (%)	20 (87.0)	12 (75.0)		7 (77.8)		
BMI, kg/m^2 (mean (SD))	**25.33 (3.30)**	**29.09 (5.63)**	**0.012**	28.27 (6.16)	0.089	0.739
IgE Assay Atopy, positive, n/N (%)	**0/9**	**3/5 (60)**	**0.034**	2/4 (50)	0.058	0.767
Disease duration (mean (SD)), years		26.69 (17.53)		16.75 (8.17)		0.145

Age of diagnosis (median [IQR]), years		8.50 [3.75, 29.75]		37.50 [16.50, 42.75]		0.066
1+ exacerbation last year = Yes (%), n/N (%)		9/15 (60.0)		5/9 (55.6)		0.729
ICS usage, yes, n/N (%)		14/14 (100)		7/8 (87.5)		0.396
Acq6 score, (median [IQR])		7.00 [4.75, 10.50]		8.00 [7.00, 13.00]		0.409
Blood eosinophils, 10^9/L (median [IQR])	0.10 [0.10, 0.19]	0.10 [0.10, 0.10]	0.305	**0.60 [0.42, 0.70]**	**<0.001**	**<0.001**
Blood neutrophils, 10^9/L	**2.86 (1.40)**	**5.56 (2.73)**	**<0.001**	**4.20 (1.73)**	**0.029**	0.193
Blood monocytes, 10^9/L,	0.45 (0.15)	0.45 (0.13)	0.999	0.59 (0.25)	0.065	0.077
Sputum eosinophils, % of non-squamous cells (median [IQR])	**0.00 [0.00, 0.00]**	**0.40 [0.00, 4.13]**	**0.028**	**4.73 [3.65, 5.62]**	**0.001**	0.231
Sputum neutrophils, % of non-squamous cells (median [IQR])	**30.64 (20.58)**	**67.33 (18.54)**	**<0.001**	**64.97 (20.51)**	**0.009**	0.844
FEV_1, % predicted (mean (SD))	**106.74 (9.31)**	**87.57 (19.39)**	**<0.001**	**78.92 (21.96)**	**<0.001**	0.318
FeNO (median [IQR])	**15.50 [12.75, 26.50]**	13.00 [11.62, 17.25]	0.071	**84.00 [60.00, 90.00]**	**<0.001**	<0.001
RV/TLC, (mean (SD))	**0.24 (0.07)**	**0.30 (0.07)**	**0.020**	0.28 (0.11)	0.292	0.580
Nasal Polyps Diagnosed, yes, n/N (%)	3/22 (13.6)	1/16 (6.3)	0.534	2/8 (25.0)	0.597	0.173
Allergic Rhinitis Diagnosed, yes, n/N (%)	1/22 (4.5)	**6/16 (37.5)**	**0.024**	**6/9 (66.7)**	**0.001**	0.325

Supplementary Table 13. *Baseline clinical characteristics of the participants of U-BIOPRED cohort. BMI: Body mass index, ppb: parts per billion, FEV_1: Forced expiratory volume in 1 second; mean and standard deviation (SD) were calculated for normally distributed data, median and quartile 1 and quartile 3 for non-normally distributed data. T-test for normally distributed variables, Mann-Whitney U test for non-normally distributed variables and chi-square test for categorical variables was used. A comparison was performed between T2-low and non-asthmatics controls and T2-high and non-asthmatic controls.*

Gene	logFC	PValue	FDR	Gene name
ENSG00000124215	0.914491936275884	2.5067215574244803e-20	5.352351869412749e-16	CDH26
ENSG00000133110	2.32958365747408	7.50189832282875e-20	8.00902664945198e-16	POSTN
ENSG00000179914	3.18339010135167	5.77784927513618e-17	4.11228792409026e-13	ITLN1
ENSG00000170373	3.16246401390681	3.48360829192312e-14	1.8595501062285601e-10	CST1
ENSG00000258667	1.32910004010418	1.3459635329936199e-12	5.7478026712959505e-09	HIF1A-AS3
ENSG00000148053	1.0080280065753	2.0399205190595398e-12	7.259397153826541e-09	NTRK2
ENSG00000287059	1.38139602238639	6.305551431576389e-11	1.92337334524313e-07	AC090004.2
ENSG00000121797	0.988605298044685	8.718906477172489e-11	2.32707613875734e-07	CCRL2
ENSG00000161905	0.628061760687389	1.13294724564736e-10	2.68785439878473e-07	ALOX15
ENSG00000112877	0.431629428386669	5.7961211293105e-10	1.23758778353038e-06	CEP72
ENSG00000181634	0.814984757936401	2.13462688626976e-09	4.14350484323926e-06	TNFSF15
ENSG00000006606	1.39717185420536	4.393889337945761e-09	7.81819376198482e-06	CCL26
ENSG00000269028	1.94744081979777	2.06030772834283e-08	3.38397620119817e-05	MTRNR2L12
ENSG00000268758	0.627054306100112	1.15241016134775e-07	0.0001757590126078	ADGRE4P
ENSG00000271781	0.892160679478718	1.68627669320532e-07	0.0002400358663554	AC026740.1
ENSG00000140479	0.636153000752363	2.25167313355721e-07	0.0003004857796732	PCSK6
ENSG00000244694	1.01129187294075	4.76020264990645e-07	0.0005978814528282	PTCHD4
ENSG00000121898	1.01334959344579	7.60930913923514e-07	0.0008720231701212	CPXM2
ENSG00000179348	1.32058185190808	7.75966665057314e-07	0.0008720231701212	GATA2
ENSG00000188242	0.583978174614723	8.65181121950983e-07	0.0009236673657948	PP7080
ENSG00000171097	0.470805654287801	9.50101164394381e-07	0.0009660266696261	KYAT1

ENSG00000164038	0.554646621214479	1.01645946068845e-06	0.0009865201093009	SLC9B2
ENSG00000143147	0.658623711567281	1.09410709408967e-06	0.0010157119423044	GPR161
ENSG00000181523	0.264383256894246	1.2694360491393e-06	0.0011293749383842	SGSH
ENSG00000187775	1.14378668860692	1.41035137089003e-06	0.0012045528988497	DNAH17
ENSG00000158669	0.154462841093136	2.14690135870523e-06	0.0017203426407566	GPAT4
ENSG00000130222	0.355629963154474	2.17540517517934e-06	0.0017203426407566	GADD45G
ENSG00000135272	-0.303122135271727	2.37213178982835e-06	0.0018089199277291	MDFIC
ENSG00000148700	-0.191593614244966	4.05148022649457e-06	0.0029830070964176	ADD3
ENSG00000160271	0.254004395298525	5.16432686634102e-06	0.0036604783181109	RALGDS
ENSG00000099994	0.816542663951747	5.31448238391898e-06	0.0036604783181109	SUSD2
ENSG00000197253	1.56289930421471	5.87145459816236e-06	0.0039177280806238	TPSB2
ENSG00000198324	0.267686103541778	6.32335400137701e-06	0.0040914016556788	PHETA1
ENSG00000183196	0.374643122177908	8.48910448520834e-06	0.0053311576167108	CHST6
ENSG00000172236	1.45368952184013	9.43300864351189e-06	0.0057546743016076	TPSAB1
ENSG00000066230	1.10104709637044	1.24513420960054e-05	0.0073850293453863	SLC9A3
ENSG00000225138	0.451153600942865	2.19659959200673e-05	0.0126036753732651	SLC9A3-AS1
ENSG00000120314	0.127301648461709	2.32342972430324e-05	0.0126036753732651	WDR55
ENSG00000163751	1.42777428331994	2.35244047923519e-05	0.0126036753732651	CPA3
ENSG00000274114	0.419592982202581	2.39202439849031e-05	0.0126036753732651	ALOX15P1
ENSG00000087494	0.908191276646457	2.42015122847448e-05	0.0126036753732651	PTHLH
ENSG00000185340	0.269610219472566	2.57521091526544e-05	0.0128818339295865	GAS2L1
ENSG00000162337	0.242374611486927	2.59422470481557e-05	0.0128818339295865	LRP5
ENSG00000086696	0.515645312538829	2.70010835105721e-05	0.0129766001805534	HSD17B2
ENSG00000279970	0.840698642988965	2.73485859931109e-05	0.0129766001805534	AC023024.2
ENSG00000122545	-0.134636159138352	2.90798330868626e-05	0.0133649132605277	SEPTIN7
ENSG00000277632	1.05634082424012	2.94188330481829e-05	0.0133649132605277	CCL3
ENSG00000148180	0.269477045685744	3.10526099154088e-05	0.013813235977371	GSN
ENSG00000107281	0.442643685760298	3.32182064076775e-05	0.0144750029227904	NPDC1

ENSG00000175087	-0.179970795748844	3.54196389029774e-05	0.0150257557380731	PDIK1L
ENSG00000105523	0.208010292011024	3.60069953990318e-05	0.0150257557380731	FAM83E
ENSG00000280800	0.91965963690714	3.65932605086081e-05	0.0150257557380731	FP671120.6
ENSG00000171566	-0.163533749267461	3.79487182018657e-05	0.015288321340495	PLRG1
ENSG00000149534	1.32375245037998	3.88980315086539e-05	0.0153805697920885	MS4A2
ENSG00000158373	-0.256816110185701	4.14721536460355e-05	0.0161002440845482	H2BC5
ENSG00000110429	-0.146121175695642	4.30796580552995e-05	0.0164256581927992	FBXO3
ENSG00000164309	0.412889272623044	4.62075495566583e-05	0.0169651775963266	CMYA5
ENSG00000185274	0.932589486012526	4.67493954112801e-05	0.0169651775963266	GALNT17
ENSG00000130234	-0.27846962382069	4.68783007766611e-05	0.0169651775963266	ACE2
ENSG00000185800	0.185049160975594	5.0818921190862e-05	0.0180847600877881	DMWD
ENSG00000177225	0.206922133931555	5.34284171684261e-05	0.0187016977603317	GATD1
ENSG00000163002	-0.175062963800365	5.48988105147373e-05	0.0189064419695269	NUP35
ENSG00000134283	-0.101288138714634	5.62278063744219e-05	0.0190567638366136	PPHLN1
ENSG00000131446	0.248421669421763	5.72347389198573e-05	0.0190949397721374	MGAT1
ENSG00000177548	0.201629418944862	6.12505158531875e-05	0.0201158966584375	RABEP2
ENSG00000119408	0.312514439249004	6.21791485320756e-05	0.0201158966584375	NEK6
ENSG00000130822	0.810921684414092	6.4907246459613e-05	0.0206850675582934	PNCK
ENSG00000099860	0.599880359830868	6.65295264853645e-05	0.0208902713164045	GADD45B
ENSG00000131149	0.15176570874625	6.79292212972924e-05	0.0209860421767928	GSE1
ENSG00000276085	1.26883743813277	6.90526455782093e-05	0.0209860421767928	CCL3L3
ENSG00000280614	1.09601243433469	6.97831113971661e-05	0.0209860421767928	FP236383.4
ENSG00000184678	-0.195218514670586	7.38073407517246e-05	0.0218879769407059	H2BC21
ENSG00000129946	0.681909697630139	7.61757025941838e-05	0.0222808712574111	SHC2

ENSG00000166348	0.153115234203874	7.7559197996319e-05	0.0223789729137487	USP54
ENSG00000066654	-0.126332822992588	8.01451384577569e-05	0.0228167866180003	THUMPD1
ENSG00000122692	-0.126123595333463	8.22030599270142e-05	0.0228342652062782	SMU1
ENSG00000152382	-0.159410464729139	8.30880474793791e-05	0.0228342652062782	TADA1
ENSG00000152061	-0.158184517599117	8.44017084204891e-05	0.0228342652062782	RABGAP1L
ENSG00000168827	-0.135975394462192	8.44842146541765e-05	0.0228342652062782	GFM1
ENSG00000129460	-0.155778905476297	8.81625086064998e-05	0.0235305735470748	NGDN
ENSG00000022567	0.301698207905007	8.96433982742639e-05	0.0236304424685442	SLC45A4
ENSG00000233251	0.69778376886006	9.78665549169802e-05	0.0254511662381707	AC007743.1
ENSG00000110931	0.212622691125497	9.89343760663249e-05	0.0254511662381707	CAMKK2
ENSG00000168621	0.863101630821217	0.0001002485277222	0.0254822209991022	GDNF
ENSG00000198087	-0.146846118323021	0.0001028272215553	0.0258301980547151	CD2AP
ENSG00000066422	-0.116156177738559	0.0001104508302188	0.0272710236143334	ZBTB11
ENSG00000102970	0.535160638812401	0.0001111174154386	0.0272710236143334	CCL17
ENSG00000133619	0.291151678104392	0.0001138671834795	0.0276283193369982	KRBA1
ENSG00000138821	0.278621356670021	0.0001202062498039	0.0286301691734924	SLC39A8
ENSG00000132386	0.391615412920738	0.0001209231455526	0.0286301691734924	SERPINF1
ENSG00000182601	0.960457209044022	0.0001230485976194	0.0286301691734924	HS3ST4
ENSG00000150540	-0.290141339235564	0.0001233596648539	0.0286301691734924	HNMT
ENSG00000086065	-0.179063298087884	0.0001320524159782	0.0303180987738517	CHMP5
ENSG00000169750	0.318299958653698	0.0001338682433609	0.0304080290664063	RAC3

ENSG00000009844	-0.137887983817916	0.0001390600065635	0.031254834317322	VTA1
ENSG00000140287	0.549903437530749	0.0001414451583525	0.031459760636923	HDC
ENSG00000281181	1.08399778369869	0.0001432866187268	0.0315407822995477	FP236383.5
ENSG00000113430	1.11321137674879	0.0001474183253937	0.0321191437123299	IRX4
ENSG00000121966	0.76253959543275	0.0001530939874296	0.032239044352575	CXCR4
ENSG00000105641	0.932670940878883	0.000153132913481	0.032239044352575	SLC5A5
ENSG00000131620	0.608027972481408	0.0001532571140756	0.032239044352575	ANO1
ENSG00000112365	-0.145017057013081	0.0001543742869773	0.032239044352575	ZBTB24
ENSG00000128271	1.10488675682285	0.0001555180577142	0.032239044352575	ADORA2A
ENSG00000159788	0.146726105647483	0.0001665234411956	0.0341885434270103	RGS12
ENSG00000006530	-0.123086068236383	0.0001738887273132	0.0353606867199294	AGK
ENSG00000115756	0.330965190490244	0.00018240946617	0.0364970861804186	HPCAL1
ENSG00000077254	-0.109120365207736	0.0001828956641675	0.0364970861804186	USP33
ENSG00000117266	0.224410182056896	0.0001848693622947	0.0365493576270106	CDK18
ENSG00000130669	0.183489735295629	0.0001875443409894	0.0367380437505145	PAK4
ENSG00000163714	-0.116668218213109	0.0001913810794449	0.0371488073482574	U2SURP
ENSG00000205903	0.20107225982938	0.0001933740829479	0.0371975082802289	ZNF316
ENSG00000012660	0.185033399543031	0.0002024667880495	0.0385988469503005	ELOVL5
ENSG00000116194	-0.805883929882622	0.0002207460559233	0.041711237044911	ANGPTL1
ENSG00000113851	-0.146386318297083	0.0002259359299736	0.0423174033052312	CRBN
ENSG00000163870	0.198504769976449	0.0002388721471007	0.0439831312169798	TPRA1
ENSG00000127993	-0.120162793823324	0.0002408222171226	0.0439831312169798	RBM48
ENSG00000026508	0.330041762282656	0.0002415326012303	0.0439831312169798	CD44

ENSG00000164830	-0.162848089813398	0.0002430690091609	0.0439831312169798	OXR1
ENSG00000123268	-0.18503620467137	0.0002469012621902	0.0443011407587052	ATF1
ENSG00000102189	-0.154557540201494	0.0002495036938435	0.0443950239245548	EEA1
ENSG00000145780	-0.21293804007531	0.0002520762989605	0.0444820920281513	FEM1C
ENSG00000165458	0.162794286245275	0.0002568616314464	0.044954996349545	INPPL1
ENSG00000187257	-0.106392073911923	0.0002654223932403	0.0457734887495834	RSBN1L
ENSG00000250385	1.05386863354247	0.0002675862955144	0.0457734887495834	AC106772.2
ENSG00000004766	-0.144517316537455	0.000267969562275	0.0457734887495834	VPS50
ENSG00000154803	0.172646977879658	0.0002725854348181	0.0461924143193498	FLCN
ENSG00000188612	-0.144106590264289	0.0002774062849576	0.046477441722351	SUMO2
ENSG00000065485	0.217270481163732	0.0002789522944017	0.046477441722351	PDIA5
ENSG00000115604	0.351925613386932	0.0002807975825301	0.046477441722351	IL18R1
ENSG00000128563	0.140604823364543	0.0002877538784347	0.0472624677872258	PRKRIP1
ENSG00000169251	-0.137158690000331	0.0002940404478135	0.0479263484100306	NMD3

Supplementary Table 14. *Differentially expressed genes between Th3-high asthma and non-asthmatic controls in the BAMSE cohort.*

Gene Symbol	logFC	AveExpr	t	P.Value	adj.P.Val	B
LIMCH1	-4.779350121485561	0.0551349412237421	-5.459716643387226	4.630842923008618e-07	0.0091156766446479	5.919624332168832

CLC	1.9025614085023204	-0.0066478527446401	5.299021075731429	9.010701966735504e-07	0.0091156766446479	5.331165678899675
MUCL1	-1.7308892097091482	0.0222287018020976	-5.141138522898834	1.7180351688237194e-06	0.0108342738740312	4.760868520479393
PTHLH	6.041873256099357	-0.1496735896757035	5.086680602971475	2.141901620922499e-06	0.0108342738740312	4.566067011974178
DCXR	1.7151794658578368	0.0205334641193332	4.965491693066225	3.4848235508240417e-06	0.0132150318481022	4.136247456498349
PROC	-1.779642033198374	0.0049458247222982	-4.8983018331791355	4.553160270362384e-06	0.0132150318481022	3.900212169827503
FSCN1	2.9821941505760416	-0.0250922955558429	4.866147687714482	5.171466820533058e-06	0.0132150318481022	3.7878452240190104
S100A2	1.7764271425734788	0.0066626314476524	4.863535806271192	5.225139859873379e-06	0.0132150318481022	3.778734636676621
ADAMTSL3	2.36686546470032	-0.0026438730752535	4.790525303706987	6.965598593526289e-06	0.0147886091953283	3.52511161602461
ODAM	1.605248744445202	-0.0217596240526877	4.758971217509809	7.881829705069916e-06	0.0147886091953283	3.4161340545875873
TSPAN6	-3.2944526172394815	0.0113736616968229	-4.75388579697794	8.040068262176242e-06	0.0147886091953283	3.3986070157056165
SLC6A8	4.8532431642143505	-0.0288720348391354	4.689523951388974	1.0330190415584072e-05	0.017417561889876	3.1776655455936886

C12orf54	3.2115796549517137	-0.0530523218998892	4.610018034138317	1.4044539280205649e-05	0.0194635108340726	2.9070410476141237
UBASH3B	4.656782851946089	-0.0281073320701454	4.589902742488765	1.5172908119880892e-05	0.0194635108340726	2.838983939476611
ST8SIA4	-7.088057400239759	0.0451256051697724	-4.5660320572283615	1.6626293329518593e-05	0.0194635108340726	2.758440128654744
RBM4	-2.389704071869143	0.0639347237955807	-4.561587163043704	1.6911445268909754e-05	0.0194635108340726	2.7434687093661765
GNS	-3.614482039784688	0.0594557747473251	-4.559109569283644	1.7072441981473415e-05	0.0194635108340726	2.7351272213109237
ABCB11	-2.7298562282241146	0.0338830817633921	-4.5450344424588955	1.8015646939782614e-05	0.0194635108340726	2.6877887242131635
TK1	2.718350013459264	0.0010097965409784	4.541254930236096	1.827740354111498e-05	0.0194635108340726	2.675091482198292
RALY	1.6698187594359624	0.0011448658830533	4.496333486383272	2.1685015781792475e-05	0.0219376462156503	2.524645035850577
MARCH6	-3.6240323239253502	0.0334400227896848	-4.38845374103756	3.257083432091526e-05	0.0307022866329309	2.166928431253993
PROM1	-1.596342085805577	0.0094547689067484	-4.365960778457932	3.542981850467341e-05	0.0307022866329309	2.0929951462371834
HMGB2	2.194404805917456	0.00117474872476	4.359498561964961	3.629507927899519e-05	0.0307022866329309	2.0717963362997063
A4GALT	1.6012662836988398	7.097036363708777e-15	4.3585891743714615	3.6418468798020175e-05	0.0307022866329309	2.068814675939312
FAM5C	1.5495842986917734	-0.0197843168754972	4.330754553703908	4.039742666386412e-05	0.0314845757744486	1.9777334614320925

C19orf55	-2.744025158509923	0.0221978395644701	-4.3303475942563665	4.045860574979807e-05	0.0314845757744486	1.9764044166736907
CLDND1	3.2465375414420423	-0.0024434736635131	4.269415584078958	5.07017788801834e-05	0.0326387178044979	1.7782735681155757
HIST1H1B	1.567541948503917	-0.0004781460689399056	4.265825369848044	5.137762223724329e-05	0.0326387178044979	1.7666530911218397
C6orf105	-2.524808102533546	0.0055905042570158	-4.256463019320238	5.318116321089408e-05	0.0326387178044979	1.736378245901042
FAM83D	1.5782135401854602	-0.0014655639543529	4.250820101808663	5.429752254165853e-05	0.0326387178044979	1.7181506960535344
AASS	-2.5537486050954037	0.0239530760080502	-4.250645375942895	5.4332447292632086e-05	0.0326387178044979	1.7175865413971003
CD109	3.9374175472738178	-0.0093443404247517	4.248911542971314	5.468018694680508e-05	0.0326387178044979	1.7119891209752636
BTC	-2.4502064377216524	0.0269066916676285	-4.248885341885386	5.468545828876842e-05	0.0326387178044979	1.711904545522776
SAMHD1	-6.3863178620494265	0.0972673201635671	-4.234219214710037	5.771427584873707e-05	0.0326387178044979	1.664613930985154
IGLV6-57	1.537790350191884	-0.0184889560102396	4.2306533964394095	5.8474757489004745e-05	0.0326387178044979	1.6531313591686
DPY19L2P2	-1.5565476196488843	0.007550982940754	-4.227442240833615	5.9167856296813395e-05	0.0326387178044979	1.6427960092742593

RABGAP1L	-3.95365534694517	0.0508545351962487	-4.222511875872436	6.0247461764801005e-05	0.0326387178044979	1.62693675460904
SLC39A8	5.722025316223425	-0.002607019226217	4.213144343424509	6.235119171143774e-05	0.0326387178044979	1.5968364548610774
LOC283547	-1.5239038411387424	-0.015838711096191	-4.210696099177612	6.291256829809818e-05	0.0326387178044979	1.588976487779635
PSME4	-3.5443282691388696	0.0354610289055058	-4.194590084241389	6.672884745595823e-05	0.0332633301815895	1.537340148367404
CAV1	2.697289480226297	-0.0233650302677803	4.1918310858093975	6.740456370509417e-05	0.0332633301815895	1.5285071557256993
AGAP2	3.081436500261385	-0.0194445595766109	4.182728769707436	6.968088343793459e-05	0.0335679360618983	1.4993917673728374
DEFB1	1.4717678065011042	-0.0040630070557087	4.172228924900729	7.239864651931711e-05	0.0340660887215196	1.4658555742465298
ENPEP	2.640587105140991	-0.0118676138357668	4.155805643073403	7.685527639525194e-05	0.034978414311045	1.4135067309173772
POSTN	5.088562839548578	-0.0450270719272491	4.142309356618318	8.071409993268206e-05	0.034978414311045	1.3705854277216636
SERPINB4	2.7889481599498587	-0.0021262338772301	4.141838040252538	8.085216997616729e-05	0.034978414311045	1.3690881327846345
VSTM1	1.4148664075344572	-0.0342524601337773	4.140475224991983	8.125267991000434e-05	0.034978414311045	1.364759300236981

USH1C	-3.747129965392578	-0.0067703689800289	-4.1307704486178665	8.416043964914174e-05	0.0354753786546059	1.3339593244401025
CD44	14.814922764644336	-0.0335073551107203	4.116242561959031	8.87010683668692e-05	0.0363794834837816	1.2879383164560965
C10orf96	1.5969109442416851	-0.0024566070007409	4.109580766035912	9.08610840265106e-05	0.0363794834837816	1.266869889645097
PALLD	-5.070020963390563	0.0165648062319213	-4.103483543330036	9.288228299162853e-05	0.0363794834837816	1.247606098547572
GSTA1	-2.3769882279375825	0.027222530563198	-4.101653216748674	9.349741220563672e-05	0.0363794834837816	1.2418268707673947
HIST1H2AL	1.5329177243180012	-0.0094046453493713	4.091741039577453	9.689725602100384e-05	0.0369909845485466	1.2105580470680437
GSTA3	-1.528034556278688	0.0041076523077729	-4.071120501003683	0.00010435458829104738	0.0373998526330898	1.1456646658371668
SERPINB13	7.162871159948992	-0.021315168358635	4.070049623528552	0.00010475655384833626	0.0373998526330898	1.1423003571330277
GCNT3	1.5501217241405822	-0.00036539893944428654	4.069607326791358	0.00010492300825132378	0.0373998526330898	1.1409109878370378
SLC5A5	1.5330871204924348	-0.0014347112788716	4.068254083680907	0.00010543386787671604	0.0373998526330898	1.1366607038941652
RGS4	2.808008191199937	-0.0147131261621725	4.063596216164011	0.00010721056950127066	0.0373998526330898	1.1220381910298487

TAGLN3	-1.5051910342994652	0.0083473486433759	-4.026332583877839	0.00012249654593900346	0.0414773376144599	1.0054470898748349
GGN	2.22234412484314	-0.0062218123714062	4.02518483406907	0.00012299907363552584	0.0414773376144599	1.0018670798852645
PATL2	-1.4693822585161724	-6.26527543952792e-16	-3.998690524468161	0.00013516429323479697	0.044411057566187	0.9194126795136236
CST4	1.5424214226323718	-0.0003147038198456326	3.9929263987950647	0.00013795970680304906	0.044411057566187	0.9015210792807586
EML1	-2.499149263190268	0.0083966604157091	-3.991184047833463	0.00013881554070755965	0.044411057566187	0.896116231622611
C16orf74	1.5149574061557234	-0.0063680384164978	3.9878271371176655	0.00014047880612049485	0.044411057566187	0.8857073204318473
BEST2	1.4212739511146042	-0.0153808470223395	3.981708576086063	0.0001435596549918638	0.044686807683852	0.8667500653866789
LMTK3	1.5271397099766206	5.96832586473571le-16	3.971574283265444	0.00014880544499178597	0.0456178874018	0.8353929772547195
CST1	1.5353497959610618	-5.590779867773118e-16	3.9481121883526886	0.00016166393235921255	0.0481643085019538	0.7630002921518368
GALNT13	-3.76658905709779	-0.0064991032235133	-3.947745976827544	0.00016187283043210912	0.0481643085019538	0.7618725922702723
SUSD2	2.1659696146974468	-0.0347644804726923	3.938226680062592	0.00016739448447089804	0.0484763072291108	0.7325835458115693

LOC728175	-2.262631174387389	0.0076770826770087	-3.928990621204769	0.0001729238697996889	0.0484763072291108	0.7042109100953642
GPR125	-2.2754985900163485	0.001780370357671	-3.92726377257433	0.0001739769019436763	0.0484763072291108	0.6989110533282483
BLVRB	1.442264679229229	0.0088219122483431	3.925749529884409	0.00017490533429182615	0.0484763072291108	0.6942649812414867
CST2	1.524349128683319	-5.684045618700635e-16	3.920115445678517	0.00017840158956048427	0.0484763072291108	0.6769887069944573
IL1R1	2.628010410158006	0.0017597677179974	3.912707029990555	0.00018310079054130633	0.0484763072291108	0.6542968604727779
PRB1	3.461508402208741	-0.1543108155545585	3.9113556602681543	0.00018397064510056264	0.0484763072291108	0.6501607321559835
CCBL1	3.4944165985903983	-0.0262591661124348	3.911015943167192	0.0001841899364058774	0.0484763072291108	0.6491211125430381
PDXK	4.185144415840902	0.009949811349302	3.910560167006424	0.00018448453796478712	0.0484763072291108	0.6477264179205058
HRH4	2.307388494301167	-0.0406091359729956	3.897363975868021	0.00019321229748420904	0.048597725562018	0.607392702080106
ATP5D	2.8855879688896	0.0017061829726869	3.895237787577016	0.0001946549511144859	0.048597725562018	0.6009026270593196
TMEM45B	-2.313175175145388	0.0378813529425283	-3.895097912851897	0.00019475021861732896	0.048597725562018	0.6004757503067815
IMPA1	-1.4368857754047446	0.0106787481557856	-3.8932184599225854	0.000196034630165379	0.048597725562018	0.5947409387013627

CISH	3.558016603537992	-0.0792821622854447	3.890289011817363	0.00019805276847605056	0.048597725562018	0.5858059617341747
SPAG1	-1.2255511589670602	0.0398705924448412	-3.886348865207444	0.0002007985358578132	0.048597725562018	0.5737954249809825
RNASE2	1.4767469744716892	-0.0083457883130547	3.883502225902885	0.00020280487427463736	0.048597725562018	0.5651232588385611
PDHA1	2.9280916591987585	0.0393173170848137	3.8815921405765983	0.00020416184810811711	0.048597725562018	0.559306661148911
GDAP1L1	1.4920712099746092	-0.0035521965841208	3.870687939101861	0.00021207619892945736	0.0498899475902342	0.5261380653379657
SCAF1	1.4698558591886774	-0.0122102786407091	3.8655877946686616	0.00021587776352702633	0.0498899475902342	0.5106459554962024
GALR3	1.366491700664883	-0.0262151074268239	3.864114681878819	0.00021698786081849505	0.0498899475902342	0.506173820767482
CCL26	1.4438946561651993	-0.0175277483200303	3.858220875002572	0.00022148407382079665	0.049904304423504	0.4882926908417175
NOTCH2	-4.395022482636392	0.0394967542459622	-3.857573580608105	0.0002219832648700324	0.049904304423504	0.4863299990733081

Supplementary Table 15. *Differentially expressed genes between T2-high asthma and non-asthmatic controls in the U-BIOPRED cohort.*

Gene	logFC	PValue	FDR	external_gene_name
ENSG00000140465	4.804	4.406e-12	9.408e-08	CYP1A1

Supplementary Table 16. *Differentially expressed genes between T2-low asthma and non-asthmatic controls in the BAMSE cohort.*

GeneSymbol	logFC	AveExpr	t	P.Value	adj.P.Val	B
CHST2	-1.686	0.001	-5.411	5.661e-07	0.011	-3.848

Supplementary Table 17. *Differentially expressed genes between T2-low asthma and non-asthmatic controls in the U-BIOPRED cohort.*